

CLASSIFICATION RESULTS FOR CONFORMALLY KÄHLER GRAVITATIONAL INSTANTONS

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ABSTRACT. We investigate the asymptotic geometry of Hermitian non-Kähler Ricci-flat metrics with quadratic curvature decay and apply the resulting description to the classification of Hermitian non-Kähler gravitational instantons. Our main results are as follows.

- Every such end is asymptotic to an ALE, ALF-A, AF, generalized special Kasner, or ALH* model, or to a finite quotient of one of these models.
- Every such gravitational instanton of sub-Euclidean volume growth is Kerr, Chen–Teo, Taub-bolt, or reversed Taub-NUT. This confirms a conjecture of Aksteiner and Andersson.
- Every such gravitational instanton is a Zariski-open subset of an algebraic surface.

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1. INTRODUCTION

A *gravitational instanton* (M, h) in this paper is a nonflat, complete, noncompact, oriented Ricci-flat four-manifold with quadratic curvature decay,

$$|\mathrm{Rm}_h|_h = O(r^{-2}).$$

For a complete Ricci-flat four-manifold, having quadratic curvature decay is equivalent to having finite $\int |\mathrm{Rm}_h|_h^2 d\mathrm{vol}_h$. We study the geometry at infinity of Hermitian non-Kähler Ricci-flat metrics, with particular emphasis on gravitational instantons that have sub-Euclidean volume growth. Under this curvature control, we classify the possible asymptotic geometries and then use that classification to classify Hermitian non-Kähler gravitational instantons.

We first explain why the Hermitian non-Kähler condition is natural in this setting. For four-dimensional Ricci-flat metrics, it is equivalent to being *nontrivially conformally Kähler*. By saying a metric h is non-trivially conformally Kähler, we mean there is a conformal change $g = f^2 h$ with f being non-constant and a complex structure J , such that g is Kähler under J . Following Derdziński [33], oriented Einstein four-manifolds, and hence oriented gravitational instantons, can be divided into three types according to the eigenvalue structure of the self-dual Weyl tensor W^+ .

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Definition-Proposition. Regard W^+ as a trace-free symmetric endomorphism of the bundle Λ^+ of self-dual two-forms. Four-dimensional Ricci-flat metrics fall into the following three classes:

- Type I: $W^+ \equiv 0$. The metric is anti-self-dual and therefore locally hyperkähler.
- Type II: W^+ has exactly two distinct eigenvalues at every point. After passing, if necessary, to a double cover,¹ there is a compatible complex structure J for which (M, h, J) is Hermitian and $g = \lambda^{2/3}h$ is Kähler, where $\lambda := 2\sqrt{6}|W^+|_h$.
- Type III: W^+ has three distinct eigenvalues on a dense open set. With the chosen orientation, the metric is not locally conformally Kähler.

Types I and II carry additional geometric structure:

- Type I gravitational instantons are anti-self-dual and, in particular, locally hyperkähler.
- Up to a double cover, Type II gravitational instantons are the Hermitian non-Kähler ones. The conformal Kähler metric $g = \lambda^{2/3}h$ is extremal in the sense of Calabi, with scalar curvature $s_g = \lambda^{1/3}$ and extremal vector field $\mathcal{K} = J\nabla_g \lambda^{1/3} = -J\nabla_h \lambda^{-1/3}$.

Our first main result classifies the asymptotic cones and asymptotic models of the above Type II Ricci-flat metrics with quadratic curvature decay.

Theorem 1.1. *Let (M, h) be a Type II Hermitian non-Kähler Ricci-flat end in the sense specified in Section 3.4. Assume that it is complete at infinity and has quadratic curvature decay. Then (M, h) is asymptotic at a polynomial rate to an ALE, ALF-A, AF, generalized special Kasner, or ALH* model, or to a finite quotient of one of these models.*

Section 2.4 describes these models and the finite quotients that occur. Before taking further finite quotients, the corresponding asymptotic cones are given by the following table

Model	ALE	ALF-A	AF	generalized special Kasner	ALH*
Cone	$\mathbb{R}^4/\Gamma$	$\mathbb{R}^3$	$\mathbb{R}^3/\mathbb{Z}_k$ or $\mathbb{H}$	$[0, \infty)$	$[0, \infty)$

In $\mathbb{R}^3/\mathbb{Z}_k$, the group $\mathbb{Z}_k$ acts by rotations about a fixed axis. The convergence to a generalized special Kasner or ALH* model is in fact at a rate faster than every polynomial. Neither a generalized special Kasner model nor an ALH* model can occur as the end of a Hermitian non-Kähler gravitational instanton. By contrast, Type III gravitational instantons asymptotic to the zero-parameter special Kasner model do exist [41]; these examples have infinite fundamental group and are not asymptotically flat. Specializing the preceding theorem to gravitational instantons gives the following statement.

Theorem 1.2. *Every Type II Hermitian non-Kähler gravitational instanton is asymptotic to a model at infinity. More precisely:*

- The gravitational instanton is ALE, with asymptotic cone $\mathbb{R}^4/\Gamma$.
- The gravitational instanton is ALF-A, with asymptotic cone $\mathbb{R}^3$, or a further finite quotient.
- The gravitational instanton is AF, with asymptotic cone $\mathbb{R}^3/\mathbb{Z}_k$ or $\mathbb{H}$.

Theorems 1.1 and 1.2 contrast sharply with the hyperkähler case. Sun–Zhang [59] proved that a hyperkähler gravitational instanton is asymptotic to an ALE, ALF, ALG, ALH, ALG*, or ALH* model. Note that not all of these models could appear for a complete hyperkähler gravitational instanton. For example, there are restrictions on the degree for ALG* and ALH*. A priori, all of these models might also have occurred in the Hermitian non-Kähler setting; the preceding theorem rules out most of them.

We next turn to classification. The classification of hyperkähler gravitational instantons is now complete, following a substantial body of work [4, 12, 25, 26, 52, 53, 16, 17, 18, 27, 19, 28, 39, 40,

¹The need to allow a double cover was pointed out to the author by Claude LeBrun.

[43, 42, 20, 49]. The known Hermitian non-Kähler examples and their underlying topological spaces are as follows:

- ALE gravitational instantons:
 - The Eguchi-Hanson metric on $T^*\mathbb{P}^1$ with reversed orientation.
- AF gravitational instantons:
 - Kerr metrics on $\mathbb{R}^2 \times S^2$;
 - Chen-Teo metrics on $\mathbb{P}^2 \setminus S^1$.
- ALF-A gravitational instantons:
 - the Taub-NUT metric on $\mathbb{C}^2$ with reversed orientation;
 - the Taub-bolt metric on $\mathbb{P}^2 \setminus \{point\}$ with both orientations.

The usual Taub-NUT metric admits a compatible complex structure whose orientation is opposite to the standard hyperkähler orientation; this is what we call the *reversed Taub-NUT* space. The term *reversed Eguchi-Hanson* has the analogous meaning. The Schwarzschild metric is a familiar member of the Kerr family. All of the examples above are explicit; see, for instance, [35, 1, 23]. They are toric: each admits an effective T^2 -action by holomorphic isometries. Aksteiner and Andersson [1] were the first to observe that Chen-Teo spaces are Hermitian non-Kähler, and they conjectured that the ALF/AF gravitational instantons in the above list constitute the complete set of Type II Hermitian non-Kähler ALF/AF gravitational instantons. Toric Hermitian non-Kähler gravitational instantons with ALF-type asymptotics were classified by Biquard-Gauduchon [9], under an additional assumption on the decay rate δ_0 . Their notion of ALF is broader than ours and includes our ALF-A and AF models, as well as finite quotients of ALF-A models. The Chen-Teo spaces are also studied in detail in [8].

Conjecture 1.3 (Aksteiner-Andersson). *If (M, h) is a Type II Hermitian non-Kähler ALF/AF gravitational instanton, it must be one of the Kerr metrics, the Chen-Teo metrics, the Taub-bolt metric with both orientations, or the reversed Taub-NUT metric.*

Our second main result removes the symmetry assumption:

Theorem 1.4. *Any Type II Hermitian non-Kähler gravitational instanton (M, h) that is asymptotic to ALF-A, a finite quotient of ALF-A, or AF is holomorphically and isometrically toric: an effective T^2 acts on (M, h) by holomorphic isometries.*

Combining this result with Biquard-Gauduchon’s toric classification [9], Theorem 1.2, and our earlier work on the ALE case [50] gives the following theorem. It proves the Aksteiner-Andersson conjecture and classifies the Hermitian non-Kähler gravitational instantons that have sub-Euclidean volume growth.

Theorem 1.5. *Every Type II Hermitian non-Kähler gravitational instanton (M, h) belongs to one of the following classes:*

- ALE, in which case it corresponds to a special Bach-flat Kähler orbifold in the terminology of [50].
- ALF-A, in which case it is either the reversed Taub-NUT metric, or the Taub-bolt metric with both orientations.
- AF, in which case it is one of the Kerr metrics or the Chen-Teo metrics.

The conjecture in [1] was formulated based on a comparison to the compact case. A complete classification for compact Hermitian non-Kähler Einstein 4-manifolds was obtained by LeBrun through a series of works [45, 21, 46], where all the metrics turn out to be toric. Our key observation for Theorem 1.4 lies in a compactification phenomenon for Hermitian non-Kähler ALF/AF gravitational instantons. The compactified surfaces enjoy certain positivity properties. An intriguing outcome that we will establish while proving our second result is the following theorem.

Theorem 1.6. *Any Type II Hermitian non-Kähler gravitational instanton (M, h) asymptotic to ALF-A, to a finite cyclic quotient of ALF-A, or to a rational AF model admits a natural compactification obtained by adding a holomorphic sphere D at infinity. The resulting pair $(\overline{M}, D)$ is a toric log del Pezzo pair; in the finite-quotient and rational AF cases, $\overline{M}$ may have orbifold singularities.*

This compactification theorem is an intermediate step in the proof of Theorem 1.5. The latter theorem shows, in particular, that no Type II Hermitian non-Kähler gravitational instanton is asymptotic to a nontrivial finite quotient of an ALF-A model.

The compactification is natural in the following sense. In these cases the extremal vector field K generates a holomorphic S^1 -action that extends to the compactified surface, and the added sphere D is fixed by the extended action. The construction is also compatible with the Kähler metric g : the metric is asymptotic to a bundle of cusps over a sphere, and one compactifies each cusp. Thus the extremal Kähler metrics are of *Poincaré type*. Irrational AF gravitational instantons can also be compactified, but not canonically by adding a fixed curve. From this perspective, the underlying complex manifolds of the Taub-bolt, Kerr, and Chen–Teo spaces are most naturally viewed as $\mathcal{O}_{\mathbb{P}^1}(1)$ (or $\mathcal{O}_{\mathbb{P}^1}(-1)$ after reversing orientation), $\mathbb{C} \times \mathbb{P}^1$, and $\mathrm{Bl}_p(\mathbb{C} \times \mathbb{P}^1)$, respectively.

Our result also reveals a notable contrast with LeBrun’s classification of compact Hermitian non-Kähler Einstein four-manifolds. LeBrun [45] proved that compact Hermitian non-Kähler Einstein metrics occur only on del Pezzo surfaces with continuous holomorphic symmetry. He later showed [46] that the Page metric [56] and the Chen–LeBrun–Weber metric [21] exhaust the compact Hermitian non-Kähler Einstein four-manifolds. In the noncompact setting, our result shows that every Type II Hermitian non-Kähler ALF/AF gravitational instanton can be compactified to a toric log del Pezzo pair by adding one rational curve. Our earlier work [50] establishes an analogous phenomenon in the ALE case. As a consequence of Theorem 1.2, Theorem 1.6, and [50], we have the following corollary.

Corollary 1.7 (Algebraicity). *Every Type II Hermitian non-Kähler gravitational instanton (M, h) is biholomorphic to $X \setminus D$ for some algebraic surface X and divisor D .*

Corollary 1.7 should be compared to a compactification conjecture proposed by S-T. Yau [62]. Yau proposed:

Let M be a complete Kähler manifold with zero Ricci curvature. Prove that M is a Zariski open set of some compact Kähler manifold. If this is true, we shall have algebraic means to classify these manifolds.

Under the hypotheses above, our corollary shows that any Type II gravitational instanton, after passing to a double cover, is a Zariski-open subset of an algebraic surface. Sun–Zhang [59], building on earlier work, proved the analogous statement for hyperkähler gravitational instantons: there exists a complex structure for which it is algebraic in the sense of Corollary 1.7.

We also mention several related developments. Aksteiner et al. [2] proved that an ALF gravitational instanton with an S^1 -action generated by a Killing field of bounded norm at infinity and with the topology of a Kerr or Taub-bolt space is necessarily Kerr or Taub-bolt; they also proved that the analogous metric with Chen–Teo topology is Hermitian. Biquard–Ozuch [13] studied the stability of conformally Kähler Einstein metrics, while Biquard–Gauduchon–LeBrun [10] studied deformations of Type II Hermitian non-Kähler ALF gravitational instantons.

Main ideas and organization of the paper.

- Collapsing geometry and classification of ends:

The analysis combines Cheeger–Fukaya–Gromov theory for collapse with bounded curvature [15] and the conformally Kähler structure. For a rescaled sequence (M, h_j, p) converging to an asymptotic cone, a small regular ball B_j collapses along infranil fibers, while its universal cover $(\tilde{B}_j, \tilde{h}_j)$ is noncollapsed. The conformally Kähler structure survives on

the smooth local cover limit and strongly restricts both the limit geometry and the local nilpotent action. This yields the classification of asymptotic cones. To obtain a full expansion of each end, we then analyze the $SU(\infty)$ Toda equation in the LeBrun ansatz; here the Toda potential plays a role analogous to the harmonic function in the Gibbons–Hawking ansatz. Section 3 carries out this analysis.

- Classification of gravitational instantons, compactification, and positivity:

Once the asymptotic models are known, the generalized special Kasner and ALH^* ends are excluded because the scalar curvature of their conformal Kähler metrics has the wrong sign. The remaining ALF/AF problem splits according to the closure of the flow generated by the extremal vector field $\mathcal{K}$:

- If $\mathcal{K}$ has closed orbits, then $\mathcal{K}$ and $J\mathcal{K}$ generate a holomorphic $\mathbb{C}^*$ -action. We compactify M by a divisor $D \simeq \mathbb{P}^1$. LeBrun’s curvature identity [45] gives a positively curved Hermitian metric $e^{-2\log s_g} g_{-K}$ on $-K_M$. Its controlled singular extension to $-(K_{\overline{M}} + D)$ proves that this log anticanonical bundle is ample. The resulting surface is holomorphically toric, except for the cases with three orbifold points, which are ruled out in the appendix; see Section 4 and Appendix A.
- If the flow of $\mathcal{K}$ is not closed, its closure supplies a second Killing field, so (M, h) is already holomorphically and isometrically toric. The method of Biquard–Gauduchon [9], combined with the expansion from Section 3, then gives the Kerr and Chen–Teo families. Such a surface still admits an algebraic compactification, although there is no canonical fixed divisor at infinity.

- Calabi-type theorem:

It remains to obtain a second isometric circle action when $\mathcal{K}$ has closed orbits. Calabi proved that the isometry group of a compact extremal Kähler manifold is a maximal compact subgroup of its automorphism group [14]. In Section 5 we prove the corresponding decomposition for holomorphic vector fields that extend to $(\overline{M}, D)$. This promotes holomorphic toricity to holomorphic isometric toricity and permits the final application of the Biquard–Gauduchon classification. The exceptional cases are excluded in Appendix A.

Notations. Throughout this paper, Hermitian non-Kähler metrics are understood to be of Type II. The orientation is the one induced by the Hermitian complex structure. We adopt the following notations:

- The pair (M, h) refers to Type II Hermitian non-Kähler gravitational instantons, with $\lambda := 2\sqrt{6}|W^+|_h$ as a fixed multiple of the norm of the self-dual Weyl curvature. The metric $g := \lambda^{2/3}h$ is the associated extremal Kähler metric. We denote the scalar curvature $\lambda^{1/3}$ of g by s_g . We put the metric in the lower index to denote the corresponding Levi-Civita connection, curvature tensor, etc. Use ρ to denote the distance function to a base point under h .
- We use $\mathcal{K}$ to denote the *extremal vector field* $\mathcal{K} := -J\nabla_h \lambda^{-1/3} = J\nabla_g \lambda^{1/3}$. It is real holomorphic and Killing with respect to both the metrics h and g . When its orbits are closed, $T_{\mathcal{K}}$ denotes its minimal period on regular orbits, and $X = (T_{\mathcal{K}}/2\pi)\mathcal{K}$ is the primitive generator of period 2π . The period always refers to the metric and covering or lattice quotient currently under consideration. By a *holomorphic vector field*, we mean a holomorphic section of $T^{1,0}M$. The *holomorphic extremal vector field* associated with $\mathcal{K}$ is $\mathcal{K} - iJ\mathcal{K}$.
- If T is a tensor on (M, h) , then we say that $T = O'(\rho^{-\tau})$ if, for any integer $k \geq 0$,

$$|\nabla_h^k T|_h = O(\rho^{-\tau-k}),$$

as ρ approaches infinity.

- We use H_k to denote the Hirzebruch surface $\mathbb{P}(\mathcal{O} \oplus \mathcal{O}(k))$ with $k \geq 0$. We use C_0 to denote a curve inside with self-intersection k while C_∞ to denote the curve at infinity with self-intersection $-k$.

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2. PRELIMINARIES

2.1. Gravitational instantons in the Hermitian non-Kähler setting. For a Type II Ricci-flat metric (M, h) , the endomorphism $W^+ : \Lambda^+ \rightarrow \Lambda^+$ has one simple eigenvalue and one eigenvalue of multiplicity two. Derdziński [33] observed that, upon setting $\lambda = 2\sqrt{6}|W^+|_h$ and $g = \lambda^{2/3}h$, a constant-length eigenform corresponding to the simple eigenvalue is parallel with respect to g . After passing to a double cover if necessary, one may choose this eigenform globally and normalize it to obtain a self-dual two-form ω compatible with the orientation. Then g is Kähler with Kähler form ω , and the pair (g, ω) determines the complex structure J . A direct calculation gives $|s_g| = \lambda^{1/3}$.

For a Type II gravitational instanton, we pass to a double cover when necessary so that ω and J are globally defined. The resulting gravitational instanton is Hermitian non-Kähler.

From the conformal change of scalar curvature, we have

$$6s_g\Delta_g s_g - 12|\nabla_g s_g|^2 + s_g^3 = 0.$$

Since $\lambda^{1/3} \rightarrow 0$ at infinity, applying the maximum principle to s_g shows that the relevant sign is $s_g = \lambda^{1/3} > 0$. The function is everywhere positive because W^+ never vanishes in the Type II case. The following is essentially due to Derdziński [33].

Proposition 2.1. *Any Type II gravitational instanton (M, h) , up to passing to a double cover, is Hermitian non-Kähler. Moreover, on the cover the conformal metric $g = \lambda^{2/3}h$ is Kähler. The scalar curvature of g is exactly the conformal factor $s_g = \lambda^{1/3}$.*

Conversely, the Riemannian Goldberg–Sachs theorem implies that, for a Hermitian Ricci-flat four-metric, W^+ has at most two distinct eigenvalues at every point, with respect to the orientation induced by the complex structure.

For a Hermitian non-Kähler gravitational instanton (M, h) , the Kähler metric g is Bach-flat because it is conformal to an Einstein metric [7]; consequently, g is extremal Kähler [47]. Since $s_g = \lambda^{1/3}$, its extremal vector field is $\mathcal{K} = J\nabla_g s_g = -J\nabla_h \lambda^{-1/3}$. The field $\mathcal{K}$ is Killing for g and therefore preserves s_g . Because $h = s_g^{-2}g$, it is Killing for h as well.

Proposition 2.2. *The Kähler metric g is extremal Kähler. The extremal vector field $\mathcal{K} = J\nabla_g s_g = -J\nabla_h \lambda^{-1/3}$ is Killing with respect to both the metrics g and h .*

2.2. Regular region and smooth region. Given a gravitational instanton (M, h) with base point p , for any sequence $r_j \rightarrow \infty$, we can consider the sequence of scaled down spaces $(M, r_j^{-2}h, p)$. Set $h_j = r_j^{-2}h$. The Gromov compactness theorem guarantees that by passing to some subsequence, (M, h_j, p) converges to a complete length space $(M_\infty, d_\infty, p_\infty)$ in the pointed Gromov-Hausdorff topology. Any such limit space is called an *asymptotic cone* of the gravitational instanton (M, h) . In the following, we will usually use (M, h_j, p) to denote the rescaled sequence $(M, r_j^{-2}h, p)$.

Definition 2.3. For a limit space $(M_\infty, d_\infty, p_\infty)$ as above

- a point $q_\infty \in M_\infty$ is defined to be *regular* if there is a constant $\delta > 0$ and a sequence of $q_j \in M_j$ converging to q_∞ , such that $\sup_{B_\delta(q_j)} |Rm_{h_j}| \leq C$ for all j ;
- a regular point $q_\infty \in M_\infty$ is defined to be *smooth* if the length space (M_∞, d_∞) is a smooth Riemannian manifold in a neighborhood of q_∞ .

The *regular region* $\mathcal{R}$ is the set of all regular points and the *smooth region* $\mathcal{G}$ is the set of all smooth points.

Both the regular region and the smooth region are open.

For asymptotic cones of gravitational instantons, the rescaled metrics have uniformly bounded curvature on every ball of fixed radius that stays a fixed positive distance from the base point. Thus the regular region is $\mathcal{R} = M_\infty \setminus \{p_\infty\}$. Points of $\mathcal{R}$ need not, however, be smooth points of the quotient limit.

Example 2.4 (Gromov's skew rotation). This example originates from Section 8.9 in [37] and was discussed in previous works [52] and [22]. Consider the flat space $\mathfrak{M}_{\mathbf{a}, \mathbf{b}}$ that is defined as the quotient of the Euclidean space $\mathbb{R}^3 \times \mathbb{R}^1$ by the skew rotation

$$(\rho, \alpha, \beta, t) \sim (\rho, \alpha + \mathbf{a}, \beta, t + \mathbf{b}).$$

Here (ρ, α, β) are spherical coordinates for $\mathbb{R}^3$, with ρ being the radius, $0 \leq \alpha < 2\pi$, $0 \leq \beta \leq \pi$. When $\mathbf{a}/2\pi$ is irrational, $\mathfrak{M}_{\mathbf{a}, \mathbf{b}}$ has the upper-half-plane $\mathbb{R}_+^1 \times \mathbb{R}^1$ as its asymptotic cone. The asymptotic cone consists entirely of regular points; however, it does have non-smooth points, namely points on the boundary.

2.3. Local ansatzes for Hermitian non-Kähler Ricci-flat metrics. We have a local ansatz for 4-dimensional Hermitian non-Kähler Ricci-flat metrics, essentially due to LeBrun [44]. Consider a complex 2-dimensional Kähler metric g with a Hamiltonian Killing field ∂_t . By taking the Kähler reduction with respect to the Hamiltonian Killing field, away from zeros of ∂_t , the metric locally can always be written as

$$(1) \quad g = W d\xi^2 + W^{-1} \eta^2 + W e^v (dx^2 + dy^2).$$

Here η is a connection one-form such that $\eta(\partial_t) = 1$, ξ is the moment map for ∂_t , and $W e^v (dx^2 + dy^2)$ is the induced metric on the Kähler reduction. The complex structure J is given by $d\xi \rightarrow -W^{-1}\eta$, $dx \rightarrow -dy$, and the Kähler form ω is given by $d\xi \wedge \eta + W e^v dx \wedge dy$. By the calculation of [45], the integrability condition for the complex structure J is

$$(2) \quad d\eta = (W e^v)_\xi dx \wedge dy + W_x dy \wedge d\xi + W_y d\xi \wedge dx.$$

We also have

$$(3) \quad (W e^v)_{\xi\xi} + W_{xx} + W_{yy} = 0,$$

which is exactly the equation $d(d\eta) = 0$. The scalar curvature of g can be calculated as

$$(4) \quad s_g = -\frac{(e^v)_{\xi\xi} + v_{xx} + v_{yy}}{W e^v}.$$

The following local ansatz is taken from Proposition 3.3 in [9].

Theorem 2.5 (Local ansatz). *For the Kähler metric g given by equation (1), when $W^+ \neq 0$, the conformal metric $h = \xi^{-2}g$ is Ricci-flat if and only if, up to replacing ∂_t by $-\partial_t$ in the Kähler reduction,*

$$(5) \quad W = k^2 \left(\frac{12}{\xi^3} - \frac{6v_\xi}{\xi^2} \right)$$

and v satisfies the twisted $SU(\infty)$ Toda equation

$$(6) \quad (e^v)_{\xi\xi} + v_{xx} + v_{yy} = -k^{-2}\xi W e^v$$

for some positive constant k .

Any complex 2-dimensional Hermitian non-Kähler Ricci-flat metric h is canonically conformally related to such a Kähler metric $g = \lambda^{2/3}h$.

For the canonical choice $g = \lambda^{2/3}h$ and $\xi = s_g$, the constant k equals one and $\partial_t = \mathcal{K} = J\nabla_g s_g$. In the positive-scalar-curvature case considered for gravitational instantons, $\xi = \lambda^{1/3}$. This canonical normalization does not prescribe the global period of $\mathcal{K}$. One can perform the transformation

$$(7) \quad \varrho = \frac{1}{\xi}, \quad V = \xi^2 W, \quad u = v - 4 \log |\xi|$$

and the above ansatz becomes

Theorem 2.6 (Local ansatz, second version). *After performing the change of variables in (7), the equations in Theorem 2.5 with $k = 1$ become*

$$(8) \quad V = -12\varrho + 6\varrho^2 u_\varrho$$

and the usual $SU(\infty)$ Toda equation

$$(9) \quad (e^u)_{\varrho\varrho} + u_{xx} + u_{yy} = 0.$$

The metrics g and h become

$$(10) \quad h = \varrho^2 g = V(d\varrho^2 + e^u(dx^2 + dy^2)) + V^{-1}\eta^2,$$

with

$$d\eta = \left(\frac{2Ve^u}{\varrho} - \partial_\varrho(Ve^u) \right) dx \wedge dy - V_x dy \wedge d\varrho - V_y d\varrho \wedge dx.$$

Note that if the Hermitian non-Kähler Ricci-flat metric is assumed to have a toric symmetry, preserving both the metric and the complex structure, the ansatz above further reduces to axisymmetric harmonic functions on $\mathbb{R}^3$ by the work of Ward [61]. See also the work of Tod [60].

2.4. Asymptotic models. In this subsection, we introduce the ALE, ALF-A, AF, generalized special Kasner, and ALH* asymptotic models that appear in Theorem 1.1.

Definition 2.7. A gravitational instanton (M, h) is ALE (asymptotically locally Euclidean), if:

- Outside a compact set K , the manifold $M \setminus K$ is diffeomorphic to $(\mathbb{R}^4/\Gamma) \setminus \overline{B_N(0)}$ via a diffeomorphism Φ , where $\Gamma \subset \mathrm{SO}(4)$ is a finite subgroup acting freely on $\mathbb{S}^3$.
- The metric h satisfies

$$\left| \nabla_h^k (h - \Phi^* g_{\mathbb{R}^4/\Gamma}) \right|_h = O(\rho^{-\delta_0 - k})$$

for a $\delta_0 > 0$ and for any $k \geq 0$, where ρ denotes the distance function to the origin on $\mathbb{R}^4/\Gamma$.

Note that as in [50], the ALE asymptotic model is $(\mathbb{C}^2/\Upsilon) \setminus \{0\}$ with $\Upsilon \subset U(2)$, but equipped with the flat metric $\frac{1}{\rho_\Upsilon^4} g_{\mathbb{C}^2/\Upsilon}$. Here ρ_Υ is the distance function to the origin in $\mathbb{C}^2/\Upsilon$. For a Hermitian non-Kähler ALE gravitational instanton the complex structure is asymptotic to the standard complex structure on $\mathbb{C}^2/\Upsilon$.

Definition 2.8. A gravitational instanton (M, h) is ALF (asymptotically locally flat) with rate $\delta_0 > 0$ if:

- Outside a compact set K , there is a diffeomorphism that identifies $M \setminus K$ with $(N, \infty) \times L$, where L is the total space of a nontrivial smooth S^1 -bundle over S^2 or $\mathbb{RP}^2$. Hence $M \setminus K$ under this diffeomorphism is the total space of an S^1 -bundle π over the complement of a compact ball in $\mathbb{R}^3$ or $\mathbb{R}^3/\{\pm \text{id}\}$.
- There are a connection one-form θ on L , normalized by $\theta(Y) = 1$ for the primitive circle generator Y of period 2π , and a constant $\ell > 0$. Put $\gamma = \ell\theta$. Then under the above diffeomorphism, the metric h satisfies

$$\left| \nabla_h^k (h - \pi^* g_{\mathbb{R}^3} - \gamma^2) \right|_h = O(\rho^{-\delta_0 - k})$$

for any $k \geq 0$, where ρ denotes the standard distance function on the base $\mathbb{R}^3$ or $\mathbb{R}^3/\{\pm \text{id}\}$. We denote the unit vertical field of the model by $\partial_t = \ell^{-1}Y$. Thus $\gamma(\partial_t) = 1$ and its regular period is $2\pi\ell$.

We call an ALF gravitational instanton *ALF-A* when the base of π is $\mathbb{R}^3 \setminus \overline{B_N(0)}$, and *ALF-D* when the base is $(\mathbb{R}^3/\{\pm \text{id}\}) \setminus \overline{B_N(0)}$. Further finite quotients of ALF-A models also occur. For $k \geq 2$, let $\mathbb{Z}_k$ act on $\mathbb{R}^3$ by rotation about the z -axis, so that

$$g_{\mathbb{R}^3/\mathbb{Z}_k} = d\rho^2 + \rho^2 g_{\mathbb{S}^2/\mathbb{Z}_k}.$$

Take an orbifold S^1 -bundle of nonzero Euler number over $\mathbb{S}^2/\mathbb{Z}_k$, equipped with an orbifold connection θ and $\gamma = \ell\theta$ with the same normalization as above, and suppose its total space L is a smooth lens space. By an orbifold S^1 -bundle with smooth total space L , we mean a smooth effective locally free S^1 -action on L whose quotient, with its induced orbifold structure, is the specified base orbifold. An S^1 -action is locally free if every point has a finite stabilizer. The metric

$$\pi^* g_{\mathbb{R}^3/\mathbb{Z}_k} + \gamma^2$$

is then a finite quotient of an ALF-A model: its $\mathbb{Z}_k$ -cover is ALF-A in the preceding sense. There could also be noncyclic finite quotients. We call these models *finite quotients of ALF-A*. Under this convention, ALF-D is not such a quotient.

There are more asymptotic models when the end is diffeomorphic to $(\mathbb{R}^3 \setminus \overline{B_N(0)}) \times S^1$.

Definition 2.9. A gravitational instanton (M, h) is AF (asymptotically flat) with rate $\delta_0 > 0$ if:

- Outside a compact subset $K \subset M$, there is a diffeomorphism

$$\Phi : M \setminus K \rightarrow \mathfrak{M}_{\mathbf{a}, \mathbf{b}} \setminus \overline{B_N(0)}$$

onto the complement of a compact ball in the model space $\mathfrak{M}_{\mathbf{a}, \mathbf{b}}$ with $0 \leq \mathbf{a} < 2\pi$ and $\mathbf{b} \neq 0$.

- The model space $(\mathfrak{M}_{\mathbf{a}, \mathbf{b}}, h_{\mathbf{a}, \mathbf{b}})$ is the quotient of $(\mathbb{R}^3 \times \mathbb{R}^1, h_{\mathbb{R}^3 \times \mathbb{R}^1})$ by the following relation:

$$(\rho, \alpha, \beta, t) \sim (\rho, \alpha + \mathbf{a}, \beta, t + \mathbf{b}).$$

Here we are using the spherical coordinates (ρ, α, β) for $\mathbb{R}^3$ with $0 \leq \alpha < 2\pi$, $0 \leq \beta \leq \pi$, and ρ is the distance function on $\mathbb{R}^3$ to the origin.

- The metric h is asymptotic to the quotient metric $h_{\mathbf{a}, \mathbf{b}}$ on the model space:

$$|\nabla_h^k (h - \Phi^* h_{\mathbf{a}, \mathbf{b}})|_h = O(\rho^{-\delta_0 - k})$$

for any $k \geq 0$.

For the model space the quotient map will be denoted by $\pi : \mathbb{R}^3 \times \mathbb{R}^1 \rightarrow \mathfrak{M}_{\mathbf{a}, \mathbf{b}}$. An AF gravitational instanton is said to be $\text{AF}_{\mathbf{a}, \mathbf{b}}$ if it is asymptotic to the model space $(\mathfrak{M}_{\mathbf{a}, \mathbf{b}}, h_{\mathbf{a}, \mathbf{b}})$.

The Kerr family [35] and Chen-Teo family [23] of gravitational instantons are examples of AF gravitational instantons, which are not hyperkähler. We call a gravitational instanton *rational* AF, if it is $\text{AF}_{\mathbf{a}, \mathbf{b}}$ with $\mathbf{a}/2\pi = m/n$ rational, and *irrational* AF, if it is $\text{AF}_{\mathbf{a}, \mathbf{b}}$ with $\mathbf{a}/2\pi$ irrational.

There are more asymptotic models that cannot occur as the end of a complete Hermitian non-Kähler gravitational instanton.

Definition 2.10. Let $\beta \in \mathbb{R}$ and choose $N > 2|\beta|$. On $(-\infty, -N) \times \mathbb{R}^3$, set

$$(11) \quad u_\beta = \log(-\varrho + \beta), \quad V_\beta = -6\varrho \frac{\varrho - 2\beta}{\varrho - \beta}, \quad \theta_\beta = dt - 12\beta x dy,$$

$$(12) \quad h_{K,\beta} = V_\beta(d\varrho^2 + (-\varrho + \beta)(dx^2 + dy^2)) + V_\beta^{-1}\theta_\beta^2.$$

The zero-parameter member is the usual *special Kasner metric*

$$(13) \quad h_{K,0} = -6\varrho \left(d\varrho^2 - \varrho(dx^2 + dy^2) \right) - \frac{1}{6\varrho} dt^2.$$

On the same region, define

$$(14) \quad h_{\text{ALH}^*} = -12\varrho \left(d\varrho^2 + dx^2 + dy^2 \right) - \frac{1}{12\varrho} (dt - 12x dy)^2.$$

To describe the compact quotients of (12), let $\mathcal{G}_\beta$ be $\mathbb{R}^3$ with multiplication

$$(x, y, t) \cdot_\beta (x', y', t') = (x + x', y + y', t + t' + 12\beta xy').$$

The forms dx , dy , and θ_β are left invariant. The group $\mathcal{G}_0$ is abelian, whereas $\mathcal{G}_\beta$ is the Heisenberg group when $\beta \neq 0$. Hence, for every cocompact lattice $\Gamma \subset \mathcal{G}_\beta$, the metric descends to a metric $h_{K,\beta,\Gamma}$ on $(-\infty, -N) \times (\Gamma \backslash \mathcal{G}_\beta)$. We call the resulting metrics *generalized special Kasner models*. For $\beta = 0$ they are the skewed special Kasner models; the word “special” distinguishes this Type II family from the general Kasner family. The ALH* metric is invariant under $\mathcal{G}_1$, and its quotients by cocompact lattices are denoted by $h_{\text{ALH}^*-\Gamma}$. When $\beta \neq 0$, ∂_t is automatically periodic. For $\beta = 0$, its flow may instead be nonclosed in a skew lattice.

The generalized special Kasner and ALH* models are Hermitian non-Kähler Ricci-flat. Indeed, they arise from the exact Toda solutions $u = \log(-\varrho + \beta)$ and $u = 0$, respectively; in the first case $d\theta_\beta = -12\beta dx \wedge dy$. The metrics $h_{\text{ALH}^*-\Gamma}$ are precisely the ALH* asymptotic models for hyperkähler gravitational instantons, taken here with the opposite orientation. The corresponding conformal extremal Kähler metrics include

$$(15) \quad g_{K,\beta} = \frac{V_\beta}{\varrho^2} d\varrho^2 + 6 \left(1 - \frac{2\beta}{\varrho} \right) (dx^2 + dy^2) + \frac{1}{\varrho^2 V_\beta} \theta_\beta^2,$$

$$g_{\text{ALH}^*-\Gamma} = \frac{1}{\varrho^2} \left(-12\varrho d\varrho^2 - \frac{1}{12\varrho} (dt - 12x dy)^2 \right) - \frac{12}{\varrho} (dx^2 + dy^2).$$

Their scalar curvature is $s_g = \varrho^{-1} < 0$. This sign is the obstruction that will rule out these models as ends of Hermitian non-Kähler gravitational instantons.

These models also admit finite quotients preserving the Hermitian non-Kähler structure. For the zero-parameter model with lattice $\mathbb{Z}^3$ in the coordinates (x, y, t) , for example, one may quotient by the free involution

$$[x, y, t] \mapsto [-x, -y, t + \tfrac{1}{2}].$$

It preserves the complex structure associated with (13). The resulting link is an orbifold S^1 -bundle over $S^2(2, 2, 2, 2)$, the quotient of $\mathbb{R}^2/\mathbb{Z}^2$ by $[x, y] \mapsto [-x, -y]$. Similarly, for the ALH* model one may take the lattice on which x , y , and $t/6$ are integral and then quotient by

$$[x, y, t] \mapsto [-x, -y, t + 3].$$

The central period is then 6 and the bundle has even Euler number, so the involution is free. Its link is again an orbifold S^1 -bundle over $S^2(2, 2, 2, 2)$.

More generally, when ∂_t is periodic, a structure-preserving finite quotient commutes with its circle action, and the link is an orbifold circle bundle over an orientable Euclidean two-orbifold.

These quotient links are infranilmanifolds. The list of orientable Euclidean two-orbifolds and the corresponding quotient geometries is described in [58, Chapter 4].

In the following, by saying a Hermitian non-Kähler gravitational instanton is ALF/AF, we mean it is asymptotic to ALF- A or its finite quotients, or AF.

3. COLLAPSING GEOMETRY AND CLASSIFICATION OF ENDS

In this section we classify all possible asymptotic cones as well as ends. Consider a d -dimensional asymptotic cone $(M_\infty, d_\infty, p_\infty)$ of (M, h, p) , which is the limit of some sequence (M, h_j, p) with $h_j = r_j^{-2}h$ and $r_j \rightarrow \infty$. In the following we study the geometric structures on the regular region $\mathcal{R} = M_\infty \setminus \{p_\infty\}$. Note that $\mathcal{R}$ is connected, unless (M_∞, d_∞) splits as $\mathbb{R}$.

Fix $q_\infty \in \mathcal{R}$ and choose $q_j \in (M, h_j, p)$ converging to q_∞ . For sufficiently small $\delta > 0$, the ball $B_\infty := B_\delta(q_\infty)$ lies in $\mathcal{R}$, and Cheeger–Fukaya–Gromov theory [15] implies that the universal covers $\tilde{B}_j$ of $B_j := B_\delta(q_j)$ are noncollapsed. Let G_j be the deck group of $\pi_j : \tilde{B}_j \rightarrow B_j$. After passing to a subsequence, we have the equivariant convergence picture

$$\begin{array}{ccc} (\tilde{B}_j, \tilde{h}_j, G_j, \tilde{q}_j) & \xrightarrow{\text{equivariant GH-convergence}} & (\tilde{B}_\infty, \tilde{h}_\infty, G_\infty, \tilde{q}_\infty) \\ \pi_j \downarrow & & \downarrow \pi_\infty \\ (B_j, h_j) & \xrightarrow{\text{GH-convergence}} & (B_\infty, d_\infty). \end{array}$$

Here, G_∞ is a subgroup of the isometry group $\text{Isom}(\tilde{B}_\infty, \tilde{h}_\infty)$ of $\tilde{B}_\infty$ and $B_\infty = \tilde{B}_\infty/G_\infty$. Moreover, G_∞ is a finite extension of its identity component G_∞^0 , which is a nilpotent Lie group, and it acts properly and faithfully on $\tilde{B}_\infty$ (Theorem 2.1 in [54]). Because $\tilde{h}_j$ is Ricci-flat, regularity for noncollapsed Einstein limits upgrades the convergence to smooth pointed convergence. We use this local cover picture throughout Sections 3.1–3.3. Section 3.4 then treats general Ricci-flat ends that are complete at infinity.

3.1. Limit rescaled conformal factor. To retain the Kähler geometry while taking an asymptotic cone of (M, h, p) , we rescale the conformal factor $\lambda^{1/3}$ simultaneously with the metric. Since $s_g = \lambda^{1/3}$ and $s_h = 0$, the conformal scalar-curvature formula gives

$$(16) \quad 6\Delta_h \lambda^{1/3} + \lambda^{4/3} = 0.$$

Note that in the complete instanton case $s_g = \lambda^{1/3} > 0$. Later in the end situation in Section 3.4, it is possible that $s_g = -\lambda^{1/3} < 0$. As above we take $q_\infty \in \mathcal{R} \subset M_\infty$ and pick a sequence of points $q_j \in (M, h_j)$ with $q_j \rightarrow q_\infty$. Renormalize $\lambda^{1/3}$ as $\lambda_j^{1/3} := \lambda^{-1/3}(q_j)\lambda^{1/3}$. We will take limits of λ_j ; the uniqueness argument below will show that changing q_∞ only changes the eventual normalization. After the renormalization, $\lambda_j^{1/3}$ satisfies

$$(17) \quad 6\Delta_{h_j} \lambda_j^{1/3} + r_j^2 \lambda(q_j) \lambda_j^{4/3} = 0.$$

The local Cheng–Yau gradient estimate argument, applied on the noncollapsed local covers, gives the estimate needed below.

Proposition 3.1. *Suppose that $B_{2a}(x_j, h_j)$ stays a definite distance from the base point in (M, h_j, p) . Then*

$$(18) \quad \sup_{B_a(x_j, h_j)} \frac{|\nabla_{h_j} \lambda_j^{1/3}|_{h_j}}{\lambda_j^{1/3}} \leq C_a.$$

The constant is uniform in j , depending on the definite distance to p .

Proof. Put $f_j = \lambda_j^{1/3}$. The equation (17) may be written as the linear equation

$$\Delta_{h_j} f_j + b_j f_j = 0, \quad b_j = \frac{1}{6} r_j^2 \lambda.$$

Quadratic curvature decay bounds b_j in L^∞ on the doubled ball since it stays away from the base point. The Harnack inequality for a positive solution of this equation, followed by the interior $W^{2,p}$ estimate with $p > 4$, gives

$$\sup_{B_a} f_j \leq C_a \inf_{B_a} f_j, \quad \sup_{B_a} |\nabla f_j| \leq C_a \sup_{B_a} f_j.$$

Their combination proves (18). This is the standard local gradient estimate method [24]. Passing to the cover removes any dependence on the collapsing injectivity radius. $\square$

Since $\limsup_{\rho \rightarrow \infty} \rho^2 |\text{Rm}_h|_h < C$, balls whose doubled balls remain in a compact subset of the regular region satisfy the hypotheses of Proposition 3.1. Therefore, for large j ,

$$(19) \quad |\nabla_{h_j} \lambda_j^{1/3}|_{h_j} \leq C \lambda_j^{1/3}$$

with C depending on a . We obtain the following Harnack inequality on a single ball.

Proposition 3.2. *For $j > j_0$ with j_0 depending on the choice of q_∞ and a , on $B_{a/2}(q_j, h_j)$, there is a constant C depending on a such that*

$$(20) \quad \sup_{B_{a/2}(q_j, h_j)} \lambda_j^{1/3} \leq C \inf_{B_{a/2}(q_j, h_j)} \lambda_j^{1/3}.$$

The Harnack inequality and the normalization $\lambda_j^{1/3}(q_j) = 1$ give uniform upper and lower bounds for $\lambda_j^{1/3}$ on $B_{a/2}(q_j, h_j)$. Chaining overlapping balls gives the following global Harnack inequality.

Proposition 3.3. *For any path-connected compact set $K \Subset \mathcal{R} \subset M_\infty$ and any sequence of path-connected $K_j \subset (M, h_j)$ that converges to K , the following Harnack inequality for $\lambda_j^{1/3}$ on K_j holds uniformly when j is large enough*

$$(21) \quad \sup_{K_j} \lambda_j^{1/3} \leq C_K \inf_{K_j} \lambda_j^{1/3}.$$

Therefore, for any such sequence of K_j , we have uniform upper and lower bounds on $\lambda_j^{1/3}$. Now the standard elliptic theory can be applied to the equation (17) to derive bounds on derivatives of $\lambda_j^{1/3}$. Passing to a subsequence of (M, h_j, p) , we can make sense of the smooth convergence of $\lambda_j^{1/3}$ on $\mathcal{R}$ in the following sense.

Definition 3.4. For an asymptotic cone $(M_\infty, d_\infty, p_\infty)$ and a sequence of smooth functions f_j on (M, h_j, p) , we say it converges smoothly to a limit function f_∞ on the regular region $\mathcal{R}$, if for any $B_\delta(q_\infty) \Subset \mathcal{R}$ and any sequence of balls $B_\delta(q_j) \subset (M, h_j, p)$ converging to $B_\delta(q_\infty)$ such that the sequence of universal covers $\tilde{B}_\delta(q_j)$ is non-collapsed converging to $\tilde{B}_\infty$, the pullback sequence of functions $\pi_j^* f_j$ converges smoothly to $\pi_\infty^* f_\infty$ on $\tilde{B}_\infty$.

Proposition 3.5. *By passing to a subsequence, $\lambda_j^{1/3}$ converges smoothly to a limit function $\lambda_\infty^{1/3}$ on the regular region $\mathcal{R}$. The limit rescaled conformal factor $\lambda_\infty^{1/3}$ is nowhere vanishing.*

Proof. On any ball $B_\delta(q_j) \subset (M, h_j, p)$ of definite size and definite positive distance from p in the rescaled metric, elliptic estimates for (17) apply on the cover $\tilde{B}_\delta(q_j)$. A diagonal subsequence gives smooth convergence on all compact subsets of $\mathcal{R}$. Positivity of the limit follows from the Harnack inequality. $\square$

We call $\lambda_\infty^{1/3}$ a *limit rescaled conformal factor*. Changing the normalizing point in $\mathcal{R}$ multiplies it by a positive constant. It may nevertheless depend on the blow-down sequence, and even for a fixed asymptotic cone a further subsequence may initially be needed. A constant limit is not yet excluded; this will be done below. On every local cover, the conformal Kähler metrics

$$g_j := \lambda_j^{2/3} h_j$$

converge smoothly to $\tilde{g}_\infty = \lambda_\infty^{2/3} \tilde{h}_\infty$.

3.2. Analysis of asymptotic cones. We now investigate gravitational instantons that have sub-Euclidean volume growth; the noncollapsed case is treated by the standard ALE situation. Thus every blow-down considered in this subsection has dimension strictly less than four, and the identity component of the local cover collapse group has a positive-dimensional orbit through every lifted base point. We begin by controlling the zeros of the extremal vector field $\mathcal{K}$.

Proposition 3.6. *On a gravitational instanton that collapses at infinity, the extremal vector field $\mathcal{K}$ has no zeros outside a compact set. Consequently, its moment map $s_g = \lambda^{1/3}$ has no critical points sufficiently far out on the end.*

Proof. Because $\mathcal{K}$ is Hamiltonian and real holomorphic, $\lambda^{1/3}$ is Morse–Bott: a critical component is either an isolated point or a holomorphic curve. Equation (16) and the maximum principle exclude local minima. A critical curve is a local maximum, while an isolated critical point is either a maximum or an index-two saddle. Thus the only Morse–Bott indices are 2 and 4.

There is exactly one maximum critical component. Indeed, since $\lambda^{1/3} \rightarrow 0$ at infinity, every positive superlevel set is compact. As the level is lowered, each local maximum creates a new component of the superlevel set. The Morse–Bott property shows that every later operation either creates another component at a local maximum or attaches a path-connected handle along a path-connected attaching region. Neither operation can merge two existing components. Thus two maximum critical components would remain in different components of every lower positive superlevel set. But any path joining them is compact and, since $\lambda^{1/3} > 0$, lies in some positive superlevel set, a contradiction. This is the superlevel-set version of the argument in [47, Proposition 3].

The unique maximum critical component is compact. Consequently, every critical point sufficiently far out is an isolated point of index 2. It remains to show that these points also stay in a compact set. Suppose, to the contrary, that there is a sequence of index-2 critical points q_j tending to infinity. Fix a base point p , and consider the rescaled spaces $(M, h_j, p) := (M, r_j^{-2} h, p)$, where $r_j = d_h(p, q_j)$.

For sufficiently small δ , the balls $B_j := B_{h_j}(q_j, \delta)$ have bounded curvature and noncollapsed universal covers $(\tilde{B}_j, \tilde{q}_j)$. Passing to a subsequence, we may assume that (B_j, q_j) converges to a limit ball $B_\infty := B(q_\infty, \delta)$ and that $(\tilde{B}_j, \tilde{q}_j)$ converges smoothly to $(\tilde{B}_\infty, \tilde{q}_\infty)$. The deck groups G_j converge equivariantly to an isometric action of a positive-dimensional Lie group G_∞ on $\tilde{B}_\infty$, and

$$B_\infty = \tilde{B}_\infty / G_\infty.$$

Lift $\mathcal{K}$ to $\tilde{B}_j$ and multiply it by a positive constant so that

$$|\nabla \mathcal{K}_j|_{\tilde{h}_j}(\tilde{q}_j) = 1.$$

The Killing equation, the identity $\nabla^* \nabla \mathcal{K}_j = \text{Ric}(\mathcal{K}_j) = 0$, and smooth convergence of the metrics give uniform bounds for $\mathcal{K}_j$ on a smaller ball. After passing to a subsequence,

$$\mathcal{K}_j \longrightarrow \mathcal{K}_\infty$$

smoothly, where $\mathcal{K}_\infty$ is a nonzero real holomorphic Killing field vanishing at $\tilde{q}_\infty$.

Since h is complete, $\mathcal{K}$ generates a global isometric $\mathbb{R}$ -action. Let T be the closure of the $\mathbb{R}$ -action in $\text{Isom}(M, h)$, where the group T fixes q_j . The stabilizer of q_j is closed and compact, so T is compact and also fixes q_j . Let $H_j := T_{q_j}$ be its isotropy group at q_j ; thus $H_j = T$, although its isotropy representation at q_j depends on j . The action by H_j lifts uniquely to $\tilde{B}_j$ by requiring it to fix $\tilde{q}_j$. We use the same notation for this lifted action. It commutes with the deck group. Equivariant compactness gives a limiting group H_∞ fixing $\tilde{q}_\infty$ and commuting with G_∞ . Since every H_j fixes every deck translate of $\tilde{q}_j$, the group H_∞ fixes the orbit

$$G_\infty^0 \cdot \tilde{q}_\infty$$

pointwise. This orbit has positive dimension.

It remains to analyze the two weights of $\mathcal{K}_j$ at q_j . After choosing a unitary frame at $\tilde{q}_j$, the infinitesimal isotropy representation of $\mathcal{K}_j$ has weights $a_j > 0 > -b_j$. If a_j/b_j remains bounded above and below, both limiting weights are nonzero. The zero of $\mathcal{K}_\infty$ at $\tilde{q}_\infty$ is then isolated, contradicting the positive-dimensional fixed orbit above.

Suppose instead, after interchanging the two complex lines if necessary, that $a_j/b_j \rightarrow \infty$. If the one-parameter group is nonclosed, its closure in $U(2)$ is already a two-torus, and the equivariant limit contains a T^2 . Otherwise, reparametrize the circle subgroup by its primitive integral weight pair (m_j, n_j) . Both entries are nonzero and $|m_j/n_j| \rightarrow \infty$, so these primitive vectors leave every finite subset of $\mathbb{Z}^2$. The corresponding circle subgroups of the maximal torus in $U(2)$ converge to the full two-torus. Thus in either case H_∞ contains an effective T^2 fixing $G_\infty^0 \cdot \tilde{q}_\infty$ pointwise.

This is impossible. The fixed set of a compact group of holomorphic isometries is a complex submanifold. Since it contains a positive-dimensional orbit, its tangent space at $\tilde{q}_\infty$ contains a complex line. The T^2 -isotropy representation is therefore trivial on that line and acts on the remaining complex line through a single character, which has a circle kernel. An isometry fixing a point and acting trivially on its tangent space is the identity, so the isotropy representation of an effective isometric action is faithful. This contradiction also excludes $a_j/b_j \rightarrow 0$ and completes the proof. $\square$

Since $\lambda^{1/3}$ has no critical points on $M \setminus K$, Kähler reduction is available throughout the end. Thus the ansatzes in Theorems 2.5 and 2.6 apply in compatible local quotient charts throughout the end. When the $\mathcal{K}$ -orbits are closed, these local reductions assemble to the global quotient used below. We distinguish two cases.

(♣) $\mathcal{K}$ induces an S^1 -action. Note that the S^1 -action on the end could either be free or have discrete isotropy.

(♠) $\mathcal{K}$ only induces an $\mathbb{R}$ -action, whose closure gives a higher rank torus action.

We deal with (♣) and (♠) in Sections 3.2.1 and 3.2.2, respectively.

3.2.1. $\mathcal{K}$ induces an S^1 -action. Suppose $\mathcal{K}$ generates an S^1 -action. Its moment map on the end is $\xi = \lambda^{1/3}$. Since $\xi \rightarrow 0$ and has no critical points there, every sufficiently small level $L_{\xi_0} = \{\xi = \xi_0\}$ is compact. Let Σ_{ξ_0} be its Kähler reduction; this is an orbifold if the circle action is only locally free. The field $\nabla_g \xi = -J\mathcal{K}$ is real holomorphic, so its flow identifies the reduced surfaces biholomorphically. We therefore write all of them as Σ . Throughout this subsection, the reduced surface Σ is regarded as a compact complex orbifold, including the smooth case. Integrals are orbifold integrals, and $\chi(\Sigma)$ denotes the orbifold Euler characteristic. Smooth convergence of orbifold metrics means equivariant smooth convergence in compatible uniformizing charts. The end is diffeomorphic to $(N_0, \infty) \times L$, where L is an S^1 -orbifold-bundle over Σ with smooth total space. In a holomorphic coordinate $x + iy$ on an open set in Σ , the metric is

$$(22) \quad h = \varrho^2 g = V(d\varrho^2 + e^u(dx^2 + dy^2)) + V^{-1}\eta^2.$$

Here we are applying the ansatz Theorem 2.6, so $\varrho = \xi^{-1} = \lambda^{-1/3} > 0$. Note that $V^{-1} = |\mathcal{K}|_h^2$ is a globally well-defined function, but u depends on the particular choice of holomorphic coordinates. However, the metric $Ve^u(dx^2 + dy^2)$ is well-defined on the Kähler reduction Σ . We have the following observation.

Lemma 3.7. *Let K_Σ be the Gauss curvature of the metric $e^u(dx^2 + dy^2)$ on Σ . Then the $SU(\infty)$ Toda equation $(e^u)_{\varrho\varrho} + u_{xx} + u_{yy} = 0$ is equivalent to*

$$\frac{1}{2}(e^u)_{\varrho\varrho}e^{-u} = K_\Sigma.$$

Proof. Since $K_\Sigma = -\frac{1}{2}e^{-u}(u_{xx} + u_{yy})$, the assertion follows immediately from the Toda equation. $\square$

By Gauss-Bonnet, for the metric $e^u(dx^2 + dy^2)$ on Σ we have $\int_\Sigma K_\Sigma d\text{vol}_\Sigma = \frac{1}{2} \int_\Sigma (e^u)_{\varrho\varrho} dx dy = 2\pi\chi_{\text{orb}}(\Sigma)$ and

$$(23) \quad \int_\Sigma e^u dx dy = 2\pi\chi(\Sigma)\varrho^2 + a\varrho + b$$

for constants a, b . A second integration gives

$$(24) \quad \begin{aligned} \int_\Sigma Ve^u dx dy &= \int_\Sigma (-12\varrho + 6\varrho^2 u_\varrho) e^u dx dy = \int_\Sigma \varrho^4 \left(\frac{6e^u}{\varrho^2}\right)_\varrho dx dy \\ &= -6a\varrho^2 - 12b\varrho. \end{aligned}$$

Positivity of both integrals for all large $\varrho > 0$ forces $\chi_{\text{orb}}(\Sigma) > 0$; hence the underlying Riemann surface is $\mathbb{P}^1$. Let $T_\mathcal{K} > 0$ denote the minimal period of $\mathcal{K}$ on its regular orbits. We obtain the following annular volume formula.

Lemma 3.8. $\text{Vol}(\{D_0 \leq \varrho \leq D_1\}) = -2T_\mathcal{K}a(D_1^3 - D_0^3) - 6T_\mathcal{K}b(D_1^2 - D_0^2)$.

Proof. Integrate (24) in ϱ and around the circle orbit. $\square$

Consider an asymptotic cone $(M, h_j, p) \rightarrow (M_\infty, d_\infty, p_\infty)$ along which $\lambda_j^{1/3} \rightarrow \lambda_\infty^{1/3}$, and set $\varrho_j := \lambda_j^{-1/3} \rightarrow \varrho_\infty := \lambda_\infty^{-1/3}$. Proposition 3.3, applied after a further rescaling by r^{-2} , gives

$$\sup_{A_{r,2r}} \lambda_\infty^{1/3} \leq C \inf_{A_{r,2r}} \lambda_\infty^{1/3},$$

where C is independent of $r > 0$ and depends only on $\limsup_{\rho \rightarrow \infty} \rho^2 |\text{Rm}_h|_h$.

By definition, $\lambda_\infty^{1/3}$ is defined on the entire $\mathcal{R}$.

Lemma 3.9. *The limit rescaled conformal factor $\lambda_\infty^{1/3}$ has range $(0, \infty)$.*

Proof. Write $\ell = \inf_{\mathcal{R}} \varrho_\infty$ and $L = \sup_{\mathcal{R}} \varrho_\infty$. We use two consequences of the fact that ϱ has no critical point on the original end. If $0 < t < \ell$, then, for every $\epsilon > 0$,

$$(25) \quad \{\varrho_j < t\} \subset B_\epsilon(p, h_j)$$

for all large j . Otherwise a point in this sublevel can be moved by the decreasing gradient flow through a fixed rescaled annulus; after passage to the limit this gives a point of $\mathcal{R}$ with value at most t . The same argument, with the flow orientation reversed, shows that if $L < T < \infty$ then every fixed ball $B_R(p, h_j)$ is contained in $\{\varrho_j < T\}$ for all large j .

We now use the exact volume identity. In the normalized variable $t = \varrho_j$, put $c_j = \lambda(q_j)^{1/3}$, so that the original coordinate is $\varrho = t/c_j$. Lemma 3.8 says that the h_j -volume of a level slab $\{t_0 < \varrho_j < t_1\}$ is

$$(26) \quad A_j(t_1^3 - t_0^3) + B_j(t_1^2 - t_0^2).$$

Here

$$A_j = -2T_{\mathcal{K}} a r_j^{-4} c_j^{-3}, \quad B_j = -6T_{\mathcal{K}} b r_j^{-4} c_j^{-2}.$$

If $a < 0$, then $A_j > 0$ and $B_j/A_j = 3bc_j/a \rightarrow 0$; if $a = 0$, positivity forces $b < 0$ and $B_j > 0$. Thus ratios between any two fixed positive level slabs are bounded above and below independently of j .

If $\ell > 0$, choose $0 < t < \ell$ and choose $0 < c_0 < c_1$ containing the values of ϱ_j on $A_{1,2}(p, h_j)$ for large j . Equations (25) and (26), applied to the slab $\{t/2 < \varrho_j < t\} \subset B_\epsilon(p, h_j)$, give

$$\frac{\text{Vol}_{h_j} B_\epsilon(p, h_j)}{\text{Vol}_{h_j} A_{1,2}(p, h_j)} \geq c(t, c_0, c_1) > 0.$$

The renormalized limit measure is non-atomic. Hence its relative mass on $B_\epsilon(p)$ tends to zero as $\epsilon \rightarrow 0$, a contradiction. Thus $\ell = 0$.

If $L < \infty$, take $T > L$ and choose $t > 0$ below the values of ϱ_∞ on $A_{1,2}(p_\infty, d_\infty)$. The flow observation gives $\{\varrho_j < t\} \subset B_1(p, h_j)$ and $B_R(p) \subset \{\varrho_j < T\}$ for large j . Comparing the fixed slab $\{t/2 < \varrho_j < t\}$ with the latter sublevel set in (26) gives

$$\frac{\text{Vol}_{h_j} B_R(p, h_j)}{\text{Vol}_{h_j} B_1(p, h_j)} \leq C(T, t),$$

where the right-hand side is independent of R . Passing first to the measured limit and then letting $R \rightarrow \infty$ contradicts the fact that a complete noncompact positive-dimensional Ricci limit has infinite total measure. Hence $L = \infty$. Since $\lambda_\infty^{1/3} = \varrho_\infty^{-1}$, its range is $(0, \infty)$. $\square$

The flow argument in the proof, followed by the scale-independent annular Harnack inequality, gives the stronger endpoint statements

$$\varrho_\infty \longrightarrow 0 \quad \text{at } p_\infty, \quad \varrho_\infty \longrightarrow \infty \quad \text{at infinity.}$$

Hence ϱ_∞ is proper on $\mathcal{R}$: for $0 < c_1 < c_2$,

$$\{c_1 < \varrho_\infty < c_2\} \Subset \mathcal{R}.$$

Since the limit rescaled conformal factor is nonconstant, unique continuation shows that $\nabla_{\tilde{h}_\infty} \lambda_\infty^{-1/3}$ is nonzero on a nonempty open subset of each noncollapsed limit $\tilde{B}_\infty$. After lifting to $\tilde{B}_j$, the vector fields

$$\mathcal{K}_j := -J\nabla_{h_j} \lambda_j^{-1/3}$$

converge smoothly to $\mathcal{K}_\infty := -J\nabla_{\tilde{h}_\infty} \lambda_\infty^{-1/3}$. On this open subset set

$$(27) \quad V_j := |\nabla_{h_j} \lambda_j^{-1/3}|_{h_j}^{-2} = \lambda^{-2/3}(q_j) r_j^{-2} V \rightarrow V_\infty := |\nabla_{\tilde{h}_\infty} \lambda_\infty^{-1/3}|_{\tilde{h}_\infty}^{-2}.$$

Lemma 3.10 will show that this definition extends across the entire local cover. Now we go back to the ansatz applied to the end. After passing to h_j the metric becomes

$$(28) \quad \begin{aligned} h_j &= r_j^{-2} V(d\varrho^2 + e^u(dx^2 + dy^2)) + r_j^{-2} V^{-1} \eta^2 \\ &= V_j(d\varrho_j^2 + \lambda^{2/3}(q_j) e^u(dx^2 + dy^2)) + r_j^{-4} \lambda^{-2/3}(q_j) V_j^{-1} \eta^2. \end{aligned}$$

We first prove

Lemma 3.10. *In each non-collapsed limit $\tilde{B}_\infty$, $\mathcal{K}_\infty$ has no zeros. Therefore $\lambda_\infty^{1/3}$ has no critical points.*

Proof. Suppose, to the contrary, that $\mathcal{K}_\infty$ vanishes. Since $\mathcal{K}_\infty = J\nabla_{\tilde{g}_\infty} \lambda_\infty^{1/3}$, each component of its zero set is either an isolated point or a holomorphic curve. The lifted field commutes with G_∞^0 . Hence, if it vanished at $\tilde{x}$, it would vanish along $G_\infty^0 \cdot \tilde{x}$. This orbit has positive dimension, so an isolated zero is impossible.

We now make precise the terminology on the collapsed limit. A point $x \in \mathcal{R}$ is called a *critical point* of $\lambda_\infty^{1/3}$ if, on a noncollapsed local cover $\pi_\infty : \tilde{B}_\infty \rightarrow B_\infty$ containing x , some lift $\tilde{x} \in \pi_\infty^{-1}(x)$ is a critical point of the lifted function. This definition is independent of the local cover and of the lift, because the lifted functions agree under the local deck transformations and on overlaps. Let $\text{Crit}_{\mathcal{R}}(\lambda_\infty^{1/3})$ denote the resulting subset of $\mathcal{R}$, and let C denote one of its connected components. By the preceding paragraph, every such C is locally covered by a holomorphic critical curve $\tilde{C} \subset \tilde{B}_\infty$.

Quadratic curvature decay bounds $r_j^2 \lambda(q_j)$. After passing to a subsequence, either this quantity tends to zero or it tends to a constant $c > 0$.

Suppose $r_j^2 \lambda(q_j) \rightarrow 0$. Passing (17) to the limit gives $\Delta_{\tilde{h}_\infty} \lambda_\infty^{1/3} = 0$. Along a Morse–Bott critical curve the Hessian vanishes in the tangent directions and is nondegenerate on the real two-dimensional normal bundle. At a critical point the Hessians with respect to the conformal metrics $\tilde{h}_\infty$ and $\tilde{g}_\infty$ agree, and the Killing equation for $J\nabla_{\tilde{g}_\infty} \lambda_\infty^{1/3}$ makes this Hessian J -invariant. Thus the two normal eigenvalues are equal. They therefore have the same nonzero sign, so their sum cannot vanish. This contradicts $\Delta_{\tilde{h}_\infty} \lambda_\infty^{1/3} = 0$ along the curve. Thus no critical curve occurs in this case.

Next suppose $r_j^2 \lambda(q_j) \rightarrow c > 0$. Passing (17) to the limit shows that every critical curve in $\tilde{B}_\infty$ is a local maximum of $\lambda_\infty^{1/3}$. The function is Morse–Bott on each local cover. Its gradient flow descends to the base and the descended local flows agree on overlaps, so they define a flow on $\mathcal{R}$. Recall that $\lambda_\infty^{1/3}$ has range $(0, \infty)$ and that $\varrho_\infty = \lambda_\infty^{-1/3}$ is proper.

For each component C of $\text{Crit}_{\mathcal{R}}(\lambda_\infty^{1/3})$, we set

$$D(C) = \{x \in \mathcal{R} \mid \text{the gradient flow starting from } x \text{ converges to } C\}.$$

We claim that $D(C)$ is open. Choose a small basin neighborhood U_C of C using the Morse–Bott normal form and the fact that C is a local maximum. If the trajectory from $x \in D(C)$ enters U_C at time T , divide its initial segment into finitely many pieces, each lifting to a local cover without critical points. ODE dependence on initial data makes the time- T map continuous near x . Every sufficiently nearby trajectory also enters U_C , proving that $D(C)$ is open.

For a constant $c > 0$ we set

$$D(c) = \{x \in \mathcal{R} \mid \text{the gradient flow starting from } x \text{ enters } \{\varrho_\infty < c\}\}.$$

The set $D(c)$ is also open. If the trajectory from $x \in D(c)$ enters $\{\varrho_\infty < c\}$ at time T , divide its initial segment into finitely many pieces and lift them successively to local covers. Continuous dependence on initial data shows that every trajectory starting sufficiently near x also lies in $\{\varrho_\infty < c\}$ at time T . Hence $D(c)$ is open.

We justify that these exhaust $\mathcal{R}$. Reparametrize a nonconstant gradient trajectory by

$$X := \frac{\nabla \lambda_\infty^{1/3}}{|\nabla \lambda_\infty^{1/3}|^2},$$

so that the value of $\lambda_\infty^{1/3}$ increases at unit speed. On every compact slab of regular values, properness and the local cover gradient bounds make this flow complete. A forward trajectory either enters $\{\varrho_\infty < c\}$ or remains in a compact slab. In the latter case its accumulation set is critical; the Morse–Bott normal form implies convergence to one critical component. Hence every point belongs to exactly one of the open sets above.

The value of ϱ_∞ is constant on each C . If there is a critical component C , choose c small enough that C is not contained in $\{\varrho_\infty < c\}$. We then have the disjoint decomposition

$$\mathcal{R} = D(c) \sqcup \coprod_{C \notin \{\varrho_\infty < c\}} D(C).$$

Each term is open and $D(c)$ is nonempty by the range statement. This disconnects $\mathcal{R}$, a contradiction. $\square$

As before, set $(M, h_j, p) = (M, r_j^{-2}h, p)$ and reduce at the level $\varrho_j = 1$. The moment map of $\mathcal{K}_j$ with respect to g_j is $\lambda_j^{1/3} = \varrho_j^{-1}$. Denote the reduced Kähler metric by $g_{\Sigma, j}$ and set

$$h_{\Sigma, j} := \lambda^{2/3}(q_j)e^u(dx^2 + dy^2) = \varrho_j^2 V_j^{-1} g_{\Sigma, j}.$$

We use q_j also for the image of the original point in Σ .

Lemma 3.11. *The sequence $(\Sigma, h_{\Sigma, j})$ has uniformly bounded curvature.*

Proof. Given $o_j \in (\Sigma, h_{\Sigma, j})$, choose a lift to the level set $\{\varrho_j = 1\} \subset (M, h_j, p)$ and denote the lift by the same o_j . Set $s_j := d(o_j, p)$.

We now instead consider the asymptotic cone arising as the limit

$$(M, s_j^{-2}h, p) \rightarrow (M'_\infty, d'_\infty, p'_\infty).$$

The point $o_j \in (M, s_j^{-2}h, p)$ converges to a limit point $o_\infty \in (M'_\infty, d'_\infty, p'_\infty)$. Take the limit renormalized conformal factor $\lambda_\infty^{1/3}$ obtained by normalizing $\lambda^{1/3}$ at o_j . Since $\lambda^{-1/3}(q_j) = \lambda^{-1/3}(o_j)$, as functions on M , $\lambda_j = \lambda'_j$. Our previous argument for $(M, h_j, p) \rightarrow (M_\infty, d_\infty, p_\infty)$ carries over. Hence we have the non-collapsed convergence $\tilde{B}'_j \rightarrow \tilde{B}'_\infty$, the smooth convergence $V'_j \rightarrow V'_\infty$, and the convergence of the Kähler metric $\tilde{g}'_j \rightarrow \tilde{g}'_\infty$.

Since the limiting Killing field has no zeros, the Kähler reduction of $\tilde{g}'_j$ over $\tilde{B}'_j$ at the level $\varrho'_j = 1$ converges smoothly to the corresponding reduction of $\tilde{g}'_\infty$. The reduced metrics, conformally scaled by $\varrho_j^2 V_j^{-1}$, also converge smoothly. Since the limit has bounded curvature, this shows that

$$(\Sigma, \lambda^{2/3}(o_j)e^u(dx^2 + dy^2))$$

at o_j has bounded curvature that is uniform in j .

Since $\lambda^{-1/3}(q_j) = \lambda^{-1/3}(o_j)$ on the level under consideration and o_j was arbitrary, the metric $\lambda^{2/3}(o_j)e^u(dx^2 + dy^2) = h_{\Sigma, j}$ has uniformly bounded curvature everywhere. $\square$

Lemma 3.12. *The sequence $(\Sigma, h_{\Sigma, j})$ is globally noncollapsed, has uniformly bounded diameter, and converges smoothly to a metric of curvature one.*

Proof. There are $\epsilon, v > 0$ and points $o_j \in \Sigma$ such that

$$\text{Area}_{h_{\Sigma, j}} B_\epsilon(o_j) \geq v.$$

Otherwise the bounded curvature sequence would collapse everywhere. The resulting positive-rank $\mathcal{F}$ -structure on Σ would force $\chi_{\text{orb}}(\Sigma) = 0$, contrary to (23); see Cheeger–Fukaya–Gromov [15] and an orbifold version [34].

Consider an asymptotic cone $(M, s_j^{-2}h, p) \rightarrow (M_\infty, d_\infty, p_\infty)$, where $s_j = d(o_j, p)$ as above. Let $B_\epsilon(o_\infty, h_{\Sigma, \infty})$ be the noncollapsed limit of $B_\epsilon(o_j, h_{\Sigma, j})$. After decreasing the radius, the Kähler reductions in the corresponding balls $B_j = B_j(o_j)$ are noncollapsed. Choose a local holomorphic coordinate $x + iy$ on Σ . As in (28),

$$(29) \quad h_j := s_j^{-2}h = V_j(d\varrho_j^2 + \lambda^{2/3}(o_j)e^u(dx^2 + dy^2)) + s_j^{-4}\lambda^{-2/3}(o_j)V_j^{-1}\eta^2.$$

Now we use the equation $V = -12\varrho + 6\varrho^2 u_\varrho$. The rescaled V_j satisfies

$$V_j = \lambda^{-1}(o_j)s_j^{-2}(-12\varrho_j + 6\varrho_j^2 \partial_{\varrho_j} u_j)$$

with $u_j = u + \log \lambda^{2/3}(o_j)$. Note that although we choose coordinates (x, y) on Σ to write (29), $-12\varrho_j + 6\varrho_j^2 \partial_{\varrho_j} u_j$ intrinsically is a function on the quotient of the end by S^1 , which does not depend

on the choice of the coordinate. Quadratic curvature decay gives $\lambda^{-1}(o_j)s_j^{-2} \geq c > 0$. On $\tilde{B}_j$, since $V_j \rightarrow V_\infty > 0$, there are therefore two situations to consider after passing to a subsequence.

- (1) $\lambda^{-1}(o_j)s_j^{-2}$ tends to a positive constant and $-12\varrho_j + 6\varrho_j^2\partial_{\varrho_j}u_j$ converges to a nonzero function.
- (2) $\lambda^{-1}(o_j)s_j^{-2} \rightarrow \infty$ and $-12\varrho_j + 6\varrho_j^2\partial_{\varrho_j}u_j$ converges to zero.

In case (1), the Ricci-flat metric $\tilde{h}_\infty$ on each $\tilde{B}_\infty$ is Type II; in case (2), it is Type I.

Claim. *The case (1) cannot happen.*

If (1) happened, then we would have $\lambda^{-1}(o_j) \sim s_j^2$. We consider the annuli A_j around p with inner and outer radii $1/2$ and 2 , which converge to $A_\infty \subset \mathcal{R}$. After shrinking B_j , the bounded length of the circle orbit through p keeps every orbit through B_j inside A_j . Formula (29) shows that such an orbit has length comparable to $s_j^{-2}\lambda^{-1/3}(o_j)$. Since the Kähler reductions in B_j are noncollapsed and V_j converges smoothly, Fubini's theorem in the local bundle chart gives the lower bound

$$\text{Vol}(A_j) \geq Cs_j^{-2}\lambda^{-1/3}(o_j) \sim Cs_j^{-4/3}.$$

On the other hand, because $\lambda^{-1}(o_j) \sim s_j^2$, the values of the unscaled coordinate ϱ on A_j are $O(s_j^{2/3})$. Lemma 3.8 therefore gives

$$\text{Vol}(A_j) \leq Cs_j^{-4}(-2T_Kas_j^2 - 6T_Kbs_j^{4/3}),$$

since $A_j \subset \{1/2 \inf_{A_\infty} \varrho_\infty < \varrho_j < 2 \sup_{A_\infty} \varrho_\infty\}$ when j is large. This leads to a contradiction and the claim is proved.

Hence we must be in case (2). In the limit $\tilde{B}_\infty$, after choosing a holomorphic coordinate $x + iy$ on the Kähler reduction, we can pass the ansatz to the limit and obtain

$$\tilde{g}_\infty = \varrho_\infty^{-2}(V_\infty(d\varrho_\infty^2 + e^{u_\infty}(dx^2 + dy^2)) + V_\infty^{-1}\eta_\infty^2).$$

It follows that $-12\varrho_\infty + 6\varrho_\infty^2\partial_{\varrho_\infty}u_\infty = 0$. Hence, in the chosen conformal chart, $u_\infty = 2 \log \varrho_\infty + \varpi(x, y)$, where $e^\varpi(dx^2 + dy^2)$ is the local expression of the limiting reduced metric. Passing the $SU(\infty)$ Toda equation to the limit shows

$$(u_\infty)_{xx} + (u_\infty)_{yy} + (e^{u_\infty})_{\varrho_\infty\varrho_\infty} = 0,$$

which shows this metric has curvature one on every noncollapsed limit.

We now propagate this conclusion over the entire pointed limit. The curvature bound in Lemma 3.11 gives a uniform lower Ricci curvature bound. Since $B_\epsilon(o_j, h_{\Sigma,j})$ is noncollapsed, relative volume comparison gives, for every fixed $R < \infty$, a constant $v_R > 0$ such that

$$\text{Area}_{h_{\Sigma,j}} B_1(x) \geq v_R \quad \text{whenever } d_{h_{\Sigma,j}}(o_j, x) \leq R.$$

The curvature bound and the two-dimensional injectivity radius estimate therefore give a positive injectivity radius bound on every fixed ball about o_j . A diagonal argument consequently gives a complete pointed smooth limit $(\Sigma_\infty, h_{\Sigma,\infty}, o_\infty)$.

Every point of this limit lies at finite distance from o_∞ and is thus represented by a noncollapsed sequence. Repeating the preceding Type I argument at that sequence shows that $K_{h_{\Sigma,\infty}} \equiv 1$. The limit is therefore a complete connected spherical surface. In the free-circle case it is the round $\mathbb{S}^2$; in the locally free case it is the corresponding round orbifold $\mathbb{S}^2/\Gamma$. In particular, the pointed limit is compact.

Compactness of the limit upgrades pointed smooth convergence to global smooth convergence. Choose R larger than its diameter. For large j , the Cheeger–Gromov embedding of $B_R(o_\infty) = \Sigma_\infty$ into Σ is a local diffeomorphism. Its image is open and, because the domain is compact, closed; connectedness of Σ makes it surjective. It follows that the diameter is uniformly bounded and the injectivity radius is uniformly positive (in the orbifold sense when appropriate). $\square$

Thus $(\Sigma, h_{\Sigma,j})$ converges to the round $\mathbb{S}^2$, or to a round orbifold $\mathbb{S}^2/\Gamma$ when the circle action has finite isotropy. The group $\Gamma \subset \mathrm{SO}(3)$ is cyclic, dihedral, tetrahedral, octahedral, or icosahedral. When Γ is a nontrivial cyclic group, there are two orbifold points in $\mathbb{S}^2/\Gamma$, while in the noncyclic cases there are three orbifold points. The corresponding orders of the orbifold points in these cases are

$$\mathbb{S}^2(n, n), \quad \mathbb{S}^2(2, 2, n), \quad \mathbb{S}^2(2, 3, 3), \quad \mathbb{S}^2(2, 3, 4), \quad \mathbb{S}^2(2, 3, 5).$$

The link is an orbifold S^1 -bundle over such a round orbifold. It is then either $S^2 \times S^1$ or covered by S^3 . Consequently,

Lemma 3.13. *The asymptotic cone is three-dimensional and carries the limit metric*

$$(30) \quad h_\infty = V_\infty(d\varrho_\infty^2 + \varrho_\infty^2 g_{\mathbb{S}^2/\Gamma}).$$

Proof. The radial direction and the globally noncollapsed round reduction give three independent limit directions. If the circle coefficient in (28) did not collapse, the local cover convergence would give a fourth noncollapsed direction, contrary to the collapsed hypothesis. Thus precisely the $\mathcal{K}$ -orbits collapse, and passing the first two terms of (28) to the limit gives (30). $\square$

We replace the distance function d_∞ in the tuple $(M_\infty, d_\infty, p_\infty)$ by the metric h_∞ .

Lemma 3.14. *When $\mathcal{K}$ induces an S^1 -action, each non-collapsed limit $(\tilde{B}_\infty, \tilde{h}_\infty)$ is flat. Any such Hermitian non-Kähler gravitational instanton has a unique asymptotic cone $\mathbb{R}^3/\Gamma$, with link $\mathbb{S}^2/\Gamma$ being a round orbifold.*

Proof. By (28) and Lemma 3.13, we have

$$\tilde{h}_\infty = V_\infty(d\varrho_\infty^2 + \varrho_\infty^2 g_{\mathbb{S}^2}) + V_\infty^{-1}\eta_\infty^2,$$

where the connected local collapse group is $G_\infty^0 = \mathbb{R}$. Since the local cover geometry is Type I, $W_{h_\infty}^+ \equiv 0$. Consequently, the conformal metric $\tilde{g}_\infty = \varrho_\infty^{-2}\tilde{h}_\infty$ is scalar-flat Kähler, with the complex structure defined by $J_\infty : d\varrho_\infty \rightarrow V_\infty^{-1}\eta_\infty$ and the standard complex structure on $\mathbb{S}^2/\Gamma$. The ansatz from Theorem 2.5 then takes the form

$$(31) \quad \tilde{g}_\infty = W_\infty d\xi_\infty^2 + W_\infty^{-1}\eta_\infty^2 + W_\infty e^{v_\infty}(dx^2 + dy^2)$$

where the scalar-flat LeBrun equations are

$$\partial_{\xi_\infty}^2(e^{v_\infty}) + \partial_x^2 v_\infty + \partial_y^2 v_\infty = 0, \quad \partial_{\xi_\infty}^2(W_\infty e^{v_\infty}) + \partial_x^2 W_\infty + \partial_y^2 W_\infty = 0.$$

The variables $W_\infty, \xi_\infty, v_\infty$ are related to $V_\infty, \varrho_\infty, u_\infty$ as in (7). The conformal scalar-curvature formula for $\tilde{h}_\infty = \xi_\infty^{-2}\tilde{g}_\infty$ shows that ξ_∞^{-1} is $\tilde{g}_\infty$ -harmonic. In the LeBrun ansatz this becomes

$$\Delta_{\tilde{g}_\infty}(\xi_\infty^{-1}) = -\frac{1}{W_\infty e^{v_\infty}} \partial_{\xi_\infty}(e^{v_\infty} \xi_\infty^{-2}).$$

Therefore, $e^{v_\infty} \xi_\infty^{-2}$ is independent of ξ_∞ . Hence, locally, $v_\infty = 2 \log \xi_\infty + \varpi(x, y)$. This agrees with the local expression $u_\infty = 2 \log \varrho_\infty + \varpi(x, y)$ above.

Now we turn to the equation

$$(32) \quad \partial_{\xi_\infty}^2(W_\infty e^{v_\infty}) + \partial_x^2 W_\infty + \partial_y^2 W_\infty = 0.$$

Since $e^\varpi(dx^2 + dy^2)$ is the local expression of the round metric $g_{\mathbb{S}^2}$ (after passing to the cover in the orbifold case), (32) can be rewritten globally as

$$(33) \quad \partial_{\xi_\infty}^2(W_\infty \xi_\infty^2) + \Delta_{\mathbb{S}^2} W_\infty = 0.$$

In the metric h_∞ , this is the Gibbons–Hawking equation

$$\partial_{\varrho_\infty}(\varrho_\infty^2 \partial_{\varrho_\infty} V_\infty) + \Delta_{\mathbb{S}^2} V_\infty = 0,$$

which says that $V_\infty > 0$ is harmonic on $\mathbb{R}^3 \setminus \{0\}$ for the flat metric $d\rho_\infty^2 + \rho_\infty^2 g_{\mathbb{S}^2}$. Bôcher's theorem gives

$$V_\infty = c + \frac{d}{\rho_\infty}$$

for constants $c, d \geq 0$.

Finally we claim $V_\infty = c$. To see this, consider the ratio between the integral formulas (23) and (24)

$$(34) \quad \frac{\int_\Sigma V e^u dx dy}{\int_\Sigma e^u dx dy} = \frac{-6a\rho^2 - 12b\rho}{2\pi\chi(\Sigma)\rho^2 + a\rho + b}.$$

Because the reduced metrics converge to the round metric, after renormalizing by constants, the left hand side converges to the spherical average of V_∞ . The expression $c + d/\rho_\infty$ is radial, so (34) determines it pointwise. If $a < 0$, then

$$V_\infty \equiv \frac{-6a}{2\pi\chi(\Sigma)};$$

if $a = 0$, then

$$V_\infty \equiv \frac{-12b}{2\pi\chi(\Sigma)\rho_\infty}.$$

The second alternative would imply $\rho V \rightarrow -12b/(2\pi\chi(\Sigma))$ along the end. Equation (22) and Lemma 3.8 would then give maximal volume growth, contrary to the collapse assumption. Hence V_∞ is constant. More explicitly, when $V_\infty = d/\rho_\infty$, $\sqrt{\rho}$ is comparable to the distance function to the base point. Therefore the ball $B_R(p)$ in (M, h, p) has order R^4 volume. $\square$

As a corollary, setting ρ_∞ to be the distance function to the cone vertex in the asymptotic cone we have

Corollary 3.15. *On the asymptotic cone, V_∞ is constant and ρ_∞ is a constant multiple of the distance function ρ_∞ .*

3.2.2. $\mathcal{K}$ induces a higher rank torus action. We now repeat the arguments from Section 3.2.1 for the case where $\mathcal{K}$ induces only an $\mathbb{R}$ -action.

Since $\lambda^{1/3}$ tends to zero at infinity, the level set $\{\lambda^{1/3} = \epsilon\}$ is compact for small ϵ . The closure of the $\mathbb{R}$ -flow in the isometry group of this level is a torus T^k . Its action preserves the Kähler form and is locally Hamiltonian; only local moment coordinates are used below. The end remains diffeomorphic to $(N_0, \infty) \times L$, with T^k acting on the link L . Kähler reduction by $\mathcal{K}$ is available on small open subsets of L on which the induced foliation is trivial, and the metric is locally of the form (22).

Lemma 3.16. *The closure of the flow of $\mathcal{K}$ is not a T^3 -action. Hence it is a T^2 -action.*

Proof. If the closure were T^3 , then each link would be a single T^3 -orbit, so u and V in (22) would depend only on ρ . The Toda equation would reduce to $(e^u)_{\rho\rho} = 0$, whence $e^u = A\rho + B$. Substitution into $V = -12\rho + 6\rho^2 u_\rho$ gives

$$V = -6\rho \frac{A\rho + 2B}{A\rho + B}.$$

Here $\rho > 0$ and the formula must hold for all sufficiently large ρ . Positivity of $e^u = A\rho + B$ rules out $A < 0$. If $A > 0$ the displayed expression is eventually negative, and if $A = 0$ then $B > 0$ and it equals -12ρ . Both contradict $V > 0$. Thus a T^3 -closure is impossible. $\square$

Thus the metric is toric in this case. Up to a finite cover, a closed three-manifold with this T^2 -action is S^3 , $S^2 \times S^1$, or T^3 , equipped with the standard action; see, for example, [58, Theorem 3.8]. So in the following we pass the end to a finite cover, so that the link L of the end is one of S^3 , $S^2 \times S^1$, or T^3 .

If $L = T^3$, one may see the contradiction directly by passing to the universal cover of a T^2 -orbit and quotienting by a lattice for which $\mathcal{K}$ is integral. The resulting end falls under the S^1 -case with reduced torus Σ ; formulas (23) and (24) cannot both be positive for large ϱ when $\chi(\Sigma) = 0$.

We therefore consider $L = S^3$ or $S^2 \times S^1$. The T^2 -action is the standard one on S^3 or $S^2 \times S^1$, and the end is diffeomorphic to $(N, \infty) \times L$. Since the quotient of the end by T^2 is simply-connected, the T^2 -action is Hamiltonian, and there are global moment maps. We now derive the analogue of Lemma 3.8. Choose a basis $(\partial_{\phi_1}, \partial_{\phi_2})$ of $\text{Lie}(T^2)$ with $\partial_{\phi_1} = \mathcal{K}$. This is a basis of the Lie algebra, not necessarily an integral basis of the torus lattice; in fact, ∂_{ϕ_1} is irrational because its flow is not closed. Let x_α be a Hamiltonian for ∂_{ϕ_α} , normalized so that $x_1 = \xi$. In the corresponding local action-angle coordinates, the Kähler metric $g = \xi^2 h$ is

$$(35) \quad g = g_{\alpha\beta} dx_\alpha dx_\beta + g^{\alpha\beta} d\phi_\alpha d\phi_\beta$$

where $g_{\alpha\beta} = \partial_{x_\alpha x_\beta}^2 f$, $(g^{\alpha\beta}) = (g_{\alpha\beta})^{-1}$, and f is the symplectic potential. Hence

$$(36) \quad h = \frac{1}{x_1^2} (g_{\alpha\beta} dx_\alpha dx_\beta + g^{\alpha\beta} d\phi_\alpha d\phi_\beta).$$

On a level set of x_1 , reduction by ∂_{ϕ_1} gives the metric

$$(37) \quad \frac{1}{x_1^2} \left(g_{22} dx_2^2 + \left(g^{22} - \frac{(g^{12})^2}{g^{11}} \right) d\phi_2^2 \right).$$

Since $V^{-1} = |\partial_{\phi_1}|_h^2 = x_1^{-2} g^{11}$, comparison with $V e^u (dx^2 + dy^2)$ gives

$$(38) \quad e^u (dx^2 + dy^2) = \frac{1}{x_1^4} (g^{11} g_{22} dx_2^2 + \det(g^{\alpha\beta}) d\phi_2^2).$$

We may take $y = \phi_2$ on a local quotient. The isothermal coordinate x transverse to the circle orbits then satisfies

$$(39) \quad \frac{dx}{dx_2} = \sqrt{\frac{g^{11} g_{22}}{\det(g^{\alpha\beta})}}.$$

and $e^u = x_1^{-4} \det(g^{\alpha\beta})$. Because all coefficients are independent of y , the $SU(\infty)$ Toda equation reduces to

$$(e^u)_{\varrho\varrho} + u_{xx} = 0.$$

Although x , y , and u are only locally defined, the three-form $e^u dx \wedge dy \wedge \eta$ is global on L .

The moment polytope of the end therefore has two boundaries P_1 and P_2 . The orbit space of a regular level $L_0 = \{\varrho = \varrho_0\}$ is an interval, whose endpoints are contained in P_1 and P_2 . Choose ∂_{ϕ_2} to be the primitive isotropy field along P_1 , and let x_2 be its affine moment coordinate. The moment image of L_0 is the segment $x_1 = \varrho_0^{-1}$ in the polytope. By contrast, the isothermal coordinate x on the regular part of the reduced two-sphere has range $\mathbb{R}$: after orienting x , the poles on P_1 and P_2 correspond to $x = -\infty$ and $x = +\infty$, respectively.

Let

$$C_\Lambda := \int_{T^2} dy \wedge \eta > 0.$$

This constant records the torus lattice in the possibly nonintegral basis $(\mathcal{K}, \partial_{\phi_2})$. Integrating the reduced Toda equation gives

$$\frac{d^2}{d\varrho^2} \int_L e^u dx dy \eta = -C_\Lambda (u_x(+\infty) - u_x(-\infty)).$$

To evaluate the boundary terms, recall that smoothness of the toric Kähler metric along P_1 gives

$$(40) \quad f(x_1, x_2) = \frac{1}{2} x_2 \log x_2 + f_S(x_1, x_2)$$

where f_S is smooth across P_1 . Thus

$$\det(\text{Hess} f) = \det(g_{\alpha\beta}) = \frac{\delta_S(x_1, x_2)}{x_2}$$

for a smooth positive function δ_S . Equations (39) and (38) yield, near P_1 ,

$$(41) \quad \frac{dx}{dx_2} = g_{22} = \frac{1}{2x_2} + O(1), \quad \det(g^{\alpha\beta}) = \delta_S^{-1} x_2, \quad u = \log \det(g^{\alpha\beta}) - 4 \log x_1 = \log x_2 + O(1).$$

Consequently $x = \frac{1}{2} \log x_2 + O(1)$ and $u_x(-\infty) = 2$. Write the primitive inward isotropy field at P_2 as $Z_2 = \alpha \mathcal{K} + \beta \partial_{\phi_2}$, with $\beta < 0$. For its affine defining function $\ell = \alpha x_1 + \beta x_2 - c$, the same calculation gives $g_{22} = \beta^2 / (2\ell) + O(1)$ and $x = (\beta/2) \log \ell + O(1)$, hence $u_x(+\infty) = 2/\beta = -2/|\beta|$. The slice expansions also give exponential convergence to these limits, so integration by parts over $x \in \mathbb{R}$ is justified. The following is the analogue of (23)–(24).

Lemma 3.17. *There are constants a, b such that*

$$(42) \quad \int_L e^u dx dy \eta = C_\Lambda (1 + |\beta|^{-1}) \varrho^2 + a\varrho + b$$

and

$$(43) \quad \int_L V e^u dx dy \eta = -6a\varrho^2 - 12b\varrho.$$

Lemma 3.18. *The volume of a ϱ -annulus can be computed explicitly*

$$\text{Vol}\{D_0 \leq \varrho \leq D_1\} = -2a(D_1^3 - D_0^3) - 6b(D_1^2 - D_0^2).$$

The local arguments above carry over, but the geometry of the collapse must be described carefully. Reduction by the dense $\mathbb{R}$ -flow of $\mathcal{K}$ is only local, whereas the orbit space of the full T^2 -action is global. We will identify the lifted $\mathcal{K}$ -direction as a pure translation after proving below that the local cover limit is flat.

The range argument in Lemma 3.9 also applies here, with Lemma 3.18 replacing Lemma 3.8; the two volume polynomials have exactly the same scaling alternatives. More importantly, the proof of Lemma 3.10 is entirely local and nowhere uses the closed-orbit assumption. Thus $\mathcal{K}_\infty$ has no zero on any noncollapsed local cover limit. Replace the global surface $(\Sigma, h_{\Sigma,j})$ by local quotients by the $\mathcal{K}_j$ -flow. On a small regular ball and its noncollapsed local cover we have

$$\tilde{h}_j = V_j(d\varrho_j^2 + \lambda^{2/3}(q_j)e^u(dx^2 + dy^2)) + r_j^{-4}\lambda^{-2/3}(q_j)V_j^{-1}\eta^2,$$

where y parametrizes the other torus direction locally. Since the lifted Kähler metrics and the nowhere-zero vector fields $\mathcal{K}_j$ converge smoothly to $\mathcal{K}_\infty$, the regular level sets and their local Kähler quotients converge smoothly. After conformal rescaling by $\varrho_j^2 V_j^{-1}$, the reduced metric is

$$\lambda^{2/3}(q_j)e^u(dx^2 + dy^2),$$

and consequently has uniformly bounded curvature.

Since ∂_{ϕ_2} is a rotational Killing field with zero along P_1 , it converges near its zero to a nontrivial rotational Killing field on the local quotient. Its norm shows that $u_j = u + \log \lambda^{2/3}(q_j)$ converges smoothly. Recall also that

$$\mathcal{K}_j = r_j^2 \lambda^{1/3}(q_j) \partial_{\phi_1}.$$

In particular, the moment map of ∂_{ϕ_2} under $\tilde{g}_j$, which is $\mu_j = r_j^{-2} c_j^{-2} x_2$ after scaling, converges smoothly to a nontrivial function on each $\tilde{B}_\infty$. Here $g_j = r_j^{-2} c_j^{-2} g$ with $c_j = \lambda(q_j)^{1/3}$.

If a local cover limit were Type II, then $r_j^2 c_j^3 \rightarrow \ell > 0$, hence $r_j^{-2} c_j^{-2} \rightarrow 0$. However, on the original end x_2 is bounded: the moment map x_2 vanishes along P_1 , while the other moment map $\alpha x_1 + \beta x_2$, which corresponds to the Killing field that vanishes along P_2 , is constant along P_2 . The constant $\beta \neq 0$ since $\mathcal{K}$ is irrational inside T^2 . Since $x_1 = \xi \rightarrow 0$ and x_2 is affine in the polytope of the end, this indicates that x_2 is bounded. This contradicts the convergence of μ_j and $r_j^{-2} c_j^{-2} \rightarrow 0$.

Thus every local cover limit is Type I, and

$$(44) \quad \varrho_j^{-2} \lambda^{2/3}(q_j) e^u (dx^2 + dy^2)$$

converges to the unit round metric on $\mathbb{S}^2$. The normalized field ∂_{ϕ_2} converges to a rotational Killing field. This conclusion propagates throughout the regular set: join q_∞ to a point of any other ball B'_∞ by a path in $\mathcal{R}$ and cover the path by finitely many overlapping balls. Smooth equivariant convergence on one ball determines the normalization on the next overlap. A nonzero Killing field cannot vanish on an open set, so the limiting rotational field remains nontrivial along the chain.

Consequently, every local reduction has the same round limit. The radial direction and the round reduction give three noncollapsed directions on the local cover. The h_j -norm of the unnormalized field $\mathcal{K}$ must tend uniformly to zero on each compact regular annulus. Otherwise a local foliation chart, together with these three transverse directions, would give four-dimensional noncollapse on a regular ball, contrary to the collapsed hypothesis.

This implies collapse of the whole torus orbit. The quotient retains the radial variable and the interval quotient of the round sphere and is therefore two-dimensional.

Because V_∞ is invariant under the limiting collapse group, $\hat{h}_\infty := V_\infty^{-1} h_\infty$ is defined on the two-dimensional quotient. Together with the radial direction, the quotient of the round local reduction by its limiting rotation is locally a Euclidean half-plane; the two fixed poles determine its boundary rays. Hence $(M_\infty, \hat{h}_\infty, p_\infty)$ is a flat wedge whose two boundary rays are the limits of the two isotropy components of the T^2 -action.

It remains to determine the wedge angle. We use the continuation argument from the proof of [51, Lemma 2.1]. Choose points q_+ and q_- at the same radial level on the two boundary components and join nearby interior points by a path in the regular annular quotient. Cover the path by finitely many balls $D_1, \dots, D_N$ with consecutive overlaps. On each local quotient the round reduction supplies the unit-speed meridional coordinate θ , and $\hat{h}_\infty = d\varrho_\infty^2 + \varrho_\infty^2 d\theta^2$. On an overlap, the common rotational field ∂_{ϕ_2} identifies the meridional coordinate, up to the same choice of orientation. Continuing along the chain therefore identifies the interval between the two isotropy components with the quotient of a unit round sphere by its rotation. This interval has length π , so the two boundary rays are opposite. Hence the wedge angle is π , and

$$(M_\infty, \hat{h}_\infty, p_\infty) \cong \mathbb{H}.$$

Thus the asymptotic cone is the half-plane equipped with

$$h_\infty = V_\infty (d\varrho_\infty^2 + e^{u_\infty} dx^2),$$

and $V_\infty^{-1} h_\infty$ is the Euclidean metric on $\mathbb{H}$.

Finally, apply the argument of Lemma 3.14. The LeBrun function W_∞ is globally defined on $\mathbb{H}$, and its pullback under the rotational quotient $\mathbb{R}^3 \rightarrow \mathbb{H}$ satisfies (33). Equivalently, the pulled-back

function V_∞ is a positive harmonic function on $\mathbb{R}^3 \setminus \{0\}$, so Bôcher's theorem gives $V_\infty = c + d/\varrho_\infty$. The ratio of the two integral formulas above rules out the d/ϱ_∞ alternative in the collapsed case, exactly as in Lemma 3.14; hence $V_\infty \equiv c > 0$. The same ratio fixes c from the end constants a and C_Λ , independently of the chosen rescaling sequence. Thus the cone is unique, and the local cover limits are flat.

We summarize the argument as follows.

Lemma 3.19. *When $\mathcal{K}$ induces only an $\mathbb{R}$ -action, the local covers of the asymptotic cone $(M_\infty, h_\infty, p_\infty)$ are flat. Any such Hermitian non-Kähler gravitational instanton has a unique asymptotic cone $\mathbb{H}$.*

Similarly, as a corollary we have

Corollary 3.20. *On the asymptotic cone, V_∞ is constant and ϱ_∞ is a constant multiple of ρ_∞ .*

Including the noncollapsed case, we have proved the following.

Theorem 3.21 (Classification of asymptotic cones). *Every Hermitian non-Kähler gravitational instanton has a unique asymptotic cone. The cone is one of the following:*

- $\mathbb{R}^4/\Gamma$ with $\Gamma \subset \mathrm{SO}(4)$ acting freely on $\mathbb{S}^3$ in the noncollapsed situation.
- $\mathbb{R}^3/\Gamma$, with round orbifold link $\mathbb{S}^2/\Gamma$, or $\mathbb{H}$ in the collapsed situation.

In the noncollapsed case, uniqueness and the ALE asymptotic expansion follow from the standard coordinate construction at infinity of Bando–Kasue–Nakajima [5]. In the collapsed cases, uniqueness is the content of Lemmas 3.14 and 3.19. In the collapsed cases, collapse occurs along S^1 -orbits when the cone is $\mathbb{R}^3/\Gamma$ and along T^2 -orbits when it is $\mathbb{H}$. Since all local cover limits are flat, $|\mathrm{Rm}_h|_h = o(\rho^{-2})$, where ρ is the distance from a fixed base point.

For brevity, call a one-ended four-manifold $\mathcal{AF}$ if its curvature is $o(\rho^{-2})$ and it has a unique asymptotic cone which is a metric cone. Note that this is different from AF that we introduced in Section 2.4. Petrunin–Tuschmann [57] proved that if such an end is simply connected, then its asymptotic cone is $\mathbb{R}^4$, $\mathbb{R}^3$, or the half-plane $\mathbb{H}$, and conjectured that the last possibility cannot occur. This conjecture was proved in [51].

Theorem 3.22 ([51]). *For an $\mathcal{AF}$ 4-manifold that is simply-connected at infinity, its asymptotic cone is either $\mathbb{R}^4$ or $\mathbb{R}^3$.*

It follows that an end with cone $\mathbb{R}^3/\Gamma$ where Γ is nontrivial, or with cone $\mathbb{H}$, is not simply connected at infinity. The preceding analysis then shows that its link L is either $S^2 \times S^1$ or finitely covered by S^3 . In the first case, the action generated by $\mathcal{K}$ combines translation on S^1 with a rotation of S^2 . A rational slope gives an S^1 -action and cone $\mathbb{R}^3/\mathbb{Z}_k$, whereas an irrational slope gives an $\mathbb{R}$ -action and cone $\mathbb{H}$. If L is covered by S^3 , a finite cover reduces the analysis to the case $L = S^3$.

We end this subsection with an observation. The limit rescaled conformal factor ϱ_∞^{-1} is a constant multiple of ρ_∞^{-1} , and V_∞ is constant. Thus, for some $k > 0$, the conformal metric is

$$\begin{aligned} (45) \quad \tilde{g}_\infty &= \frac{k^2}{\rho_\infty^2} (d\rho_\infty^2 + \rho_\infty^2 g_{\mathbb{S}^2}) + \frac{k^2}{\rho_\infty^2} \eta_\infty^2 \\ &= \frac{k^2}{\rho_\infty^2} (d\rho_\infty^2 + \eta_\infty^2) + k^2 g_{\mathbb{S}^2}, \end{aligned}$$

which splits as the product of a sphere with a cusp.

3.3. Classification of ends. We proceed to improve the previous classification of asymptotic cones to a classification of ends. By saying that a gravitational instanton has a specific kind of asymptotic geometry, we mean it is asymptotic to the corresponding asymptotic model with rate $O(\rho^{-\delta_0})$ for some $\delta_0 > 0$.

Theorem 3.23 (Classification of ends). *A Hermitian non-Kähler gravitational instanton (M, h) has one of the following asymptotic geometries:*

- *ALE with asymptotic cone $\mathbb{R}^4/\Gamma$.*
- *ALF-A with asymptotic cone $\mathbb{R}^3$, or a finite quotient of ALF-A with asymptotic cone $\mathbb{R}^3/\Gamma$.*
- *Rational AF with asymptotic cone $\mathbb{R}^3/\mathbb{Z}_k$ or irrational AF with asymptotic cone the half-plane $\mathbb{H}$.*

We prove the theorem in the remainder of this subsection. It suffices to treat non-maximal volume growth. Fix a $\mathcal{K}$ -invariant end $(M \setminus K, h)$ and make the following reduction.

- (1) If the link is covered by S^3 , pass to a finite cover with link S^3 . The induced S^1 -action is then free on the end. Set $(M_0, h_0) = (M \setminus K, h)$ and denote its circle quotient by (M_b, h_b) .
- (2) If the link is $S^2 \times S^1$, then the end is diffeomorphic to $(N_0, \infty) \times S^2 \times S^1$, with $\mathcal{K}$ generating a skew rotation. Take the universal cover and then quotient by an isometric $\mathbb{Z}$ -subgroup of the $\mathbb{R}$ -action generated by $\mathcal{K}$. Denote the resulting end by (M_0, h_0) . This operation makes the collapsing direction integral, so $\mathcal{K}$ generates a free circle action on (M_0, h_0) and its asymptotic cone is $\mathbb{R}^3$. Again denote the circle quotient by (M_b, h_b) .

It remains to prove that (M_0, h_0) is ALF-A in case (1) and AF_{0,b_0} in case (2). Once the asymptotics of (M_0, h_0) have been established, the isometric actions involved in the above reductions can be identified with isometric actions on the corresponding asymptotic model. Conjugating the asymptotic diffeomorphism by these actions and averaging via the center-of-mass construction [38] yields an equivariant asymptotic diffeomorphism with the same decay estimates. Reversing the covering and unwrapping operations therefore shows that the original end $(M \setminus K, h)$ is asymptotic to a finite quotient of an ALF-A model in case (1), or to an $\text{AF}_{a,b}$ model in case (2). We suppress the subscript in (M_0, h_0) for the remainder of this subsection.

Lemma 3.24. *There is a constant $k_0 > 0$ such that $V \rightarrow k_0^2$ as we approach infinity.*

Proof. Consider any rescaling sequence (M, h_j, p) converging to an asymptotic cone. Since V_∞ is constant on every local cover limit, suitable constants $C_j > 0$ satisfy $C_j V \rightarrow 1$ locally smoothly. By (23) and (24), the average of V over a reduced sphere is

$$\frac{\int_{\Sigma} V e^u dx dy}{\int_{\Sigma} e^u dx dy} = \frac{-6a\varrho^2 - 12b\varrho}{4\pi\varrho^2 + a\varrho + b}.$$

If $a = 0$, this average is asymptotic to a nonzero multiple of ϱ^{-1} . After normalization at $\varrho = \varrho_j$, its value at $t\varrho_j$ tends to t^{-1} , contradicting the constant local limit. Hence $a \neq 0$. Positivity of the average for large ϱ gives $a < 0$, and the same formula yields

$$V \longrightarrow -\frac{3a}{2\pi} =: k_0^2.$$

□

Corollary 3.15 and Lemma 3.24 give

$$\lambda^{-1/3}(q_j) \sim r_j.$$

Use the flow of $\nabla \varrho$ to identify all sufficiently large reduced spheres with one fixed S^2 . On a conformal chart of this fixed sphere the Toda function is denoted by u . If $z = x + iy$ and $\hat{z}$ are two conformal

coordinates, then

$$\widehat{u} = u + \log \left| \frac{dz}{d\widehat{z}} \right|^2.$$

Write the unit round metric locally as $g_{\mathbb{S}^2} = e^{\varpi_z} |dz|^2$. Its local coefficient obeys the same transformation law. Thus neither u nor ϖ_z is a global function on S^2 , but their difference $u - \varpi_z$ is global.

The convergence to the unique cone is smooth on local covers and uniform on dyadic annuli. In the fixed spherical trivialization this gives uniform scale-invariant bounds for all derivatives of V . It also shows that $e^{u-2\log \varrho} |dz|^2$ converges to the round metric just chosen; in particular, its limiting local coefficient is e^{ϖ_z} . The identity

$$u_\varrho - \frac{2}{\varrho} = \frac{V}{6\varrho^2}$$

therefore shows, after integration from infinity, that the following is a global estimate, written in any conformal chart:

$$u - \varpi_z = 2 \log \varrho + O'(\varrho^{-1}).$$

Set, in any conformal chart,

$$\varepsilon := u - 2 \log \varrho - \varpi_z.$$

The transformation law above makes ε a global function on each fixed reduced sphere and hence on the product end. Since $e^{\varpi_z} |dz|^2$ has curvature one, substitution in the Toda equation gives the global identity

$$(46) \quad \Delta_{\mathbb{S}^2} \varepsilon + \partial_\varrho^2 (\varrho^2 \varepsilon) = -\partial_\varrho^2 [\varrho^2 (e^\varepsilon - 1 - \varepsilon)].$$

Expand $\varepsilon = \sum \varepsilon_\lambda f_\lambda$ in an orthogonal eigenbasis of $L^2(\mathbb{S}^2)$, and write the right-hand side as $-\sum a_\lambda f_\lambda$. Projecting (46) gives

$$\partial_\varrho^2 (\varrho^2 \varepsilon_\lambda) + \lambda \varepsilon_\lambda + a_\lambda = 0.$$

The eigenvalues of $\Delta_{\mathbb{S}^2}$ are $-k(k+1)$, $k \geq 0$. The corresponding indicial roots are $k-1$ and $-k-2$. The $O'(\varrho^{-1})$ estimate eliminates every nondecaying indicial mode (including the root 0 for $k=1$). The only indicial term of order ϱ^{-1} is the constant spherical mode c/ϱ ; the identity $V = -12\varrho + 6\varrho^2 u_\varrho$ and $V \rightarrow k_0^2$ determine $c = -k_0^2/6$. After subtracting this term, the right-hand side of (46) is quadratic and is $O'(\varrho^{-2})$. The scale-invariant Schauder estimate on overlapping dyadic annuli, applied with a weight $-1 - \delta_0$ strictly between the two roots -1 and -2 , gives $O'(\varrho^{-1-\delta_0})$ for every $0 < \delta_0 < 1$. No other spherical mode has an indicial root in this interval. Consequently,

$$u - \varpi_z = 2 \log \varrho - \frac{k_0^2}{6\varrho} + O'(\varrho^{-1-\delta_0}), \quad V = k_0^2 + O'(\varrho^{-\delta_0}).$$

It remains to control the connection term. Choose a radially constant model connection η_0 on the same circle bundle, with $\eta_0(\mathcal{K}) = 1$, and put $a = \eta - \eta_0$, viewed as a one-form on the base. Substituting the preceding estimates into (2) gives $da = O'(\varrho^{-1-\delta_0})$. In radial gauge, $a(\partial_\varrho) = 0$, so for each s , $a(s)$ is a one-form on the sphere. We have

$$a(\varrho) = a(R) + \int_R^\varrho \iota_{\partial_s} da ds.$$

In fixed spherical coordinates the integrand has coefficients $O(s^{-\delta_0})$, with corresponding derivative estimates. Since $0 < \delta_0 < 1$ and angular covectors have norm $O(\varrho^{-1})$, integration and differentiation give $a = O'(\varrho^{-\delta_0})$ in the model metric. Before normalizing k_0 , the unit vertical model one-form is $\gamma = k_0^{-1} \eta_0$.

This proves Theorem 3.23. A homothety $h \mapsto c^2 h$, with $\lambda, g, \mathcal{K}$ recomputed canonically, sends $k_0 \mapsto c^{1/3} k_0$ and $\mathcal{K} \mapsto c^{-4/3} \mathcal{K}$; in the closed-orbit case, $T_{\mathcal{K}} \mapsto c^{4/3} T_{\mathcal{K}}$. Taking $c = k_0^{-3}$ therefore

normalizes k_0 to one. We adopt this normalization for the ALF/AF analysis below, retaining the actual period $T_{\mathcal{K}}$. In the resulting ALF/AF coordinates of Definitions 2.8 and 2.9, we choose the direction of t so that $\mathcal{K} = \partial_t$, and η is asymptotic to the model one-form γ (to dt in the AF case). The decay exponent δ_0 may be chosen arbitrarily close to, but strictly less than, one. We fix such a choice for the rest of the paper.

Finally we have

Theorem 3.25. *For a Hermitian non-Kähler ALF/AF gravitational instanton (M, h) with a decay rate of δ_0 , after the constant homothety just described, the conformal factor $\lambda^{1/3}$ has an asymptotic expansion*

$$\lambda^{1/3} = \frac{1}{\rho} + O'(\rho^{-1-\delta_0}).$$

Proof. Recall that $\varrho = \lambda^{-1/3}$. From $V = 1 + O'(\varrho^{-\delta_0})$ and the corresponding expansion of the full metric, the h -distance satisfies $\rho = \varrho + O'(\varrho^{1-\delta_0})$. Inverting this relation gives

$$\varrho^{-1} = \rho^{-1} + O'(\rho^{-1-\delta_0}),$$

which is the asserted expansion. $\square$

Remark 3.26. *The real holomorphic vector field $JK = \nabla_h \lambda^{-1/3}$ is complete. Indeed, the expansion above gives $|\nabla_h \lambda^{-1/3}|_h = 1 + o(1)$ at infinity. The field is therefore bounded on the complete manifold (M, h) , so its flow exists for all time.*

3.4. Asymptotic models that cannot occur. We next study Hermitian non-Kähler asymptotic models that cannot occur as the end of a Hermitian non-Kähler gravitational instanton. For this purpose, a *Hermitian non-Kähler Ricci-flat end* means an exterior region $E = M_0 \setminus K$ in a complete pointed manifold (M_0, h, p) , with the following properties:

- The restriction of h to E is Hermitian non-Kähler and Ricci-flat.
- For a fixed base point p and all large $R > 0$, $M_0 \setminus B_R(p)$ has only one unbounded component.
- For every sequence $r_j \rightarrow \infty$, the rescaled spaces $(M_0, r_j^{-2}h, p)$ admit a pointed Gromov–Hausdorff convergent subsequence, and the regular region $\mathcal{R} = M_\infty \setminus \{p_\infty\}$ of each limit is connected.
- For every $\delta > 0$, there is $N(\delta)$ such that $A_{R, 2R}(p)$ can be covered by at most $N(\delta)$ balls of radius δR , uniformly in R , and the balls may be chosen with a connected overlap chain inside $A_{R/2, 4R}(p)$.

These conditions hold for the gravitational instantons considered above and ensure the existence of asymptotic cones. The last two conditions are guaranteed by Gromov’s compactness and the splitting theorem in the complete situation. For notational simplicity, we continue to denote the exterior region by (M, h) .

Remark 3.27. *It is actually pointed out in [36], Proposition 7.10, that under the last three conditions above, volume comparison holds. Thus most of the arguments in Section 3.2 carry over.*

Let (M, h) be a Hermitian non-Kähler Ricci-flat end with quadratic curvature decay. Since $\lambda^{1/3}$ is positive and tends to zero at infinity, the level set $\{\lambda^{1/3} = \epsilon\}$ is compact for small ϵ . After enlarging the compact complement, the Killing field $\mathcal{K}$ preserves the end and induces an isometric $\mathbb{R}$ -action there.

We shall also use that $\mathcal{K}$ has no zeros sufficiently far out. In the collapsed situation considered below, Proposition 3.6 is an end argument: the relative superlevel-set proof confines maximal critical components to a compact set, and the local cover weight argument confines the remaining index-two points. When $s_g < 0$, the same proof uses sublevel sets and reverses the gradient-flow

orientation; the weight argument is unchanged. Thus, after enlarging the compact boundary, all reductions below take place on a critical-point-free exterior region.

In this subsection we prove

Theorem 3.28. *A Type II Hermitian non-Kähler Ricci-flat end (M, h) with quadratic curvature decay is asymptotic, at a polynomial rate, to an ALE, ALF-A, AF, generalized special Kasner, or ALH* model, or to a finite quotient of one of these models.*

Fix $q_\infty \in \mathcal{R}$ in an asymptotic cone and choose $q_j \rightarrow q_\infty$. Normalize λ at q_j and denote the result by λ_j .

The scalar curvature s_g is nowhere zero. On a general end, however, it may have either sign. For $f_j = \lambda_j^{1/3}$ both conformal scalar-curvature equations take the linear form

$$\Delta_{h_j} f_j \pm \frac{1}{6} r_j^2 \lambda f_j = 0.$$

The coefficient is uniformly bounded on every fixed rescaled annulus, so the proof of Proposition 3.1 applies with either sign. It gives the Harnack inequality and, after passing to a subsequence, smooth convergence to $\lambda_\infty^{1/3}$ on local covers. Recall that $\varrho := \pm \lambda^{-1/3}$, with the sign chosen so that ϱ has the same sign as s_g .

As before, $\mathcal{K}$ either generates an S^1 -action or has nonclosed $\mathbb{R}$ -orbits. In the latter case, its closure on the compact level set $\{\lambda^{1/3} = \epsilon\}$ is a torus T^k . If $k = 3$, the Toda equation reduces to $(e^u)_{\varrho\varrho} = 0$, so $e^u = a\varrho + b$. For $a < 0$ this gives a generalized special Kasner model, whereas $a = 0$ gives an ALH* model. The connection equation gives $d\eta|_{T^3} = -12b dx \wedge dy$. Since the T^3 -orbit is abelian and all the data are invariant, this curvature vanishes; hence $b = 0$. Thus only the zero-parameter special Kasner model occurs in the T^3 case. We may therefore assume below that the closure is T^2 .

We first choose a blow-down sequence for which $\lambda_\infty^{1/3}$ is nonconstant. Let $\gamma : [0, \infty) \rightarrow M_0$ be a minimizing ray from p and consider the dyadic scales $r_k = 2^k$. If every normalized limit along these scales were constant, then comparison of the points $\gamma(2^{k-1})$ and $\gamma(2^k)$ in the $2^{-2k}h$ rescaling would give

$$\frac{\lambda(\gamma(2^k))}{\lambda(\gamma(2^{k-1}))} \rightarrow 1.$$

Consequently,

$$\frac{1}{k} \log \lambda(\gamma(2^k)) \rightarrow 0.$$

On the other hand, $\lambda = O(r^{-2})$ gives

$$\frac{1}{k} \log \lambda(\gamma(2^k)) \leq -2 \log 2 + \frac{C_0}{k},$$

a contradiction. We henceforth fix such a sequence. Both $\lambda_\infty^{1/3}$ and $\varrho_\infty = \pm \lambda_\infty^{-1/3}$ are nonconstant on $\mathcal{R}$.

We say that a point $x \in \mathcal{R}$ is a critical point for $\lambda_\infty^{1/3}$, if in a local cover $\tilde{B}_\infty$, there is a point $\tilde{x}$ that is a critical point of $\lambda_\infty^{1/3}$ and x is the image of $\tilde{x}$. As in the proof of Lemma 3.10, the function $\lambda_\infty^{1/3}$ is Morse–Bott on each local cover $\tilde{B}_\infty$. Its critical components can only be curves $\tilde{C}$, and such curves can occur only when $r_j^2 \lambda(q_j) \rightarrow c > 0$. We write C for a component of the critical set in $\mathcal{R}$.

Lemma 3.29. *We have*

$$\liminf_{x \rightarrow p_\infty} \lambda_\infty(x) > 0, \quad \limsup_{x \rightarrow \infty} \lambda_\infty(x) < \infty.$$

Proof. Suppose the first assertion fails. We can then find $o_k \rightarrow p_\infty$ with $\lambda_\infty(o_k) < 1/k$. Choose points $o_{j,k} \in M_0$ converging to o_k as $j \rightarrow \infty$. For $j > j_k$, with j_k depending on k , one has

$$\lambda(o_{j,k})/\lambda(q_j) < \frac{2}{k}.$$

Starting at $o_{j,k}$, follow the negative gradient flow of $\lambda^{1/3}$. There are no critical points on the end, and λ tends to zero at infinity. The trajectory therefore reaches a point $o'_{j,k}$ in the sphere of radius r_j satisfying

$$\lambda(o'_{j,k})/\lambda(q_j) < \frac{2}{k}.$$

A diagonal subsequence converges at radius one to a point $o'_\infty \in \mathcal{R}$ with $\lambda_\infty(o'_\infty) = 0$, contradicting the smooth positive limit on local covers.

For the second assertion, suppose that there are points $z_k \rightarrow \infty$ in $\mathcal{R}$ with $\lambda_\infty(z_k) > k$. Choose approximating points $z_{j,k}$ in the original end. Follow the positive gradient flow of $\lambda^{1/3}$ from $z_{j,k}$. Since it contains no critical point on the exterior region, it reaches a point $z'_{j,k}$ with radius r_j ,

$$\lambda(z'_{j,k})/\lambda(q_j) > \frac{k}{2}$$

for all sufficiently large j . A diagonal subsequence would give a radius one point of $\mathcal{R}$ at which the smooth limit λ_∞ is infinite, a contradiction. $\square$

Lemma 3.30. *There are constants*

$$0 \leq \alpha < \beta \leq \infty$$

such that

$$\lim_{x \rightarrow \infty} \lambda_\infty(x) = \alpha, \quad \lim_{x \rightarrow p_\infty} \lambda_\infty(x) = \beta.$$

Proof. Put $f_j = \lambda_j^{1/3}$ and $f = \lambda_\infty^{1/3}$. We use the same Harnack argument as in Proposition 3.3, but apply it also to the distance from an endpoint of the range. The annular covering assumption allows one to chain the local estimates over $A_{R,2R}$ with a constant independent of R .

Let

$$A = \liminf_{x \rightarrow \infty} f(x) < \infty.$$

If $A = 0$, choose radii $R_m \rightarrow \infty$ and points $x_m \in A_{R_m,2R_m}$ with $f(x_m) \rightarrow 0$. The ordinary annular Harnack inequality for the positive function f gives $\sup_{A_{R_m,2R_m}} f \rightarrow 0$. Starting at x_m and following the decreasing gradient flow supplies a point with value at most $f(x_m)$ on every more distant annulus; the same Harnack inequality controls that entire annulus. Hence $f \rightarrow 0$ at infinity.

Suppose $A > 0$. After passing to a subsequence, write $c_j = r_j^2 \lambda(q_j) \rightarrow c \geq 0$. Here $c = 0$ for a simple reason. Given $R > 1$, the definition of A gives $f_j \geq A/2$ on the part of the end converging to $A_{R,2R}$, for all sufficiently large j . Thus

$$(A/2)^3 \lambda(q_j) \leq \lambda \leq \frac{C}{R^2 r_j^2}$$

there, where the last inequality follows from quadratic curvature decay. Hence $c_j \leq C_A R^{-2}$. Letting first $j \rightarrow \infty$ and then $R \rightarrow \infty$ proves

$$(47) \quad r_j^2 \lambda(q_j) \longrightarrow 0 \quad \text{whenever } A > 0.$$

Consequently the limiting function f is harmonic on every local cover. The strong minimum principle gives $f > A$ on $\mathcal{R}$: equality at one point would make the nonconstant harmonic function f constant. Applying Harnack to the positive function $f - A$ gives the shifted inequality

$$(48) \quad \sup_{A_{R,2R}} (f - A) \leq C \inf_{A_{R,2R}} (f - A),$$

where C is independent of R . Choose $R_m \rightarrow \infty$ and $x_m \in A_{R_m, 2R_m}$ with $f(x_m) \rightarrow A$. Following the decreasing gradient flow from x_m supplies a point on every more distant annulus where $f - A \leq f(x_m) - A$. Equation (48) therefore proves $f \rightarrow A$.

Now set

$$B = \limsup_{x \rightarrow p_\infty} f(x) \in (0, \infty].$$

If $B = \infty$, apply the ordinary Harnack inequality to f and follow the increasing gradient flow through all smaller annuli; this gives $f \rightarrow \infty$ at p_∞ . If $B < \infty$, fix $\varepsilon > 0$. Near p_∞ the function $B + \varepsilon - f$ is positive. On $A_{R/2, 4R}$ its Laplacian is uniformly bounded, and after rescaling this annulus to unit size the bound is $O(R^2)$. Harnack gives, for fixed $\varepsilon > 0$,

$$(49) \quad \sup_{A_{R, 2R}} (B + \varepsilon - f) \leq C \left(\inf_{A_{R, 2R}} (B + \varepsilon - f) + R^2 \right).$$

Choose $R_m \rightarrow 0$ and $x_m \in A_{R_m, 2R_m}$ with $f(x_m) \rightarrow B$. Following the increasing gradient flow from x_m supplies a point on every smaller annulus where $B - f \leq B - f(x_m)$. Applying (49), and then letting $m \rightarrow \infty$ and $\varepsilon \rightarrow 0$, proves $f \rightarrow B$ at p_∞ . We shall use below the associated interior gradient estimate for the shifted function; it is obtained from the same local argument.

The flow gives $A \leq B$. Equality would be positive and, by (47), f would be a nonconstant harmonic function tending to the same value at both ends. It would attain an interior extremum, contrary to the strong maximum principle. Hence $A < B$. Taking $\alpha = A^3$ and $\beta = B^3$ completes the proof. $\square$

As a consequence of Lemma 3.30, every value σ with $\alpha^{1/3} < \sigma < \beta^{1/3}$ is attained away from both p_∞ and infinity. Sard's theorem allows us to choose such a σ to be a regular value of $\lambda_\infty^{1/3}$.

3.4.1. $\mathcal{K}$ induces an S^1 -action. We first treat the S^1 -action case. Equations (23)–(24) imply $\chi_{\text{orb}}(\Sigma) \geq 0$. We work with the nonconstant blow-down sequence fixed above.

Assume first that $\chi(\Sigma)_{\text{orb}} > 0$. We write the argument for $s_g > 0$, so that $\varrho_\infty = \lambda_\infty^{-1/3} > 0$; the negative sign case is identical after reversing the sign of ϱ .

Lemma 3.31. *In the $\chi(\Sigma)_{\text{orb}} > 0$ case, $r_j^2 \lambda(q_j) \rightarrow 0$.*

Proof. If $A = \alpha^{1/3} > 0$, this is already (47). Suppose $A = 0$ and, to the contrary, that $r_j^2 \lambda(q_j) \rightarrow c > 0$. Then the local cover limits are Type II. Since $\chi(\Sigma) > 0$, the reduced surfaces cannot collapse everywhere. Starting from a noncollapsed reduced ball, repeat the volume comparison in the claim inside Lemma 3.12. The Type II scaling gives the same lower bound for the volume of a fixed rescaled annulus, while the global identity of Lemma 3.8 gives the strictly smaller upper order. That argument uses only the scale-independent annular Harnack bound and the two integral identities, all of which hold for the present end. The contradiction rules out $c > 0$ and proves the lemma. $\square$

Hence $\lambda_\infty^{1/3}$ has no critical point in $\mathcal{R}$. Choose $\alpha^{1/3} < \sigma < \beta^{1/3}$ and reduce (M_0, h_j, p) along

$$L_j = \{\lambda_j^{1/3} = \sigma\}.$$

Write $g_{\Sigma, j}$ for the reduced Kähler metric and $h_{\Sigma, j} = \varrho_j^2 V_j^{-1} g_{\Sigma, j}$. The limiting level remains in a compact subset of $\mathcal{R}$, so these reduced metrics have uniformly bounded curvature. Because $\chi(\Sigma) > 0$, they are noncollapsed and converge to a round orbifold metric of curvature σ^2 . Accordingly, the asymptotic cone is three-dimensional, with metric

$$h_\infty = V_\infty (d\varrho_\infty^2 + \varrho_\infty^2 g_{\mathbb{S}^2/\Gamma}).$$

Here ϱ_∞ takes values in $(\beta^{-1/3}, \alpha^{-1/3})$, and

$$h_b = d\varrho_\infty^2 + \varrho_\infty^2 g_{\mathbb{S}^2/\Gamma}$$

is the Euclidean background metric on $\mathcal{R}$. The function V_∞ is harmonic with respect to h_b .

On the level L_j , the rescaled identity (24) becomes

$$\int_{\Sigma} V_j d\text{vol}_{h_{\Sigma,j}} = \frac{\varrho(q_j)^2}{r_j^2} \varrho(q_j)^{-2} (-6a\varrho^2 - 12b\varrho).$$

Smooth convergence of V_j and $(\Sigma, h_{\Sigma,j})$ gives two possibilities. In the first,

$$\frac{\varrho(q_j)^2}{r_j^2} \longrightarrow a_1, \quad \int_{\mathbb{S}^2/\Gamma} V_\infty d\text{vol}_{\mathbb{S}^2/\Gamma} = -6a_1a.$$

In the second, $a = 0$ and

$$\frac{\varrho(q_j)}{r_j^2} \longrightarrow a_2, \quad \int_{\mathbb{S}^2/\Gamma} V_\infty d\text{vol}_{\mathbb{S}^2/\Gamma} = -12a_2b\sigma.$$

In the latter case, Lemma 3.8 gives maximal volume growth, and the end is ALE, which is therefore excluded. It remains to consider the first case.

Lemma 3.32. *The range of ϱ_∞ is $(0, \infty)$.*

Proof. Recall that the range is initially known to be $(\beta^{-1/3}, \alpha^{-1/3})$. If $\alpha^{-1/3} < \infty$, then the rescaled integral identity above and the finite h_b -volume of this annular region give

$$\int_{\mathcal{R}} V_\infty d\text{vol}_{h_b} < \infty.$$

Cauchy–Schwarz therefore yields

$$\int_{\mathcal{R}} \sqrt{V_\infty} d\text{vol}_{h_b} < \infty.$$

Fubini’s theorem gives $\omega_0 \in \mathbb{S}^2/\Gamma$ such that

$$\int_{\beta^{-1/3}}^{\alpha^{-1/3}} \sqrt{V_\infty(\varrho_\infty, \omega_0)} d\varrho_\infty < C_0.$$

The corresponding radial curve has finite length, contradicting the completeness of (M_∞, h_∞) . Hence $\alpha^{-1/3} = \infty$.

We denote the distance function to p_∞ by d_∞ . Suppose $\beta^{-1/3} > 0$ and put $f = \lambda_\infty^{1/3}$ and $B = \beta^{1/3}$. Lemma 3.31 shows that f is harmonic on every local cover. Moreover, $B - f$ is positive and tends to zero at p_∞ by Lemma 3.30. Applying the Harnack inequality and the Cheng–Yau gradient estimate to $B - f$ on $A_{s/2, 4s}$ gives

$$(50) \quad \sup_{A_{s/2, 4s}} (B - f) \leq C \inf_{A_{s/2, 4s}} (B - f), \quad \sup_{A_{s/2, 4s}} |\nabla f|_{h_\infty} \leq \frac{C}{s} \inf_{A_{s/2, 4s}} (B - f).$$

Near the positive endpoint, the change of variables $f \mapsto \varrho_\infty = f^{-1}$ is uniformly smooth. Thus the same estimates hold, up to a fixed multiplicative constant, for $\varrho_\infty - \beta^{-1/3}$. Set

$$a_s = \inf_{A_{s/2, 4s}} (\varrho_\infty - \beta^{-1/3}).$$

The endpoint statement in Lemma 3.30 gives $a_s \rightarrow 0$, while (50) gives

$$\sup_{A_{s/2, 4s}} (\varrho_\infty - \beta^{-1/3}) \leq Ca_s, \quad \sup_{A_{s/2, 4s}} |\nabla \varrho_\infty|_{h_\infty} \leq C \frac{a_s}{s}.$$

Since $V_\infty^{-1/2} = |\nabla \varrho_\infty|_{h_\infty}$, it follows that $\sqrt{V_\infty} \geq cs/a_s$ on this annulus.

Now we can cover $A_{s,2s}$ with at most N balls $B_\infty(x_k)$ with radius δs , where N, δ are independent of s . If x, y belong to one such ball, then they can be connected by a path with h_∞ -length less than $2\delta s$. However,

$$h_\infty = V_\infty h_b > c_0^2 \frac{s^2}{a_s^2} h_b.$$

So under the Euclidean background metric h_b , $|x - y|_{h_b} < C_0 \delta a_s$ for some $C_0 > 0$. Consequently, if $\beta^{-1/3} > 0$, under the Euclidean projection map $\Pi : x \rightarrow x/|x| \in \mathbb{S}^2/\Gamma$, the image of each $B_\infty(x_k)$ has spherical diameter at most

$$\text{diam}_{\mathbb{S}^2/\Gamma} \Pi(B_\infty(x_k)) < C_0 \frac{\delta a_s}{\beta^{-1/3}}.$$

However, each Euclidean ray starting at $\varrho_\infty = \beta^{-1/3}$ must cross $A_{s,2s}$, so $\Pi(A_{s,2s}) = \mathbb{S}^2/\Gamma$. Thus $\mathbb{S}^2/\Gamma$ is covered by at most N sets with diameter converging to zero. This is impossible since their total spherical area is bounded by

$$C'_0 N a_s^2 \rightarrow 0,$$

which is a contradiction. $\square$

Since the range of ϱ_∞ is $(0, \infty)$, V_∞ is a positive harmonic function on the punctured flat orbifold $\mathbb{R}^3/\Gamma$. Lift it to $\mathbb{R}^3 \setminus \{0\}$. Bôcher's theorem gives $V_\infty = c_0 + c_1/\varrho_\infty$. In the present $a < 0$ case, the rescaled integral identity (24) makes the spherical average independent of ϱ_∞ , and hence forces $c_1 = 0$. Thus V_∞ is constant, every local cover $(\tilde{B}_\infty, \tilde{h}_\infty)$ is flat, and the asymptotic cone is $\mathbb{R}^3/\Gamma$. Thus, when $\chi_{\text{orb}}(\Sigma) > 0$, an asymptotic cone is either the flat orbifold $\mathbb{R}^3/\Gamma$, with $\lambda_\infty^{-1/3}$ proportional to the distance from the vertex, or the limit rescaled conformal factor is constant.

There is at least one asymptotic cone that is a flat $\mathbb{R}^3/\Gamma$. From this we have

Lemma 3.33. *All asymptotic cones are the same flat $\mathbb{R}^3/\Gamma$.*

Proof. Cover the end by the dyadic annuli $A_j = A_{2^{j-1}, 2^{j+1}}$ and set $f = -\frac{1}{3} \log \lambda$. Normalize $f_j = f + C_j$ by $\inf_{A_j} f_j = 0$, and rescale the metric by 2^{-2j} . The preceding asymptotic cone analysis implies the following uniform dichotomy: there is a sequence $\epsilon_j \rightarrow 0$ such that, on A_j , either f_j is ϵ_j -close to a constant or it is ϵ_j -close to the radial function $\log(2^{1-j}r)$.

The two alternatives cannot occur on consecutive annuli once j is large. Indeed, f_j and f_{j+1} differ only by an additive constant on $A_j \cap A_{j+1}$, so their oscillations on any fixed-ratio subannulus of the overlap agree. In the constant alternative this oscillation tends to zero, whereas in the radial alternative it tends to a fixed positive number. Hence the alternative is eventually independent of j . Since a nonconstant asymptotic cone occurs along at least one sequence of scales, the eventual alternative must be the radial one. It follows that every sequence of scales has the same flat limit $\mathbb{R}^3/\Gamma$. $\square$

Once every asymptotic cone is $\mathbb{R}^3/\Gamma$, the expansion argument of Section 3.3 applies and shows that (M, h) is ALF-A, rational AF, or a finite quotient thereof.

It remains to consider the $\chi_{\text{orb}}(\Sigma) = 0$ case. Equations (23)–(24) imply that $s_g = -\lambda^{1/3} < 0$ and $\varrho = -\lambda^{-1/3} < 0$.

As before, consider a blow-down for which λ_∞ is nonconstant. Choose a regular value σ with $\alpha^{1/3} < \sigma < \beta^{1/3}$, and reduce at

$$L_j = \{\lambda_j^{1/3} = \sigma\}.$$

Denote the reduced Kähler metric by $g_{\Sigma,j}$ and set $h_{\Sigma,j} = \varrho_j^2 V_j^{-1} g_{\Sigma,j}$. Regularity of the limiting level gives a uniform curvature bound for $(\Sigma, h_{\Sigma,j})$.

After passing to a finite cover, Σ is a torus and the link L is an S^1 -bundle over it. The holomorphic gradient flow identifies all regular reductions and preserves their conformal class. Write

$$T^2 = \mathbb{R}^2/\Lambda, \quad g_0 = dx^2 + dy^2.$$

We normalize this flat representative of the conformal class to have diameter one. The functions x, y themselves are coordinates only on the universal cover $\mathbb{R}^2$, but dx, dy and every function used below are Λ -periodic. Lemma 3.12 is replaced by the following statement.

Lemma 3.34. *In the $\chi(\Sigma) = 0$ case, $\text{diam}(\Sigma, h_{\Sigma,j}) \rightarrow 0$.*

Proof. Let $\varrho(q_j)$ be the unnormalized level used in the blow-down. On the fixed normalized level $\lambda_j^{1/3} = \sigma$, the value of the original coordinate ϱ is a fixed nonzero constant multiple of $\varrho(q_j)$. Since $\chi(\Sigma) = 0$, formula (23) therefore gives

$$\text{Area}(\Sigma, h_{\Sigma,j}) = \lambda(q_j)^{2/3} \int_{\Sigma} e^u dx dy = O(|\varrho(q_j)|^{-1}) \rightarrow 0$$

when $a \neq 0$, and the decay is $O(|\varrho(q_j)|^{-2})$ when $a = 0$. Together with the uniform curvature bound, this rules out a noncollapsed ball and makes the injectivity radius tend uniformly to zero.

The reduced surfaces are all biholomorphic under the holomorphic gradient flow, so their conformal class is fixed. Cheeger–Fukaya–Gromov theory supplies a positive-rank $\mathcal{F}$ -structure with small orbits (after a bilipschitz-small invariant perturbation). Rank-one collapse on a torus makes the extremal length of a primitive homology class tend to zero and sends its conformal class to the boundary of the moduli space. The fixed conformal class excludes this. The collapse therefore has rank two, which means that the whole torus has diameter tending to zero. Bilipschitz closeness transfers the conclusion back to $h_{\Sigma,j}$. $\square$

Put

$$C_j := \text{diam}(\Sigma, h_{\Sigma,j})^{-2}.$$

Thus $(\Sigma, C_j h_{\Sigma,j})$ has diameter one. Its curvature tends to zero, and the fixed conformal class and the preceding derivative bounds give smooth convergence to the diameter-one flat metric g_0 . Next we replace Lemmas 3.13–3.14 by the following lemma.

Lemma 3.35. *When $\chi(\Sigma) = 0$ and $\mathcal{K}$ induces an S^1 -action, local covers of the asymptotic cone are Type II. The asymptotic cone is 1-dimensional with the metric $h_{\infty} = V_{\infty} d\varrho_{\infty}^2$.*

Proof. Suppose one local cover limit were Type I. Then

$$\lambda^{-1}(q_j) r_j^{-2} \rightarrow \infty,$$

so every local cover limit at this scale would be Type I. As in Lemma 3.14, the conformal scalar-curvature equation in the LeBrun ansatz would give $v_{\infty} = 2 \log \xi_{\infty} + \varpi(x, y)$. Substitution into

$$\partial_{\xi_{\infty}}^2(e^{v_{\infty}}) + \partial_x^2 v_{\infty} + \partial_y^2 v_{\infty} = 0$$

shows, on every local cover, that the curvature of $h_{\Sigma,j}$ tends to the positive constant σ^2 uniformly. This violates Gauss–Bonnet on the torus. Hence every local cover limit is Type II, and $\lambda^{-1}(q_j) r_j^{-2}$ tends to a positive constant.

For the second statement, it remains only to show that the $\mathcal{K}$ -direction collapses. This is equivalent to $r_j^{-4} \lambda^{-2/3}(q_j) \rightarrow 0$, which follows because $\lambda^{-1}(q_j) r_j^{-2}$ tends to a positive constant. $\square$

Cheeger–Fukaya–Gromov theory now shows that the entire link L , an S^1 -bundle over T^2 , collapses. The connected three-dimensional local collapse group is nilpotent, and hence is either the abelian group $\mathbb{R}^3$ or the Heisenberg group $\mathcal{H}$. The topology of the bundle alone does not determine which group occurs in the blow-down: under an anisotropic rescaling a Heisenberg bracket may contract to zero. The limiting connection below distinguishes the two possibilities.

We now justify the convergence of the rescaled coordinates. On the universal cover of the torus, translate x, y so that the lifted base point is the origin. Write

$$\ell_j := \lambda(q_j)^{1/3}, \quad \varrho_j = \ell_j \varrho,$$

and set

$$X_j = C_j^{-1/2} x, \quad Y_j = C_j^{-1/2} y, \quad Z_j = X_j + iY_j.$$

On every compact subset of the noncollapsed local cover, the smooth control of the reduced metrics and the positive upper and lower bounds for V_j give

$$|dX_j|_{h_j}^2 = |dY_j|_{h_j}^2 = \frac{1}{V_j C_j \ell_j^2 e^u} \leq C, \quad \langle dX_j, dY_j \rangle_{h_j} = 0.$$

At the lifted base point these norms are bounded below, because $C_j h_{\Sigma, j}$ converges to the diameter-one flat metric g_0 and V_j converges to a positive function. Thus the two rescaled coordinate functions are uniformly Lipschitz on compact subsets and are normalized to vanish at the base point. Integration along bounded length paths also gives their uniform local C^0 bounds.

The ansatz also shows directly that X_j, Y_j are harmonic for the full four-dimensional metric, including the connection terms. Hence interior elliptic regularity and the pointed smooth convergence of $\tilde{h}_j$ give, after passing to a subsequence,

$$(51) \quad X_j \longrightarrow x_\infty, \quad Y_j \longrightarrow y_\infty \quad \text{in } C_{\text{loc}}^\infty.$$

The base-point lower bound and the orthogonality of dX_j, dY_j show that the limiting differentials are linearly independent.

In these coordinates define

$$(52) \quad \hat{u}_j(\varrho_j, X_j, Y_j) = u(\varrho, x, y) + \log(C_j \ell_j^2), \quad (x, y) = C_j^{1/2}(X_j, Y_j).$$

Then

$$\ell_j^2 e^u (dx^2 + dy^2) = e^{\hat{u}_j} (dX_j^2 + dY_j^2).$$

Equivalently,

$$(53) \quad e^{\hat{u}_j} = \frac{1}{V_j |dX_j|_{h_j}^2} = \frac{1}{V_j |dY_j|_{h_j}^2}.$$

The smooth metric convergence and (51), therefore give $\hat{u}_j \rightarrow u_\infty$ in C_{loc}^∞ , after shrinking the coordinate neighborhood if necessary. In the same variables the Toda equation is

$$(e^{\hat{u}_j})_{\varrho_j \varrho_j} + \partial_{X_j}^2 \hat{u}_j + \partial_{Y_j}^2 \hat{u}_j = 0,$$

so it passes directly to the limit. The limit metric on $\tilde{B}_\infty$ takes the form

$$(54) \quad \tilde{h}_\infty = V_\infty (d\varrho_\infty^2 + e^{u_\infty} (dx_\infty^2 + dy_\infty^2)) + V_\infty^{-1} \eta_\infty^2.$$

The group G_∞ is either $\mathbb{R}^3$ or $\mathcal{H}$. The $SU(\infty)$ Toda equation in the limit now becomes

$$(55) \quad \partial_{\varrho_\infty}^2 (e^{u_\infty}) + \partial_{x_\infty}^2 u_\infty + \partial_{y_\infty}^2 u_\infty = 0.$$

The two translations in the limiting $\mathbb{R}^2$ -orbit are limits of the deck translations of the torus cover. They preserve the limiting metric, ϱ_∞ , and V_∞ . Equation (54) then shows that they also preserve u_∞ . Hence we have the following.

Lemma 3.36. *V_∞ and u_∞ are invariant under G_∞ . They are functions of ϱ_∞ only.*

Therefore (55) reduces to $\partial_{\varrho_\infty}^2(e^{u_\infty}) = 0$, which implies there are constants A, B such that $e^{u_\infty} = A\varrho_\infty + B$. Set $\lim_{j \rightarrow \infty} \lambda^{-1}(q_j)r_j^{-2} = c$. Since $\varrho_j = \lambda^{1/3}(q_j)\varrho$, direct rescaling of (8) gives

$$V_j = \lambda^{-1}(q_j)r_j^{-2}(-12\varrho_j + 6\varrho_j^2\partial_{\varrho_j}\widehat{u}_j).$$

The additive constant and the transverse coordinate rescaling in (52) do not affect the ϱ_j -derivative. The smooth convergence just proved therefore gives

$$V_\infty = c(-12\varrho_\infty + 6\varrho_\infty^2\partial_{\varrho_\infty}u_\infty) = -c\frac{6A\varrho_\infty^2 + 12B\varrho_\infty}{A\varrho_\infty + B}.$$

Lemma 3.37. *Either $A < 0$ and $B = 0$, or $A = 0$ and $B > 0$.*

Proof. Because $\varrho_\infty < 0$ and $e^{u_\infty} = A\varrho_\infty + B > 0$, the signs are fixed by positivity. If $a < 0$ in (23)–(24), the linear term dominates and $A < 0$, $B = 0$, giving $V_\infty = -6c\varrho_\infty$. If $a = 0$, then e^{u_∞} is a positive constant, so $A = 0$, $B > 0$, and $V_\infty = -12c\varrho_\infty$. $\square$

One can determine η_∞ via (2).

- When $a < 0$, we have $A < 0$, $B = 0$, and (2) gives $d\eta_\infty = 0$. In suitable coordinates,

$$\tilde{h}_\infty = -6\varrho_\infty(d\varrho_\infty^2 - \varrho_\infty(dx_\infty^2 + dy_\infty^2)) - \frac{1}{6\varrho_\infty}dt_\infty^2.$$

Hence we recover the zero-parameter special Kasner metric (13),

- When $a = 0$, we have $A = 0$, $B > 0$, and (2) gives

$$d\eta_\infty = -12dx_\infty \wedge dy_\infty.$$

In suitable coordinates,

$$\tilde{h}_\infty = -12\varrho_\infty(d\varrho_\infty^2 + dx_\infty^2 + dy_\infty^2) - \frac{1}{12\varrho_\infty}(dt_\infty - 12x_\infty dy_\infty)^2.$$

Hence we recover the ALH* metric (14).

Completeness now shows that the asymptotic cone is $M_\infty = (-\infty, 0]$ with metric $h_\infty = V_\infty d\varrho_\infty^2$. Thus, when $\chi(\Sigma) = 0$, an asymptotic cone is either isometric to a ray $[0, \infty)$ or the limit rescaled conformal factor is constant. In the former case, if r is the unit-speed coordinate on the ray, then $\lambda_\infty^{-1/3}$ is proportional to $r^{2/3}$.

At least one asymptotic cone is therefore a ray. The argument used in the $\chi(\Sigma) > 0$ case then shows that every limit rescaled conformal factor is nonconstant and every asymptotic cone is isometric to $[0, \infty)$.

A particular consequence of Lemma 3.37 is the uniqueness of the asymptotic cone. In summary we have

Lemma 3.38. *When $\chi(\Sigma) = 0$, the asymptotic cone is isometrically a ray $[0, \infty)$. Local covers of the asymptotic cone are either zero-parameter special Kasner models or ALH* models, determined by the value of a in (23)–(24).*

We next upgrade the asymptotic cone description to a full asymptotic expansion.

Lemma 3.39. *$-\frac{a\varrho+b}{6a\varrho^2+12b\varrho} \cdot V \rightarrow 1$ as we approach infinity.*

Proof. The ratio of the integral identities is

$$\frac{\int_\Sigma V e^u dx dy}{\int_\Sigma e^u dx dy} = -\frac{6a\varrho^2 + 12b\varrho}{a\varrho + b}.$$

Every recentered local cover limit is the corresponding homogeneous zero-parameter special Kasner or ALH* model, on which V is constant along the reduced torus. If the quotient of V by the

displayed average failed to converge uniformly to one, a recentering at a violating sequence would produce a limit with a nonconstant normalized V , a contradiction. The same recentering argument and interior elliptic estimates give convergence after any number of scale-invariant derivatives. $\square$

Since $V = -12\varrho + 6\varrho^2 u_\varrho$, we have $u_\varrho = \frac{1}{\varrho} + o(\varrho^{-1})$ when $a < 0$, and $u_\varrho = o(\varrho^{-1})$ when $a = 0$.

We record the derivative input used in both expansions. Put $R = -\varrho$ and let $A_0 = \text{Area}(T^2, g_0)$. Define

$$c(\varrho) = \begin{cases} \log((a\varrho + b)/A_0), & a < 0, \\ \log(b/A_0), & a = 0, \end{cases} \quad \phi = u - c(\varrho).$$

Then $\int_{T^2} e^\phi d \text{vol}_{g_0} = A_0$. Consider any scale-down sequence based on levels $\varrho = -R_j$, $R_j \rightarrow \infty$, and normalize $\varrho_j = R_j^{-1}\varrho$; thus $\ell_j = R_j^{-1}$. Because the coordinates X_j, Y_j and the functions $\hat{u}_j$ converge smoothly, ϕ , regarded as a function of (ϱ_j, X_j, Y_j) , has a smooth limit. The exact normalization identifies this limit with zero. Consequently, for every $k, s \geq 0$,

$$(56) \quad \partial_{\varrho_j}^k D_{Z_j}^s \phi \longrightarrow 0 \quad \text{uniformly on the rescaled fundamental domain,}$$

where $D_{Z_j}^s$ denotes any real derivative of total order s in (X_j, Y_j) .

Scaling back gives

$$\partial_\varrho^k D_z^s \phi = R_j^{-k} C_j^{-s/2} \partial_{\varrho_j}^k D_{Z_j}^s \phi = o(R_j^{-k} C_j^{-s/2}) = o(R_j^{-k}).$$

Here $C_j \rightarrow \infty$, and periodicity makes the estimate on the shrinking rescaled fundamental domain uniform on the original torus. Since the argument applies to every sequence $R_j \rightarrow \infty$, it gives, for all $k, s \geq 0$,

$$(57) \quad \|\partial_\varrho^k \phi(\varrho, \cdot)\|_{C^s(T^2, g_0)} = o(R^{-k}).$$

We first consider the case $a < 0$.

Proposition 3.40. *When $a < 0$, after passing the end to a finite cover, it is asymptotic, at a rate faster than every polynomial, to a generalized special Kasner model $h_{K, \beta_K, \Gamma}$, where $\beta_K = -b/a$ and ∂_t induces an S^1 -action.*

For generalized special Kasner metrics the distance function is comparable to $|\varrho|^{3/2}$.

Proof. Let $\bar{\phi}$ denote the g_0 -average of ϕ and put $\phi^\perp = \phi - \bar{\phi}$. Since c is constant on each torus, the Toda equation gives

$$(58) \quad \Delta_{g_0} \phi^\perp = -(e^u)_{\varrho\varrho}.$$

We claim that, for every $N, k, s \in \mathbb{N}_0$,

$$(P_N) \quad \|\partial_\varrho^k \phi\|_{C^s(T^2, g_0)} = o(R^{-N-k}).$$

The case $N = 0$ is (57). Suppose P_N holds. Writing $L(\varrho) = e^{c(\varrho)} = (a\varrho + b)/A_0$, we have

$$(e^u)_{\varrho\varrho} = L(\varrho) e^\phi (\phi_{\varrho\varrho} + (\phi_\varrho)^2) + 2L' e^\phi \phi_\varrho.$$

Here $L(\varrho) = O(R)$ and L' is constant. Differentiating this identity and using P_N gives

$$\|\partial_\varrho^k (e^u)_{\varrho\varrho}\|_{C^s(T^2)} = o(R^{-N-k-1}).$$

The right-hand side has zero integral because $\int_{T^2} e^u = a\varrho + b$ is affine. The fixed-torus Schauder estimates show from (58) that

$$\|\partial_\varrho^k \phi^\perp\|_{C^s(T^2)} = o(R^{-N-k-1}).$$

On the other hand, the exact normalization $\int_{T^2} e^\phi = A_0$ gives

$$\bar{\phi} = -\log \left(A_0^{-1} \int_{T^2} e^{\phi^\perp} d\text{vol}_{g_0} \right).$$

Since $\phi^\perp$ has zero average, Taylor expansion of the last identity, together with its ϱ -derivatives, gives $\partial_\varrho^k \bar{\phi} = o(R^{-2N-2-k})$. Thus P_{N+1} follows, and induction proves the claim.

Now set

$$\kappa = -\frac{a}{A_0} > 0, \quad \beta_K = -\frac{b}{a}, \quad X = \sqrt{\kappa}x, \quad Y = \sqrt{\kappa}y.$$

In these coordinates

$$c(\varrho) - \log \kappa = \log(-\varrho + \beta_K),$$

so $u - \log \kappa$ differs from the exact model coefficient u_{β_K} by ϕ , which decays faster than every power. Moreover,

$$V - V_{\beta_K} = 6\varrho^2 \phi_\varrho,$$

and hence, after reindexing the arbitrary power, V has the same decay to the model coefficient.

It remains to recover the connection from its curvature. In the coordinates (ϱ, X, Y) , equation (2) is

$$(59) \quad d\eta = -\varrho^2 \partial_\varrho (\varrho^{-2} V e^{u - \log \kappa}) dX \wedge dY + V_X d\varrho \wedge dY - V_Y d\varrho \wedge dX.$$

Thus $d\eta - d\theta_{\beta_K}$ decays faster than every power. Moreover, using (24) and $\text{Area}_{X,Y}(T^2) = \kappa A_0 = -a$, integration of the tangential component in (59) gives

$$\int_{T^2} d\eta = -12b = \int_{T^2} d\theta_{\beta_K}.$$

Choose the central period of the model lattice to equal T_K on the current end cover. The two circle bundles therefore have the same Euler class and may be identified. The final model lattice has not yet been fixed. Choose any preliminary compatible lattice Γ_0 , identify its model bundle with the end bundle, and denote the pulled-back standard model connection by $\theta_{\beta_K}^0$. The form $\alpha = \eta - \theta_{\beta_K}^0$ may be put in radial gauge. Thus $\alpha = \alpha_t(\varrho)$ and

$$\partial_\varrho \alpha_t = (d\alpha)_{\varrho t}, \quad d_{T^2} \alpha_t = (d\alpha)_{tt}.$$

Hence α converges to a limiting one-form h_∞ on T^2 . The second identity shows that the limit is closed; a radially independent gauge removes its exact part. The remaining form h_∞ is harmonic, which determines the final model lattice Γ . Equivalently, replace the reference connection by $\theta_{\beta_K}^0 + h_\infty$; on the universal cover a linear change of the fibre coordinate returns it to the standard form θ_{β_K} and transforms the deck group into Γ . Relative to this final model, the ansatz gives super-polynomial metric decay. This proves the proposition. $\square$

Then we consider the case $a = 0$.

Proposition 3.41. *When $a = 0$, after passing the end to a finite cover, it is asymptotic, at a rate faster than every polynomial, to an ALH^* model $h_{\text{ALH}^*-\Gamma}$.*

For $\text{ALH}^*-\Gamma$ metrics the distance function again is comparable to $|\varrho|^{3/2}$.

Proof. Again write $u = \bar{u} + u^\perp$. The initial estimate (57) now says $\|\partial_\varrho^k(u - a')\|_{C^s} = o(R^{-k})$, where $a' = \log(b/A_0)$. Consequently,

$$\|\partial_\varrho^k(e^u)_{\varrho\varrho}\|_{C^s(T^2)} = o(R^{-k-2}).$$

The Toda equation and the fixed-torus Schauder estimates on zero-average functions, using two additional tangential derivatives of the source, first give

$$\|\partial_\varrho^k u^\perp\|_{C^s(T^2)} = o(R^{-k-2}).$$

The exact identity $\int_{T^2} e^u = b$ then implies, for the same a' ,

$$\|\partial_\varrho^k(u - a')\|_{C^s(T^2)} = o(R^{-k-2}).$$

Suppose inductively that the last estimate holds with 2 replaced by N . Differentiating the Toda equation and using the product rule shows

$$\|\Delta_{g_0} \partial_\varrho^k u^\perp\|_{C^s(T^2)} = o(R^{-k-N-2}).$$

Ellipticity improves $u^\perp$ by two powers. Since $\int e^{u^\perp} = A_0 + O(\|u^\perp\|_{C^0}^2)$, the exact integral identity gives the same improvement for the average $\bar{u} - a'$. Induction in steps of two proves decay faster than every power, with arbitrary radial and tangential derivatives.

Put $\kappa_* = b/A_0$ and use the fixed coordinates $X = \sqrt{\kappa_*}x$, $Y = \sqrt{\kappa_*}y$. Then $u - \log \kappa_*$ decays faster than every power to the model value zero, and the ansatz gives the same conclusion for $V + 12\varrho$. Equation (59), with κ replaced by κ_* , shows that

$$d\eta - d(dt - 12XdY)$$

has super-polynomial decay. The model torus has (X, Y) -area $\kappa_* A_0 = b$, so the integral of its curvature is $-12b$, exactly the integral of $d\eta$. Choose the central period of the model lattice to equal T_K on the current end cover; the model connection lies on the same circle bundle. Apply the connection argument from Proposition 3.40: make the radial component of the connection difference vanish, remove the exact part of its limiting closed form, and incorporate the two limiting periods into the model lattice. The curvature equation and elliptic regularity on the fixed torus then give super-polynomial decay of the connection itself. This completes the metric expansion. $\square$

Propositions 3.40 and 3.41 complete the proof of Theorem 3.28 when $\mathcal{K}$ generates an S^1 -action.

3.4.2. $\mathcal{K}$ induces a higher rank torus action. It remains to treat the case in which the closure of the $\mathbb{R}$ -flow is a T^2 -action.

When L is a lens space or $S^2 \times S^1$, the proof of Section 3.2.2 applies with minor modifications. We again consider a blow-down for which λ_∞ is nonconstant and first treat $s_g > 0$; reversing the sign of ϱ gives the $s_g < 0$ case.

The main difference from the gravitational instanton case is that, a priori, $\mathcal{R}$ may contain critical points and the known range of ϱ_∞ is only $(\beta^{-1/3}, \alpha^{-1/3})$. Choose a regular value σ of $\lambda_\infty^{1/3}$ and a point o_∞ on that level. On a small ball centered at o_∞ , the argument of Section 3.2.2 shows that $\bar{B}_\infty$ is Type I. Repeating this on every regular component rules out isolated critical points by the collapse-orbit argument. The basin argument in Lemma 3.10 then excludes critical curves: its proof needs only connectedness of $\mathcal{R}$ and properness of every intermediate slab of $\lambda_\infty^{1/3}$, both supplied here by Lemma 3.30. Thus $\lambda_\infty^{1/3}$ has no critical points in $\mathcal{R}$. The argument of Section 3.2.2 now gives a two-dimensional asymptotic cone with metric

$$h_\infty = V_\infty(d\varrho_\infty^2 + e^{u_\infty} dx^2).$$

Moreover, $V_\infty^{-1}h_\infty$ is the flat metric on the annular region $\{\beta^{-1/3} < \varrho_\infty < \alpha^{-1/3}\}$ in the half-plane $\mathbb{H}$. The argument of Lemma 3.32 shows that the range of ϱ_∞ is $(0, \infty)$. Hence either the asymptotic cone is a flat half-plane, with $\lambda_\infty^{-1/3}$ proportional to the radial function, or the limit rescaled conformal factor is constant. Lemma 3.33 then gives uniqueness of the flat half-plane cone. After the same improvement used in Section 3.3, it follows that (M, h) is asymptotic to an ALF- A model, a finite quotient of such a model, or an AF model.

When L is T^3 , take its universal cover $\mathbb{R}^3$. Then we take the quotient of $\mathbb{R}^3$ by a different lattice in the isometric $\mathbb{R}^2$ -action induced by the cover of the T^2 -action. The lattice can be chosen such that $\mathcal{K}$ induces an S^1 -action in the new link $L' \simeq T^3$. This results in a new end (M', h') ,

which is a Hermitian non-Kähler Ricci-flat end where the extremal vector field induces an S^1 -action. Therefore it reduces to the situation discussed in Section 3.4.1. Since the link of (M', h') is T^3 , its Euler class vanishes, so Proposition 3.40 has $\beta_K = 0$. Thus (M', h') is asymptotic to the zero-parameter special Kasner model, and the original end (M, h) is asymptotic to a skewed special Kasner model, namely the $\beta = 0$ member. This completes the proof of Theorem 3.28 for the $\mathbb{R}$ -action case.

Finally, we recover the asymptotics of the original end from those obtained after the covering and lattice reductions. As in Section 3.3, we identify the relevant isometric actions with the corresponding model actions and apply the center-of-mass construction [38] to the almost-equivariant asymptotic diffeomorphism. This yields an equivariant asymptotic identification with the same decay estimates. Descending through the finite coverings and restoring the original lattices therefore gives the asserted asymptotics of the original end. This completes the proof of Theorem 3.28.

4. COMPACTIFICATION

We now return to ALF and AF Hermitian non-Kähler gravitational instantons. There are two cases.

- (♣) The field $\mathcal{K}$ generates an S^1 -action. According to its isotropy at infinity, the end is ALF-A, a finite quotient of ALF-A, or rational AF, and its cone is $\mathbb{R}^3$ or $\mathbb{R}^3/\Gamma$.
- (♠) The field $\mathcal{K}$ has nonclosed orbits and its closure is T^2 . Then (M, h) is irrational AF, already holomorphically and isometrically toric, and its cone is $\mathbb{H}$. The Biquard–Gauduchon classification will be applied directly in this case.

We henceforth focus on the closed-orbit case. The fields $\mathcal{K}$ and $J\mathcal{K}$ generate a holomorphic $\mathbb{C}^*$ -action. Sections 4.1–4.2 first treat a free circle action at infinity.

4.1. Local ansatz revisited. Assume that the S^1 -action generated by $\mathcal{K} = \partial_t$ is free on the end. We retain the normalization $k_0 = 1$ and the actual period T_K , and put $c_K = 2\pi/T_K$. The locally free case is obtained by working in orbifold charts, using the regular period; on a further end cover, T_K is the period of the lifted field. We apply the ansatz Theorem 2.5 this time. With the expansion for $\lambda^{1/3}$, the moment map ξ asymptotically is

$$(60) \quad \xi = \lambda^{1/3} = \frac{1}{\rho} + O'(\rho^{-1-\delta_0}).$$

Employing Kähler reduction, we can put the metric g on $M \setminus K$ into the form

$$(61) \quad W d\xi^2 + W^{-1}\eta^2 + g_\xi.$$

Here g_ξ is the metric on $\Sigma = \mathbb{P}^1$ arising from the Kähler reduction. The Hermitian Ricci-flat metric h is related to g via $g = \lambda^{2/3}h$, so the metric h is

$$(62) \quad h = \lambda^{-2/3}(W d\xi^2 + W^{-1}\eta^2 + g_\xi).$$

If we pick a local holomorphic coordinate function $z = x + iy$ on Σ , then the complex structure J is given by $J : d\xi \rightarrow -W^{-1}\eta$ and $J : dx \rightarrow -dy$. The metric g then further takes the form

$$g = W d\xi^2 + W^{-1}\eta^2 + W e^v(dx^2 + dy^2)$$

for a function v , as in Theorem 2.5. Let $p : M \setminus K \rightarrow \Sigma$ be the holomorphic quotient map, and normalize the end so that $0 < \xi \leq \epsilon$ and its compact boundary is $\{\xi = \epsilon\}$. The expansion of $\xi = \lambda^{1/3}$ gives the following expansions for W and v .

Lemma 4.1. *The function $W^{-1} = |\partial_t|_g^2$ has the following asymptotic expansion:*

$$(63) \quad W^{-1} = \frac{1}{\rho^2} + O'(\rho^{-2-\delta_0}).$$

Lemma 4.2. *If we pick a local coordinate function $z = x + iy$ on an open subset $U \subset \mathbb{P}^1$, then on the open set $p^{-1}(U) \subset M \setminus K$, the function v has the asymptotic expansion*

$$(64) \quad v(\rho, x, y) = \varpi(x, y) - 2 \log \rho + O'(\rho^{-\delta_0})$$

with some function ϖ that only depends on x and y . The function ϖ satisfies the equation $\varpi_{xx} + \varpi_{yy} = -2e^\varpi$.

Notice that the function ϖ depends on the choice of the coordinate function $z = x + iy$. However, $e^\varpi(dx^2 + dy^2)$ is a well-defined metric and has curvature one.

4.2. Compactification in the ALF-A case.

Theorem 4.3. *For a Hermitian non-Kähler ALF-A gravitational instanton (M, h) , the complex surface M can be naturally compactified by adding a holomorphic $\mathbb{P}^1$ at infinity to a smooth compact complex surface.*

Proof. We consider the complex structure on the open set $p^{-1}(U)$, where the S^1 -bundle is trivialized and we use the coordinate system (ξ, t, x, y) . On $p^{-1}(U)$, the connection one-form η can be written as $dt + \mathcal{X}dx + \mathcal{Y}dy + \mathcal{Z}d\xi$, and the complex structure is given by

$$J : d\xi \rightarrow -W^{-1}\eta, \quad dx \rightarrow -dy.$$

As η is ∂_t -invariant, $\mathcal{X}(\xi, x, y)$, $\mathcal{Y}(\xi, x, y)$, and $\mathcal{Z}(\xi, x, y)$ are functions of ξ, x, y only. The following two differential forms are of type $(1, 0)$, which determine the complex structure on $p^{-1}(U)$ completely

$$d\xi + iW^{-1}\eta, \quad dx + idy.$$

A straightforward calculation shows that in terms of the basis of vector fields $\partial_\xi, \partial_t, \partial_x, \partial_y$ under the system (ξ, t, x, y) , the complex structure is given by

$$(65) \quad J\partial_\xi = W\partial_t - W^{-1}\mathcal{Z}\partial_\xi + W^{-1}\mathcal{Z}^2\partial_t,$$

$$(66) \quad J\partial_x = \partial_y - W^{-1}\mathcal{X}\partial_\xi + (-\mathcal{Y} + W^{-1}\mathcal{X}\mathcal{Z})\partial_t.$$

Recall $\xi \in (0, \epsilon]$. In the coordinate system (ξ, t, x, y) , we introduce

$$(67) \quad \zeta := \int_\xi^\epsilon W d\xi,$$

$$(68) \quad \varphi := \left(-t + \int_\xi^\epsilon \mathcal{Z} d\xi \right) \mod T_K.$$

The expansion of W shows that $\zeta \in [0, \infty)$. Under the change of variables $(\xi, t, x, y) \mapsto (\zeta, \varphi, x, y)$, the coordinate fields ∂_x and ∂_y change because the integrals in (67)–(68) depend on (x, y) . For the radial and angular fields one has

$$(69) \quad \partial_\xi = -W\partial_\zeta - \mathcal{Z}\partial_\varphi, \quad \partial_t = -\partial_\varphi.$$

Using (2) to transform the remaining coordinate fields gives

$$(70) \quad J\partial_x = \partial_y + \mathcal{X}(\epsilon, x, y)\partial_\zeta + \mathcal{Y}(\epsilon, x, y)\partial_\varphi,$$

$$(71) \quad J\partial_\zeta = \partial_\varphi.$$

The coefficients are evaluated at the reference level $\xi = \epsilon$ and are independent of (ζ, φ) . This is the key point of the gauge: the real holomorphic fields ∂_φ and $\partial_\zeta = -J\partial_\varphi$ now act by translations and the complex structure is standard in these directions.

Set $w = \exp[-c_K(\zeta + i\varphi)]$. Smoothly compactify $p^{-1}(U)$ by adjoining $w = 0$ and using $(\operatorname{Re} w, \operatorname{Im} w, x, y)$ as coordinates. In these coordinates the complex structure is

$$(72) \quad \begin{aligned} J\partial_x &= \partial_y - c_K \mathcal{X}(\epsilon, x, y)(\operatorname{Re} w \partial_{\operatorname{Re} w} + \operatorname{Im} w \partial_{\operatorname{Im} w}) \\ &\quad + c_K \mathcal{Y}(\epsilon, x, y)(\operatorname{Im} w \partial_{\operatorname{Re} w} - \operatorname{Re} w \partial_{\operatorname{Im} w}), \end{aligned}$$

$$(73) \quad J\partial_{\operatorname{Re} w} = \partial_{\operatorname{Im} w}.$$

The coefficients are smooth at $w = 0$, so J extends smoothly as an almost complex structure. To see directly that the extension is integrable and to obtain holomorphic transition functions, put $\Theta = c_K(\mathcal{X}(\epsilon, x, y) + i\mathcal{Y}(\epsilon, x, y))$. Equations (72)–(73) give

$$2\bar{\partial}w = w(-i\Theta dx - \Theta dy).$$

The form on the right after division by w is the smooth $(0, 1)$ -form $-i\Theta d\bar{z}$ on the Riemann surface U . It is $\bar{\partial}$ -closed, since U has complex dimension one. The local Dolbeault lemma therefore gives, after shrinking U , a smooth function F satisfying

$$2\bar{\partial}F = -i\Theta dx - \Theta dy = -i\Theta d\bar{z}.$$

Then

$$q := we^{-F}$$

is holomorphic, vanishes simply on the added set, and together with $z = x + iy$ forms a holomorphic coordinate system. Thus the extended J is integrable and the added set is the holomorphic curve $q = 0$.

Since the quotient surface is S^2 , the added curve is biholomorphic to $\mathbb{P}^1$.

Since ζ is globally defined, $|w_\alpha| = |w_\beta|$ on overlaps. Consequently, the holomorphic ratio q_β/q_α and its reciprocal are locally bounded, because

$$\left| \frac{q_\beta}{q_\alpha} \right| = e^{\operatorname{Re}(F_\alpha - F_\beta)}.$$

By the removable singularity theorem, this ratio extends holomorphically and remains nonzero across $q_\alpha = 0$. Together with the holomorphic coordinate changes on the base, this gives biholomorphic transition maps across the added curves. The cocycle identities persist by continuity, so the local compactifications glue. $\square$

4.3. Positivity of the compactified surfaces for ALF-A gravitational instantons. We label the compactified surface as $\overline{M}$ and denote the added divisor $\mathbb{P}^1$ as D . Denote the hermitian metric induced by the Kähler metric g on the anti-canonical bundle $-K_M$ as g_{-K} . Subsequently, the metric $e^{-2\log s_g} g_{-K}$ is also a hermitian metric on the line bundle $-K_M$. This particular hermitian metric has positive curvature, as initially observed by LeBrun [45].

Proposition 4.4. *The hermitian metric $e^{-2\log s_g} g_{-K}$ on the line bundle $-K_M$ extends to a singular hermitian metric on the line bundle $-(K_{\overline{M}} + D)$ over $\overline{M}$. As a singular hermitian metric, it has the following regularity:*

- *It is smooth outside D and continuous across D : if $\tau : -(K_{\overline{M}} + D)|_V \rightarrow V \times \mathbb{C}$ is a local holomorphic trivialization near D , then $e^{-2\log s_g} g_{-K}(s, s) = |\tau(s)|^2 e^{-2\phi}$ for a continuous local weight ϕ .*
- *The Lelong number $\nu(\phi, d) := \liminf_{l \rightarrow d} \frac{\phi(l)}{\log|l-d|} = 0$ for any $d \in D$.*

Proof. We consider locally on $\overline{p^{-1}(U)}$. Equations (72) and (73) show that

$$(74) \quad \begin{aligned} Jd\text{Re}w &= -d\text{Im}w + c_K(-\mathcal{X}(\epsilon, x, y)\text{Re}w + \mathcal{Y}(\epsilon, x, y)\text{Im}w)dx \\ &\quad + c_K(-\mathcal{X}(\epsilon, x, y)\text{Im}w - \mathcal{Y}(\epsilon, x, y)\text{Re}w)dy, \end{aligned}$$

$$(75) \quad Jdx = -dy.$$

A computation, with $\Theta := c_K(\mathcal{X}(\epsilon, x, y) + i\mathcal{Y}(\epsilon, x, y))$, gives

$$2\bar{\partial}w = w(-i\Theta dx - \Theta dy).$$

The form $-i\Theta dx - \Theta dy$ has type $(0, 1)$ on $U \subset \Sigma$. Choose a smooth function F on U satisfying $2\bar{\partial}F = -i\Theta dx - \Theta dy$. Then we^{-F} is holomorphic on $\overline{p^{-1}(U)}$ and is a defining function for D . The pair of holomorphic functions we^{-F} and $x + iy$ give a holomorphic coordinate system for $\overline{p^{-1}(U)}$. Notably, the two-form $s := d(we^{-F}) \wedge (dx + idy)$ is of type $(2, 0)$ and is a holomorphic section of $K_{\overline{M}}$ over $\overline{p^{-1}(U)}$. The two-form $\frac{1}{we^{-F}}s$ constitutes a holomorphic basis of $K_{\overline{M}} + D$ since we^{-F} is a defining function of the divisor D .

It suffices to establish the extension of the hermitian metric $e^{2\log s_g}g_K$ on K_M to a singular hermitian metric on $K_{\overline{M}} + D$, with the corresponding regularity properties. Therefore, our task reduces to computing the norm of the holomorphic section $\frac{1}{we^{-F}}s$ under the $e^{2\log s_g}g_K$ metric within $\overline{p^{-1}(U)}$. Let us proceed with the calculations

$$\begin{aligned} e^{2\log s_g}g_K\left(\frac{1}{we^{-F}}s, \frac{1}{we^{-F}}s\right) &= e^{2\text{Re } F} \frac{s_g^2}{|w|^2} g_K(s, s) = e^{2\text{Re } F} \frac{s_g^2}{|w|^2} \frac{s \wedge \bar{s}}{d\text{vol}_g} \\ &= \frac{s_g^2}{|w|^2} \frac{4d\text{Re}w \wedge d\text{Im}w \wedge dx \wedge dy}{We^v d\xi \wedge \eta \wedge dx \wedge dy} \\ &= \frac{s_g^2}{|w|^2} \frac{4c_K^2 |w|^2 d\xi \wedge d\varphi \wedge dx \wedge dy}{e^v d\xi \wedge d\varphi \wedge dx \wedge dy} \\ &= 4c_K^2 s_g^2 e^{-v}. \end{aligned}$$

We have $s_g^2 e^{-v} = e^{-\varpi}(1 + O'(\rho^{-\delta_0}))$. The displayed norm therefore extends continuously across the added copy of U , with boundary value $4c_K^2 e^{-\varpi}$. The associated local weight is continuous, so its Lelong number vanishes. $\square$

The subsequent crucial observation stems from [45]. Given the fact that the hermitian metric h is Ricci-flat and conformally related to g through $g = s_g^2 h$, a straightforward computation shows that the symmetric $(1, 1)$ -form $\Omega(\cdot, J\cdot)$ associated with the curvature form Ω of $e^{-2\log s_g}g_{-K}$ is

$$(76) \quad \Omega(\cdot, J\cdot)_{ab} = \frac{s_g}{6} g_{ab} + s_g^{-2} \left(|ds_g|^2 g_{ab} - (ds_g)_a (ds_g)_b - (Jds_g)_a (Jds_g)_b \right).$$

It follows that $\Omega(\cdot, J\cdot)$ is a positive symmetric $(1, 1)$ -form on M , owing to the evident positivity of

$$\begin{aligned} (77) \quad & |ds_g|^2 g_{ab} - (ds_g)_a (ds_g)_b - (Jds_g)_a (Jds_g)_b \\ &= W^{-1}g - d\xi \otimes d\xi - (Jd\xi) \otimes (Jd\xi) \\ &= d\xi^2 + W^{-2}\eta^2 + e^v(dx^2 + dy^2) - d\xi^2 - W^{-2}\eta^2 \\ &= e^v(dx^2 + dy^2) \end{aligned}$$

as a symmetric $(1, 1)$ -form. The positivity of Ω can be extended across the added divisor D as follows.

Proposition 4.5. *On the compactified complex surface $\overline{M}$, there exists a smooth strictly positive $(1, 1)$ -form ϑ such that $\Omega \geq \vartheta$ in the sense of $(1, 1)$ -currents.*

Proof. Due to the positivity of Ω on M , it is sufficient to verify this property near the divisor D . Consequently, we only need to consider Ω locally on each $p^{-1}(U)$.

Using (76) along with the expansions for W and v , we have that on $p^{-1}(U)$:

$$\begin{aligned}\Omega(\cdot, J\cdot) &= \frac{\xi}{6}g + \xi^{-2} \left(W^{-1}g - d\xi \otimes d\xi - (Jd\xi) \otimes (Jd\xi) \right) \\ &= \frac{\xi}{6}g + \xi^{-2}e^v(dx^2 + dy^2) \\ &\geq \epsilon \frac{1}{|w|^2 |\log |w||^3} \left(d\text{Re}w^2 + d\text{Im}w^2 \right) + \epsilon(dx^2 + dy^2).\end{aligned}$$

We now justify the extension as a strict inequality of currents. On a holomorphic coordinate ball with coordinates (q, z) , let $\omega_{\text{eucl}} = idq \wedge d\bar{q} + idz \wedge d\bar{z}$. The estimate above gives a constant $c_U > 0$ such that

$$\Omega \geq c_U \omega_{\text{eucl}} \quad \text{on } U \setminus D.$$

By Proposition 4.4, $\Omega = dd^c \phi$ on U for a continuous weight ϕ . Hence $\phi - c_U(|q|^2 + |z|^2)$ is plurisubharmonic on $U \setminus D$ and is locally bounded above along D . The removable-singularity theorem for plurisubharmonic functions extends it across D ; see [32, Theorem 5.24]. Therefore the displayed current inequality holds on all of U . A finite covering of the compact divisor, together with the strict smooth positivity of Ω away from D , gives a smooth Hermitian $(1, 1)$ -form $\vartheta > 0$ on $\bar{M}$ and $\varepsilon > 0$ such that

$$\Omega \geq \varepsilon \vartheta$$

globally as currents. Renaming $\varepsilon \vartheta$ as ϑ proves the proposition. $\square$

Theorem 4.6. *For a Hermitian non-Kähler ALF-A gravitational instanton, the line bundle $-(K_{\bar{M}} + D)$ on the compactified surface $\bar{M}$ is ample. In particular, $\bar{M}$ is an algebraic surface.*

Proof. By Proposition 4.5, the first Chern class of $-(K_{\bar{M}} + D)$ contains a Kähler current whose local potentials are continuous and hence have zero Lelong numbers. Demailly's regularization theorem [31, Theorem 1.1], applied with a loss of positivity smaller than the strict lower bound in Proposition 4.5, produces a smooth positive closed $(1, 1)$ -form in the same integral class. Thus $-(K_{\bar{M}} + D)$ is a positive line bundle. Kodaira's embedding theorem makes it ample and makes $\bar{M}$ projective. $\square$

Proposition 4.7. *The holomorphic $\mathbb{C}^*$ -action induced by ∂_t and $J\partial_t$ on M extends to $\bar{M}$ and fixes D pointwise.*

Proof. In the coordinate $q = we^{-F}$ constructed above, the complexified flow acts by

$$q \mapsto aq, \quad a \in \mathbb{C}^*,$$

and leaves the coordinate on D unchanged. This formula extends holomorphically to $q = 0$ and fixes D pointwise. The formulas agree on overlaps because two defining functions differ by a nowhere-vanishing holomorphic factor. $\square$

Remark 4.8. *The compactification is natural in the sense that we are adding a holomorphic sphere D that is fixed by the extended $\mathbb{C}^*$ -action above.*

4.4. Rational AF gravitational instantons. In this subsection, we study Hermitian non-Kähler rational AF_{a,b} gravitational instantons. We retain $\mathfrak{b}$ and choose the generator of the defining deck group so that $\mathfrak{b} > 0$; replacing $(\mathfrak{a}, \mathfrak{b})$ by $(-\mathfrak{a}, -\mathfrak{b})$, with the angle taken modulo 2π , does not change the quotient. In the rational AF action formulas below, $e^{i\theta}$ denotes the action of the primitive generator $X_- := -(T_{\mathcal{K}}/2\pi)\mathcal{K}$. We represent the equivalence class associated with (ρ, α, β, t) within

the quotient by $[\rho, \alpha, \beta, t]$. When $\mathfrak{a}/2\pi = m/n$ in lowest terms, the regular period is $T_K = n\mathfrak{b}$, and there is a finite covering map

$$(78) \quad \Pi : \mathfrak{M}_{0,n\mathfrak{b}} \rightarrow \mathfrak{M}_{\mathfrak{a},\mathfrak{b}}, \quad [\rho, \alpha, \beta, t] \rightarrow [\rho, \alpha, \beta, t].$$

This is nothing but the unwrapping operation in (2) of Section 3.3. We can pull back the end $\mathfrak{M}_{\mathfrak{a},\mathfrak{b}} \setminus \overline{B_N(0)}$, alongside the Hermitian Ricci-flat metric h , the Kähler metric g , and the complex structure J , onto the cover $\mathfrak{M}_{0,n\mathfrak{b}} \setminus \overline{B_N(0)}$. Our previous treatment for Hermitian non-Kähler ALF-A gravitational instantons in Section 4.1-4.3 remains wholly applicable to the end $\mathfrak{M}_{0,n\mathfrak{b}} \setminus \overline{B_N(0)}$, with the pulled-back complex structure Π^*J and Kähler metric Π^*g . Thus

Proposition 4.9. *The end $\mathfrak{M}_{0,n\mathfrak{b}} \setminus \overline{B_N(0)}$, along with the complex structure Π^*J , can be compactified at infinity, by introducing a holomorphic sphere D' at infinity. This added curve D' has trivial self-intersection.*

Let M' denote the partial compactification of the covering end $\mathfrak{M}_{0,n\mathfrak{b}} \setminus \overline{B_N(0)}$ by its divisor at infinity. We use a holomorphic fibre coordinate z that vanishes along D' . The lifted canonical field has the same regular period $T_K = n\mathfrak{b}$, and the action of X_- on z is $z \mapsto e^{-i\theta}z$.

Proposition 4.10. *For a Hermitian non-Kähler rational $AF_{\mathfrak{a},\mathfrak{b}}$ gravitational instanton with $\mathfrak{a}/2\pi = m/n$, a suitable end $M \setminus K$ of the gravitational instanton is naturally biholomorphic to the quotient $\mathbb{D}^* \times \mathbb{P}^1 / \sim$ by the following relation*

$$\sim : z \times [x : y] \leftrightarrow e^{2\pi i \frac{1}{n}z} z \times [e^{2\pi i \frac{m}{n}}x : y].$$

The circle subgroup generated by K , parameterized by X_- , corresponds to the S^1 -action on the quotient $\mathbb{D}^ \times \mathbb{P}^1 / \sim$ descended from*

$$e^{i\theta} \cdot (z \times [x : y]) = e^{-i\theta} z \times [x : y].$$

The negative character here comes from using X_- ; the canonical field remains $K = \partial_t$.

Proof. The cover of a suitable end of the gravitational instanton is biholomorphic to $\mathbb{D}^* \times \mathbb{P}^1$. Because of Proposition 4.7, the way we compactify the cover at infinity is natural, in the sense that the added holomorphic sphere D' is fixed by the extended $\mathbb{C}^*$ -action. The pulled-back primitive field Π^*X_- corresponds to the rotation on the $\mathbb{D}^*$ factor

$$e^{i\theta} \cdot (z \times [x : y]) = e^{-i\theta} z \times [x : y].$$

Using the covering map Π , we can now recover the end holomorphically, and the end emerges as the quotient of $\mathbb{D}^* \times \mathbb{P}^1$ by the action

$$z \times [x : y] \mapsto e^{2\pi i \frac{1}{n}z} z \times [e^{2\pi i \frac{m}{n}}x : y].$$

□

Proposition 4.11. *The hermitian metric $e^{-2\log s_g} g_{-K}$ on $-K_M$ restricted to the end $\mathfrak{M}_{\mathfrak{a},\mathfrak{b}} \setminus \overline{B_N(0)}$ pulled back to the cover $\mathfrak{M}_{0,n\mathfrak{b}} \setminus \overline{B_N(0)}$ can be extended to a singular hermitian metric on $-(K_{M'} + D')$, with the regularity properties described in Proposition 4.4.*

Proof. This is just Proposition 4.4 applied to the end $\mathfrak{M}_{0,n\mathfrak{b}} \setminus \overline{B_N(0)}$. □

Theorem 4.12. *For a Hermitian non-Kähler rational $AF_{\mathfrak{a},\mathfrak{b}}$ gravitational instanton (M, h) with $\mathfrak{a}/2\pi = m/n$, the complex surface M can be naturally compactified by adding a holomorphic $D = \mathbb{P}^1$ at infinity to a compact normal algebraic surface $\overline{M}$ with at most two cyclic quotient points in D (exactly two when $m > 0$). The $\mathbb{Q}$ -line bundle $-(K_{\overline{M}} + D)$ is ample on the compactified surface $\overline{M}$.*

Proof. On the finite cover, Theorem 4.3 and Propositions 4.4–4.5 give the compactification M' and a deck-invariant Kähler current after averaging over the cyclic deck group. The deck action extends across D' : in the product description it is

$$z \times [x : y] \mapsto e^{2\pi i/n} z \times [e^{2\pi i m/n} x : y].$$

It is free away from D' near infinity. On D' its stabilizer is trivial away from the two fixed points $[0 : 1]$ and $[1 : 0]$, so the quotient has exactly two cyclic quotient singularities when $n > 1$, both on $D = D'/\mathbb{Z}_n \simeq \mathbb{P}^1$. This quotient compactifies a collar of the original end. Gluing that collar to the unchanged compact interior of M produces a compact normal complex surface $\overline{M}$ with $M = \overline{M} \setminus D$.

The averaged current descends to a strictly positive current in the class $c_1(-(K_{\overline{M}} + D))$ with continuous local orbifold potentials and zero Lelong numbers. Equivariant regularization on the uniformizing charts, followed by averaging, produces a smooth positive orbifold form in this $\mathbb{Q}$ -class. Hence some positive tensor power of $-(K_{\overline{M}} + D)$ is an ample line bundle. $\square$

Remark 4.13. *Similarly, the holomorphic $\mathbb{C}^*$ -action induced by $\mathcal{K}$ and JK extends to the compactification $\overline{M}$, fixing the holomorphic sphere D . This explains the naturality of the compactification.*

4.5. The case of finite quotients of ALF-A. It remains to treat gravitational instantons asymptotic to a nontrivial finite quotient of an ALF-A model. Their links are orbifold S^1 -bundles over the round orbifold $\mathbb{S}^2/\Gamma$.

In the cyclic case, where $\Gamma = \mathbb{Z}_k$, after passing to the finite $\mathbb{Z}_k$ -cover, the end is ALF-A and can be compactified as in Section 4.2, by adding a smooth $\mathbb{P}^1$. The quotient is therefore a compactification by a sphere with precisely two cyclic quotient points. Averaging the strict Kähler current on the cover and applying equivariant regularization in the two uniformizing charts proves that the descended log-anticanonical $\mathbb{Q}$ -bundle is ample, exactly as in the proof of Theorem 4.12. In conclusion, we have

Theorem 4.14. *If (M, h) is asymptotic to a nontrivial cyclic quotient of an ALF-A model, then M admits a natural compactification $\overline{M} = M \sqcup D$ by a holomorphic curve $D \simeq \mathbb{P}^1$. The compactification is an algebraic orbifold with two orbifold points on D , and the $\mathbb{Q}$ -line bundle $-(K_{\overline{M}} + D)$ is ample.*

In the noncyclic case we can similarly compactify by adding a holomorphic $\mathbb{P}^1$ with three orbifold points. The compactified pair $(\overline{M}, D)$ contains three orbifold points in D and is still log del Pezzo in the sense that $-(K_{\overline{M}} + D)$ is ample. The holomorphic $\mathbb{C}^*$ -action extends to $(\overline{M}, D)$ and fixes D pointwise. We will later study the noncyclic case in Appendix A and show that this case cannot happen for Hermitian non-Kähler gravitational instantons.

4.6. Structure of the compactified surfaces. In the following, the pair $(\overline{M}, D)$ always denotes the naturally compactified algebraic surface with the added divisor, arising from a Hermitian non-Kähler gravitational instanton that is asymptotic to ALF-A or its quotients, or rational AF.

4.6.1. The ALF-A case. The main conclusions of this part are Corollary 4.17 and Corollary 4.18.

Proposition 4.15. *The complex surface $\overline{M}$ is rational.*

Proof. Near D , the $\mathbb{C}^*$ -action is multiplication in the normal direction. Thus the closure of a generic orbit is a rational curve meeting D in exactly one point, and this gives the rational quotient map

$$q : \overline{M} \dashrightarrow D \simeq \mathbb{P}^1.$$

After resolving the finitely many indeterminacy points equivariantly, q becomes a ruling whose general fiber is the closure of a $\mathbb{C}^*$ -orbit. The resolved surface is therefore birational to a ruled surface over $\mathbb{P}^1$. Hence $\overline{M}$ is rational. $\square$

Recall that a rational surface must be either $\mathbb{P}^1 \times \mathbb{P}^1$, or have $\mathbb{P}^2$ as a minimal model, or have the Hirzebruch surface H_k with $k \geq 2$ as a minimal model.

Corollary 4.16. *Except possibly for D , the surface $\overline{M}$ contains no rational curve of self-intersection at most -2 . A rational curve distinct from and meeting D has nonnegative self-intersection.*

Proof. Set $A := -(K_{\overline{M}} + D)$. For an irreducible rational curve $C \neq D$, the adjunction formula for its arithmetic genus gives

$$C^2 + K_{\overline{M}} \cdot C = 2p_a(C) - 2 \geq -2.$$

If $C \cap D = \emptyset$, then $C^2 \geq -2 + A \cdot C > -2$, and integrality gives $C^2 \geq -1$. If C meets D , then $D \cdot C \geq 1$ and

$$C^2 \geq -2 + A \cdot C + D \cdot C > -1,$$

so $C^2 \geq 0$. Notice that this also covers singular irreducible rational curves, for which $p_a(C)$ may be positive; smoothness of C is not used. $\square$

Corollary 4.17. *Suppose (M, h) is a Hermitian non-Kähler ALF-A gravitational instanton. If D has nonnegative self-intersection $D^2 \geq 0$, then $\overline{M}$ is a del Pezzo surface. Moreover, the pair $(\overline{M}, D)$, up to biholomorphism, must be one of the following:*

- $(\mathbb{P}^2, D)$, where $D = \mathbb{P}^1$ is a hyperplane with self-intersection 1.
- $(\text{Bl}_p \mathbb{P}^2, D)$, where p does not lie on the line D and its proper transform has self-intersection 1.
- $(\mathbb{P}^1 \times \mathbb{P}^1, D)$, where $D = 0 \times \mathbb{P}^1$ is a curve with self-intersection 0.

Proof. Set $A := -(K_{\overline{M}} + D)$, which is ample. If $C \neq D$ is an irreducible curve, then $-K_{\overline{M}} \cdot C = A \cdot C + D \cdot C > 0$. By adjunction, $-K_{\overline{M}} \cdot D = 2 + D^2 > 0$. Since $-K_{\overline{M}} = A + D$ and $A^2 > 0$, we have

$$(-K_{\overline{M}})^2 = (A + D) \cdot A + (A + D) \cdot D = A^2 + A \cdot D + 2 + D^2 > 0.$$

The Nakai–Moishezon criterion therefore shows that $-K_{\overline{M}}$ is ample, so $\overline{M}$ is a del Pezzo surface. Its minimal model is either $\mathbb{P}^2$ or $\mathbb{P}^1 \times \mathbb{P}^1$. The $\mathbb{C}^*$ -action on $\overline{M}$ then comes from a $\mathbb{C}^*$ -action on its minimal model. Blow-ups can only be performed at fixed points of the $\mathbb{C}^*$ -action.

Suppose first that the minimal model is $\mathbb{P}^2$. A nontrivial $\mathbb{C}^*$ -action on $\mathbb{P}^2$ with a fixed curve has fixed locus consisting of a line, which is the image of D , and one point p outside that line. A blow-up on D would create a (-1) -curve meeting D , contrary to Corollary 4.16; hence the only possible first blow-up is at p . After this blow-up every fixed point outside D lies on the exceptional curve. Any further blow-up either has center on D , which is already excluded, or makes the proper transform of the exceptional curve a curve of self-intersection at most -2 . Thus there is at most one blow-up, giving the first two pairs in the statement.

If the minimal model is $\mathbb{P}^1 \times \mathbb{P}^1$, then, after exchanging the factors, the action is nontrivial on the first factor and trivial on the second. Its fixed curves are $D = \{0\} \times \mathbb{P}^1$ and $D_\infty = \{\infty\} \times \mathbb{P}^1$. A center on D is excluded as above. If a point of D_∞ were blown up, the proper transform of the transverse ruling through that point would be a (-1) -curve still meeting D , again contrary to Corollary 4.16. Hence no blow-up occurs, and the third pair is the only possibility. $\square$

Corollary 4.18. *Suppose (M, h) is a Hermitian non-Kähler ALF-A gravitational instanton. If D has negative self-intersection number $D^2 = -k$, then $\overline{M}$ up to biholomorphisms is the Hirzebruch surface $H_k = \mathbb{P}(\mathcal{O} \oplus \mathcal{O}(k))$ and D is the curve at infinity C_∞ .*

Proof. By Corollary 4.16, every (-1) -curve different from D is disjoint from D . Contract such curves until none remains; these contractions do not change $D^2 = -k$. If $k \geq 2$, the resulting minimal rational surface contains a curve of square $-k$, and hence is H_k , with D its negative section.

Suppose $k = 1$. Contract D to a point p on a smooth rational surface Y . The surface Y is minimal. Indeed, a (-1) -curve avoiding p would lift to a remaining (-1) -curve, whereas one through p would have proper transform of self-intersection at most -2 meeting D , contrary to Corollary 4.16. If Y were a Hirzebruch surface, the ruling through p would lift to a (-1) -curve meeting D , giving the same contradiction. Thus $Y = \mathbb{P}^2$, and the surface before contracting D is H_1 .

It remains only to observe that no blow-up can occur in recovering $\overline{M}$ from H_k . A center on the negative section is excluded by Corollary 4.16. If the center is away from it, the proper transform of the ruling through the center is a (-1) -curve that still meets D , again a contradiction. Therefore $(\overline{M}, D) \simeq (H_k, C_\infty)$. $\square$

4.6.2. *The rational AF case.* The main conclusion of this part is Corollary 4.23.

In product descriptions of the end below, the $\mathbb{C}^*$ factor is restricted to $0 < |z| < \varepsilon$ on the cover and $0 < |w| < \varepsilon^n$ on the quotient, for sufficiently small $\varepsilon > 0$. After the explicit coordinate inversion below, the corresponding restrictions are $|z| > \varepsilon^{-1}$ and $|w| > \varepsilon^{-n}$.

Proposition 4.19. *For a Hermitian non-Kähler rational $AF_{a,b}$ gravitational instanton (M, h) with $a/2\pi = m/n$, a suitable end $M \setminus K$ together with the circle action parameterized by X_- is biholomorphic to:*

$$(79) \quad e^{i\theta} \cdot (w, [u : v]) = (e^{-in\theta}w, [e^{im\theta}u : v]) \quad \text{on } \mathbb{C}^* \times \mathbb{P}^1.$$

Here, we use $w \times [u : v]$ to parametrize $\mathbb{C}^* \times \mathbb{P}^1$.

Proof. Recall that holomorphically, a suitable end of such a gravitational instanton appears as the quotient $\mathbb{C}^* \times \mathbb{P}^1 / \sim$ by the relation $\sim: z \times [x : y] \mapsto e^{2\pi i \frac{1}{n}} z \times [e^{2\pi i \frac{m}{n}} x : y]$. The S^1 -action parameterized by X_- is given by the rotation on $z \in \mathbb{C}^*$ on the cover of the end $\mathbb{C}^* \times \mathbb{P}^1$

$$e^{i\theta} \cdot (z \times [x : y]) = e^{-i\theta} z \times [x : y].$$

Now, the end of the gravitational instanton $\mathbb{C}^* \times \mathbb{P}^1 / \sim$ can be identified to the standard $\mathbb{C}^* \times \mathbb{P}^1$ by the following biholomorphism

$$(80) \quad \mathbb{C}^* \times \mathbb{P}^1 / \sim \rightarrow \mathbb{C}^* \times \mathbb{P}^1 \quad z \times [x : y] / \sim \mapsto w \times [u : v] := z^n \times [z^{-m}x : y].$$

Under this identification, the end of the gravitational instanton holomorphically becomes $\mathbb{C}^* \times \mathbb{P}^1$, and the S^1 -action parameterized by X_- becomes

$$e^{i\theta} \cdot (w \times [u : v]) = e^{-in\theta} w \times [e^{im\theta} u : v].$$

$\square$

Proposition 4.20. *Suppose (M, h) is a Hermitian non-Kähler rational $AF_{a,b}$ gravitational instanton with $a/2\pi = m/n$. Then M admits a smooth compactification $\overline{M}^\sharp$ obtained by adding a holomorphic sphere $D^\sharp$.*

Proof. Add $0 \times \mathbb{P}^1$ to the end $\mathbb{C}^* \times \mathbb{P}^1$ that appears in Proposition 4.19; here $w = z^n \rightarrow 0$ at infinity. $\square$

Remark 4.21. *This compactification is auxiliary rather than natural: the holomorphic sphere $D^\sharp$ is invariant under the extended $\mathbb{C}^*$ -action. If $m > 0$, it is not fixed pointwise and its complement of the two fixed points is a single $\mathbb{C}^*$ -orbit; if $m = 0$, it is fixed pointwise.*

Proposition 4.22. *Up to biholomorphism, the auxiliary compactification $(\overline{M}^\sharp, D^\sharp)$ is one of the following:*

- $(H_k, \mathbb{P}^1)$, where $H_k = \mathbb{P}(\mathcal{O} \oplus \mathcal{O}(k))$ is the Hirzebruch surface with $k \geq 0$ and $\mathbb{P}^1$ is a ruling with trivial self-intersection.

- $(\text{Bl}_p H_k, \mathbb{P}^1)$, where $p \in (C_0 \cup C_\infty) \setminus D^\sharp$ and $\mathbb{P}^1 = D^\sharp$ is a ruling with trivial self-intersection.

Proof. Since $D^\sharp \simeq \mathbb{P}^1$ and $(D^\sharp)^2 = 0$, the classical ruled-surface criterion [6, Chapter V, Proposition 4.3] shows that, after contracting finitely many (-1) -curves contained in M , one obtains a ruled surface for which $D^\sharp$ is a fiber. Since $\mathbb{C}^*$ is connected, it preserves every (-1) -curve, so these contractions may be chosen equivariantly. By (79), the induced action on the base of the ruling is nontrivial. Hence the base is $\mathbb{P}^1$, and the ruled surface is H_k for some $k \geq 0$.

Thus $\overline{M}^\sharp$ is obtained from H_k by equivariant blow-ups at fixed points away from $D^\sharp$. Besides the invariant fiber $D^\sharp$, the action on the base has one other invariant fiber F . On F the fixed locus is either all of F or two points; up to an automorphism preserving the ruling, a possible blow-up center lies on $C_0 \cup C_\infty$. A second blow-up would make the proper transform of F , or the first exceptional curve, have self-intersection at most -2 inside M , contrary to Corollary 4.16. Hence at most one blow-up occurs. $\square$

In either case, $\overline{M}$ is toric. Since $(\overline{M}^\sharp, D^\sharp)$ is H_k or its single toric blow-up, the fan $\mathbf{F}_{\overline{M}}$ of $\overline{M}$ has four or five rays. For this fan description, invert the fibre coordinates by $z_{\text{new}} = z^{-1}$ and $w_{\text{new}} = w^{-1}$, keeping $[u : v]$ unchanged, and then drop the new subscripts. Thus $w = z^n$, $[u : v] = [z^m x : y]$, and the added divisor is at $z = \infty$. The quotient model

$$(\mathbb{C}^* \cup \{\infty\}) \times \mathbb{P}^1 / (z, [x : y]) \sim (e^{-2\pi i/n} z, [e^{2\pi i m/n} x : y])$$

identifies the primitive ray of D as $(-n, -m)$. After an $SL(2, \mathbb{Z})$ change of basis, the remaining rays are exactly those shown in Figures 1 and 2.

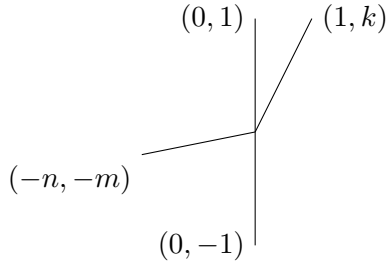


FIGURE 1. Four edges $\mathbf{F}_{\overline{M}}$.

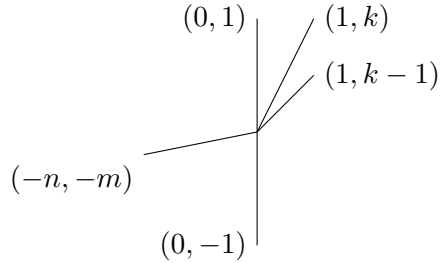


FIGURE 2. Five edges $\mathbf{F}_{\overline{M}}$.

If the corresponding polytope $\Delta_{\overline{M}}$ has four edges, the ray in direction $(-n, -m)$ corresponds to D , and k is the index of the Hirzebruch surface H_k . The five-edge case is analogous. With our explicit understanding on the possible structure of the toric surface $\overline{M}$ and the positivity of $-(K_{\overline{M}} + D)$, we can now give a more precise classification of the natural compactification $(\overline{M}, D)$.

Corollary 4.23. *Suppose (M, h) is a rational Hermitian non-Kähler $AF_{\mathfrak{a}, \mathfrak{b}}$ gravitational instanton with $\mathfrak{a}/2\pi = m/n$, where $0 \leq m < n$ and m, n are coprime. Then the natural compactification $(\overline{M}, D)$ can only be*

$$(H_{m,n}, D), \quad (H_{n-m,n}, D), \quad (\text{Bl}_p H_{m,n}, D).$$

Here $H_{m,n}$ and $\text{Bl}_p H_{m,n}$ correspond to the following fans, with D being the $\mathbb{P}^1$ given by the ray $(-n, -m)$.

Proof. It suffices to rule out the other possibilities in Figures 1 and 2. We give both intersection calculations, since the blow-up strengthens one of the inequalities.

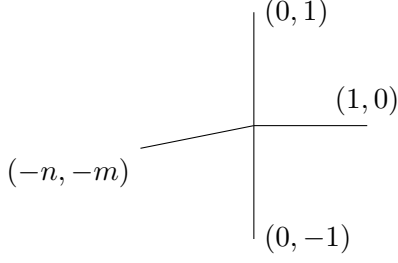


FIGURE 3. The fan $\mathbf{F}_{m,n}$.

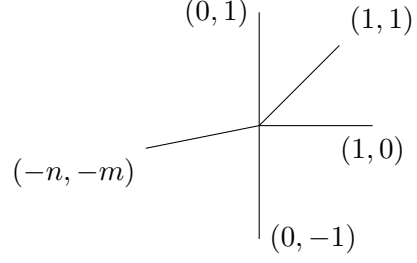
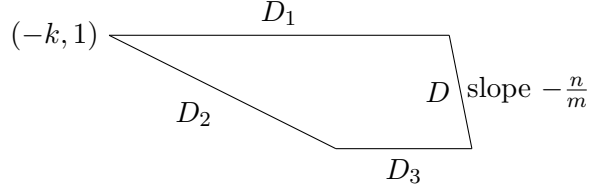


FIGURE 4. The fan $\text{Bl}_p \mathbf{F}_{m,n}$.

For the fan given by Figure 1, its polytope is



The divisors D_1, D_3, D are merely $\mathbb{Q}$ -Cartier. The intersection numbers are given by

$$(D_1)^2 = \frac{-m + nk}{n}, \quad (D_3)^2 = \frac{m - nk}{n}, \quad D_1 \cdot D = D_3 \cdot D = \frac{1}{n}.$$

For the toric surface $\overline{M}$, its canonical divisor can be written as $K_{\overline{M}} = -D_1 - D_2 - D_3 - D$. The positivity of $-(K_{\overline{M}} + D)$ therefore implies

$$-(K_{\overline{M}} + D) \cdot D_1 = (D_1 + D_2 + D_3) \cdot D_1 = \frac{-m + kn}{n} + 1 > 0,$$

$$-(K_{\overline{M}} + D) \cdot D_3 = (D_1 + D_2 + D_3) \cdot D_3 = 1 + \frac{m - kn}{n} > 0.$$

These inequalities force $(k - 1)n < m < (k + 1)n$. If $0 < m < n$, then $k = 0$ or 1 . For $k = 0$ one obtains $\mathbf{F}_{m,n}$; for $k = 1$, a $\text{GL}(2, \mathbb{Z})$ transformation identifies the fan with $\mathbf{F}_{n-m,n}$. If $m = 0$ we obtain $(\mathbb{P}^1 \times \mathbb{P}^1, D)$.

For the five-ray fan, consider the blow-up of the cone between the rays $(1, k)$ and $(0, 1)$; the new ray is $(1, k + 1)$. If D'_1 is the proper transform of D_1 (which is unaffected by this blow-up), D'_3 is the proper transform of D_3 , and E is the exceptional divisor, then

$$-(K_{\overline{M}} + D) \cdot D'_1 = \frac{(k + 1)n - m}{n},$$

$$-(K_{\overline{M}} + D) \cdot D'_3 = \frac{m - kn}{n},$$

$$-(K_{\overline{M}} + D) \cdot E = 1.$$

Ampleness therefore gives

$$kn < m < (k + 1)n.$$

Thus $k = 0$ when $0 < m < n$; if $m = 0$, the strict inequalities exclude this blow-up. If instead the cone between $(0, -1)$ and $(1, k)$ is blown up, the new ray is $(1, k - 1)$ and the identical calculation gives

$$(k - 1)n < m < kn,$$

so $k = 1$. Both cases give exactly the fan in Figure 4. Hence the only five-ray possibility is the single indicated toric blow-up $\text{Bl}_p \mathbf{F}_{m,n}$. $\square$

4.6.3. *The case of cyclic quotients of ALF-A.* The main conclusion of this part is the following proposition.

Proposition 4.24. *If (M, h) is asymptotic to a cyclic quotient of an ALF-A model, then M is holomorphically toric.*

We will actually prove the following lemma which implies the proposition.

Lemma 4.25. *Let $\overline{M}$ be a rational surface with precisely two cyclic quotient singularities, both lying on a divisor $D \simeq \mathbb{P}^1$. If $-(K_{\overline{M}} + D)$ is ample and a nontrivial holomorphic $\mathbb{C}^*$ -action fixes D pointwise, then $\overline{M}$ is toric.*

Proof. Let $l : \widetilde{M} \rightarrow \overline{M}$ be the minimal resolution, let E_1, E_2 be its two exceptional chains, and let $\widetilde{D}$ be the proper transform of D . The $\mathbb{C}^*$ -action lifts to $\widetilde{M}$. Choose a sequence of contractions from $\widetilde{M}$ to a minimal rational surface M_0 . These contractions are equivariant. Thus M_0 is $\mathbb{P}^2$ or H_k with $k = 0$ or $k \geq 2$, and $\widetilde{M}$ is recovered by successive blow-ups at fixed points.

We use the same adjunction argument as in Corollary 4.16, applied to the images of curves in the smooth locus of $\overline{M}$. A rational curve in $\widetilde{M}$ that is disjoint from $\widetilde{D} \cup E_1 \cup E_2$ has self-intersection at least -1 . If it is disjoint from $E_1 \cup E_2$ and meets $\widetilde{D}$, then it has nonnegative self-intersection.

It remains to follow the equivariant blow-ups. A blow-up at an isolated fixed point preserves toricity. On a fixed rational curve of the $\mathbb{C}^*$, one or two marked points can still be fixed by another suitable $\mathbb{C}^*$; three distinct centers on a fixed curve are therefore needed to destroy toricity. Three such blow-ups would produce three negative orbit closures meeting $\widetilde{D}$ at distinct points. Let C be the third orbit closure, meeting $\widetilde{D}$ at $p_3 \notin \widetilde{D} \cap (E_1 \cup E_2)$. Write $t \cdot x$, with $t \in \mathbb{C}^*$, for the lift to $\widetilde{M}$ of the holomorphic $\mathbb{C}^*$ -action in the hypothesis. After replacing t by t^{-1} if necessary, assume that $\widetilde{D}$ is the source, meaning that

$$\lim_{t \rightarrow 0} t \cdot x \in \widetilde{D}$$

for every x in a neighborhood of $\widetilde{D}$. If C intersects an exceptional chain, the intersection must be a fixed point. Every fixed point on the chain away from $\widetilde{D}$ is a saddle, whose unique orbit branch tending to it as $t \rightarrow \infty$ lies in the chain. Since the backward endpoint of C is p_3 , the intersection would be its forward endpoint. Thus C would coincide with an exceptional component, a contradiction. Hence $C \cap (E_1 \cup E_2) = \emptyset$. Therefore, all the blow-ups can be chosen torically, and $\widetilde{M}$ is toric.

Finally, contracting the torus-invariant chains E_1, E_2 gives the torus action on $\overline{M}$. Toricity descends since each curve in the exceptional set is negative, so it is invariant under the torus action. $\square$

Proof of Proposition 4.24. The minimal resolution of the natural compactification is rational by the same orbit-closure argument as Proposition 4.15. Indeed, the lifted $\mathbb{C}^*$ -action fixes the proper transform of $D \simeq \mathbb{P}^1$, and the closure of a generic orbit meets it once. The resulting rational quotient map to D , after equivariant resolution, is a ruling over $\mathbb{P}^1$. The natural compactification has exactly two cyclic quotient singularities on D , its log-anticanonical divisor is ample, and the action generated by $\mathcal{K}$ fixes D pointwise. The Lemma 4.25 therefore makes the compactification toric. Restricting its torus action to M proves the proposition. $\square$

We conclude the section with the following consequence.

Theorem 4.26. *Any Hermitian non-Kähler gravitational instanton that is either AF, ALF-A, or cyclic quotient of ALF-A, is holomorphically toric, with the torus action containing the $\mathcal{K}$ -action.*

Here, by saying it is holomorphically toric, we mean there is an effective holomorphic $(\mathbb{C}^*)^2$ -action with a dense open orbit.

Proof. If $\mathcal{K}$ has nonclosed $\mathbb{R}$ -orbits, the closure of its flow provides the required torus action.

If $\mathcal{K}$ generates an S^1 -action, Corollaries 4.17–4.23 and Proposition 4.24 show that the compactified surface is holomorphically toric. Restricting the torus action to M proves the closed-orbit case. $\square$

Corollaries 4.17–4.23, together with the positivity of $-(K_{\overline{M}} + D)$, prove Theorem 1.6. The topology of the model spaces gives the following natural compactifications:

- The complex structure of the reversed Taub-NUT space can be compactified to $(\mathbb{P}^2, \mathbb{P}^1)$, where $D = \mathbb{P}^1$ is a standard hyperplane in $\mathbb{P}^2$.
- The complex structures of the Taub-bolt space can be compactified to (H_1, C_0) or (H_1, C_∞) depending on the choice of orientation.
- The complex structure of the rational Kerr spaces can be compactified to $(H_{m,n}, D)$.
- The complex structure of the rational Chen-Teo spaces can be compactified to $(\text{Bl}_p H_{m,n}, D)$.

5. THE CALABI-TYPE THEOREM

In this section we prove a Calabi-type theorem for Hermitian non-Kähler ALF/AF gravitational instantons whose extremal field $\mathcal{K}$ generates an S^1 -action. For ALF-A, cyclic quotients of ALF-A, and rational AF ends, Section 4 shows that the natural compactifications are holomorphically toric: they carry a holomorphic T^2 -action preserving the divisor D . We work throughout with the Kähler metric g .

Let $\bar{\partial}^\#$ map a smooth complex-valued function f to the $(1,0)$ -vector field $(\bar{\partial}f)^\#$. The holomorphic gradient $\bar{\partial}^\# s_g$ is a nonzero constant multiple of $\mathcal{K} - iJ\mathcal{K}$ and will be used as the complex extremal field. By Proposition 4.7, it extends holomorphically to $\overline{M}$ and vanishes along D .

We use the following notation. Let $\mathbf{aut}_\mathcal{K}(\overline{M}, D)$ be the space of real holomorphic vector fields on $\overline{M}$ that are tangent to D and commute with $\mathcal{K}$, restricted to M . Let $\mathbf{hamiltoniso}_\mathcal{K}(M, g)$ be the space of Hamiltonian Killing fields of g that commute with $\mathcal{K}$ and extend holomorphically to $(\overline{M}, D)$. The following statement is inspired in part by the corresponding result for complete shrinking gradient Kähler–Ricci solitons in [29].

Theorem 5.1 (Calabi-type theorem). *Let (M, h, J) be a Hermitian non-Kähler gravitational instanton asymptotic to ALF-A, a finite quotient of ALF-A, or rational AF. Then*

$$(81) \quad \mathbf{aut}_\mathcal{K}(\overline{M}, D) = \mathbf{hamiltoniso}_\mathcal{K}(M, g) \oplus J\mathbf{hamiltoniso}_\mathcal{K}(M, g).$$

The holomorphic T^2 -action on $(\overline{M}, D)$ supplies a four-dimensional real subspace generated by the two torus fields and their J -images. Theorem 5.1 therefore implies that $\mathbf{hamiltoniso}_\mathcal{K}(M, g)$ has dimension at least two. Since $h = s_g^{-2}g$ and these fields preserve s_g , they are Killing for h as well. Thus every gravitational instanton in these three cases with closed $\mathcal{K}$ -orbits is holomorphically and isometrically toric.

We now prove the theorem.

The expansion in Section 4 shows that g is complete and has finite volume: its normal part is asymptotic to a Poincaré cusp and its tangential part converges to a metric on D .

We first prove that the sum in (81) is direct. Suppose that both $\mathfrak{k}$ and $J\mathfrak{k}$ are Hamiltonian Killing fields. Since $\mathfrak{k}$ is real holomorphic, the endomorphism $A := \nabla \mathfrak{k}$ commutes with J . The Killing equation says that A is skew-adjoint. Moreover,

$$\nabla(J\mathfrak{k}) = JA.$$

Because A commutes with J , the endomorphism JA is self-adjoint; because $J\mathfrak{k}$ is Killing, it is also skew-adjoint. Hence $JA = 0$, and therefore $A = 0$. Thus $\mathfrak{k}$ and $J\mathfrak{k}$ are parallel.

Let μ be a Hamiltonian for $\mathfrak{k}$, normalized by

$$d\mu = -g(J\mathfrak{k}, \cdot).$$

If $\mathfrak{k} \neq 0$, then $|\nabla\mu| = |\mathfrak{k}|$ is a positive constant. The complete isometric flow of $-J\mathfrak{k}$ translates μ at constant speed. Consequently, the mutually disjoint slabs $\{j \leq \mu < j+1\}$ are isometric and have the same positive volume. This contradicts $\text{Vol}_g(M) < \infty$. Hence $\mathfrak{k} = 0$, proving directness.

It remains to prove surjectivity. Fix $Y_{\text{real}} \in \text{aut}_{\mathcal{K}}(\overline{M}, D)$ and put

$$Y := Y_{\text{real}} - iJY_{\text{real}}.$$

This is a $(1,0)$ holomorphic vector field. Since Y_{real} commutes with the real holomorphic field $\mathcal{K}$, holomorphicity also gives commutation with $J\mathcal{K}$; hence $[Y, \bar{\partial}^{\#} s_g] = 0$. We use throughout the Kähler metric

$$g = Wd\xi^2 + W^{-1}\eta^2 + g_{\xi}.$$

5.1. Construction of the potential function. We construct a complex-valued function χ satisfying

$$\bar{\partial}\chi = g(Y, \cdot).$$

We first work on a finite cover of the end so that the reduced surface has no orbifold points. At the end of the argument we average over the finite deck group, so the resulting function descends to M .

Lemma 5.2. *The holomorphic vector field Y descends to a holomorphic vector field Y_{Σ} on $\Sigma = \mathbb{P}^1$.*

Proof. The fibers of the holomorphic quotient are the orbits of the complexified flow generated by $\bar{\partial}^{\#} s_g$. Since $[Y, \bar{\partial}^{\#} s_g] = 0$, the field Y preserves this foliation and is projectable. Its projection is holomorphic because both the quotient map and Y are holomorphic. $\square$

Write $\bar{\partial}_{\Sigma}$ for the Dolbeault operator on $\Sigma = \mathbb{P}^1$.

Lemma 5.3. *Every smooth $\bar{\partial}_{\Sigma}$ -closed $(0,1)$ -form ν on $\Sigma = \mathbb{P}^1$ is of the form $\nu = \bar{\partial}_{\Sigma} f$ for a smooth function f , unique up to an additive constant.*

Proof. This is the vanishing $H^{0,1}(\mathbb{P}^1) = 0$. $\square$

Recall that g_{ξ} is the metric induced on the Kähler reduction at level ξ ; locally,

$$g_{\xi} = We^v(dx^2 + dy^2).$$

Lemma 5.4. *On each (Σ, g_{ξ}) , we can find a potential function f_{ξ} for the holomorphic vector field Y_{Σ} , in the sense that*

$$(82) \quad \bar{\partial}_{\Sigma} f_{\xi} = g_{\xi}(Y_{\Sigma}, \cdot).$$

Proof. Because Y_{Σ} is holomorphic and g_{ξ} is Kähler, the $(0,1)$ -form $g_{\xi}(Y_{\Sigma}, \cdot)$ is $\bar{\partial}_{\Sigma}$ -closed. The preceding lemma gives f_{ξ} . We initially fix its additive constant by requiring

$$\int_{\Sigma} f_{\xi} d\text{vol}_{g_{\xi}} = 0.$$

It depends smoothly on ξ . $\square$

Lemma 5.5. *After changing the ξ -dependent additive constant in f_{ξ} , there is a smooth function f_{end} on the end such that*

$$\bar{\partial} f_{\text{end}} = g(Y, \cdot).$$

Moreover, f_{end} and all of its covariant derivatives with respect to g are bounded.

Proof. Set

$$\alpha := g(Y, \cdot) - \bar{\partial} f_\xi.$$

The form α is $\bar{\partial}$ -closed and invariant under the circle generated by $\mathcal{K}$. By the defining equation for f_ξ , it vanishes on every $(0, 1)$ -vector tangent to a reduced sphere. Hence α has only a component in the complexified orbit direction.

Because $d\xi$ is nonzero on the end, $\bar{\partial}\xi$ spans the vertical $(0, 1)$ -cotangent line. Thus

$$\alpha = a\bar{\partial}\xi$$

for a smooth circle-invariant function a . Since both α and $\bar{\partial}\xi$ are $\bar{\partial}$ -closed,

$$\bar{\partial}a \wedge \bar{\partial}\xi = 0.$$

Consequently, for each fixed value of ξ , the function a descends to a holomorphic function on the reduced surface $\Sigma \simeq \mathbb{P}^1$. It is therefore constant on that surface, so $a = a(\xi)$. Choose $c(\xi)$ with $c'(\xi) = a(\xi)$. Then $\alpha = \bar{\partial}c(\xi)$, and replacing f_ξ by

$$f_{\text{end}} := f_\xi + c(\xi)$$

proves the first assertion.

Let g_D denote the limiting round metric on the reduced sphere. The end expansions give

$$g_\xi = g_D + O'(\xi^{\delta_0}).$$

Elliptic estimates for the equation defining f_ξ therefore give $f_\xi = f_0 + O'(\xi^{\delta_0})$. Moreover, a direct decomposition in the coordinates (ξ, t, x, y) gives

$$a = 2i\eta(Y) - \partial_\xi f_\xi.$$

The differentiated end expansions and the fact that Y extends holomorphically and is tangent to D imply

$$\eta(Y) = O'(\xi^{-1+\delta_0}), \quad a = O'(\xi^{-1+\delta_0}).$$

Since $c' = a$ and $\delta_0 > 0$, it follows that $c = c_0 + O'(\xi^{\delta_0})$. Finally, a bounded radial field for g is comparable to $\xi\partial_\xi$, the tangential geometry stays uniformly bounded, and f_{end} is circle-invariant. Hence

$$|\nabla_g^k f_{\text{end}}|_g \leq C_k, \quad k \geq 0.$$

□

Let κ be an S^1 -invariant cut-off function that is zero on a compact subset and one sufficiently far out on the end. Define

$$\beta := g(Y, \cdot) - \bar{\partial}(\kappa f_{\text{end}}).$$

Lemma 5.6. *The form β is $\bar{\partial}$ -closed and compactly supported. Under the natural identification*

$$\mathcal{A}|_M \simeq K_M^{-1}, \quad \mathcal{A} := -(K_{\overline{M}} + D),$$

it therefore extends by zero to a smooth $\mathcal{A}$ -valued $(2, 1)$ -form on $\overline{M}$.

Proof. Both terms in the definition of β are $\bar{\partial}$ -closed. On the region where $\kappa = 1$, Lemma 5.5 gives

$$\beta = g(Y, \cdot) - \bar{\partial}f_{\text{end}} = 0.$$

Thus β is compactly supported. Let s_D be the canonical section of $\mathcal{O}(D)$. Multiplication by s_D gives

$$K_M \otimes \mathcal{A}|_M \simeq \mathcal{O}_M,$$

and hence identifies the scalar $(0, 1)$ -form β with an $\mathcal{A}$ -valued $(2, 1)$ -form $\beta_{\mathcal{A}}$. Since β vanishes near D , so does $\beta_{\mathcal{A}}$, and it extends smoothly by zero. The same argument applies in an orbifold chart. □

Proposition 5.7. *There is a smooth complex-valued function χ on M such that*

$$\bar{\partial}\chi = g(Y, \cdot).$$

It may be chosen invariant under the circle generated by $\mathcal{K}$, and it satisfies

$$(83) \quad |\nabla_g^k \chi|_g \leq C_k, \quad k \geq 0.$$

Proof. In the quotient cases, regard $\overline{M}$ as a compact Kähler orbifold. The orbifold line bundle $\mathcal{A} = -(K_{\overline{M}} + D)$ is positive by Theorem 4.6 and its quotient-case analogue. Orbifold Kodaira–Nakano vanishing [3, Theorem 1] gives

$$H^1(\overline{M}, K_{\overline{M}} \otimes \mathcal{A}) = 0.$$

Here and below, cohomology is understood in the orbifold sense in the quotient cases. Thus the global $\bar{\partial}$ -problem for smooth $\mathcal{A}$ -valued $(2, 1)$ -forms is solvable. Applying this to $\beta_{\mathcal{A}}$ from Lemma 5.6, we obtain a smooth $\mathcal{A}$ -valued $(2, 0)$ -form $u_{\mathcal{A}}$ such that

$$\bar{\partial}u_{\mathcal{A}} = \beta_{\mathcal{A}}.$$

Multiplication by the holomorphic section s_D turns $u_{\mathcal{A}}$ into a scalar function u on M and gives $\bar{\partial}u = \beta$. Hence

$$\chi := \kappa f_{\text{end}} + u$$

has the required potential equation.²

All data in the equation are S^1 -invariant, so averaging χ over the circle preserves the equation.

Finally, $K_{\overline{M}} \otimes \mathcal{A} = \mathcal{O}(-D)$. Thus the scalar $u = s_D u_{\mathcal{A}}$ extends smoothly across D and vanishes there. Locally, if q is the defining function constructed in Section 4.2, then $u = q\tilde{u}$ with $\tilde{u}$ smooth. Recall the exact relation

$$q = e^{-c\kappa(\zeta+i\varphi)-F},$$

where F is smooth in the base variables, while the normal part of g is uniformly comparable to $\zeta^{-2}(d\zeta^2 + d\varphi^2)$. Consequently, repeated derivatives of u in the bounded normal fields $\zeta\partial_{\zeta}$ and $\zeta\partial_{\varphi}$, and in fields tangent to D , remain bounded. Combining these bounds with Lemma 5.5 proves (83). $\square$

5.2. Estimates on the potential function. The construction above gives all estimates needed below. We isolate the conclusion for later reference.

Lemma 5.8. *The potential χ can be chosen $\mathcal{K}$ -invariant, and it obeys (83).*

Proof. Circle averaging gives invariance, and the estimates were proved in Proposition 5.7. $\square$

5.3. Calabi’s argument. We now adapt Calabi’s integration-by-parts argument [14] to the finite-volume Kähler metric g . Define

$$\mathfrak{L}f := \bar{\partial}\bar{\partial}^{\#}f, \quad \mathfrak{D} := \mathfrak{L}^*\mathfrak{L},$$

Define the conjugate operator by

$$\overline{\mathfrak{L}}f := \overline{\mathfrak{L}f}, \quad \overline{\mathfrak{D}}f := \overline{\mathfrak{D}f}.$$

The formal adjoint is taken with respect to g . Thus $\mathfrak{L}f = 0$ precisely when $\bar{\partial}^{\#}f$ is holomorphic. In particular,

$$\mathfrak{L}s_g = 0, \quad \mathfrak{L}\chi = 0, \quad \mathfrak{D}\chi = 0.$$

Proposition 5.9. *The operators $\mathfrak{D}$ and $\overline{\mathfrak{D}}$ commute.*

²This construction of the potential is simpler than the previous L^2 -existence argument in this paper and is suggested by ChatGPT. The same idea was already mentioned to the author by Claude LeBrun in 2024.

Proof. For an extremal Kähler metric this is the commutation identity in the corollary to [14, Proposition 3.1]. $\square$

Lemma 5.10. *In local holomorphic coordinates,*

$$(\bar{\mathfrak{D}} - \mathfrak{D})f = (s_g)^j f_j - (s_g)_j f^j = g^{j\bar{k}} \partial_{\bar{k}} s_g \partial_j f - g^{j\bar{k}} \partial_{\bar{k}} f \partial_j s_g.$$

Moreover, $\mathfrak{L}f$ is the $(0,1)$ -covariant derivative of $\bar{\partial}^\# f$.

Proof. The first formula is equation (2.3') of [14]; the second is the definition of $\mathfrak{L}$. $\square$

We shall use the following cut-off twice. Let $\varphi_\varepsilon = \varphi_\varepsilon(\xi)$ equal 0 for $\xi \leq \varepsilon/2$ and 1 for $\xi \geq \varepsilon$. It can be chosen so that

$$(84) \quad |\nabla \varphi_\varepsilon|_g + |\nabla^2 \varphi_\varepsilon|_g \leq C, \quad \text{Vol}_g\{\varepsilon/2 < \xi < \varepsilon\} \leq C\varepsilon.$$

These estimates follow directly from $W \sim \xi^{-2}$ and $We^v = O(1)$.

Lemma 5.11. *Suppose that f and its derivatives through order four are bounded and that $\mathfrak{D}f = 0$. Then $\mathfrak{L}f = 0$.*

Proof. Since $\varphi_\varepsilon f$ is compactly supported, formal adjointness gives

$$\int_M \langle \mathfrak{L}(\varphi_\varepsilon f), \mathfrak{L}f \rangle d\text{vol}_g = \int_M \varphi_\varepsilon f \bar{\mathfrak{D}}f d\text{vol}_g = 0.$$

The difference between the left-hand side and $\int_M \varphi_\varepsilon |\mathfrak{L}f|^2 d\text{vol}_g$ is supported in $\{\varepsilon/2 < \xi < \varepsilon\}$. The product rule, the derivative bounds on f , and (84) bound its absolute value by $C\varepsilon$. Letting $\varepsilon \rightarrow 0$ and using monotone convergence yields $\mathfrak{L}f = 0$. $\square$

Proposition 5.12. *If $\mathfrak{D}\chi = \bar{\mathfrak{D}}\chi = 0$, then*

$$Y_{\text{real}} = \mathfrak{k} + J\mathfrak{w}$$

for some $\mathfrak{k}, \mathfrak{w} \in \mathfrak{hamiltoniso}_{\mathcal{K}}(M, g)$.

Proof. Write $\chi = \chi_R + i\chi_I$. Since $\bar{\mathfrak{D}}\chi = 0$ is equivalent to $\mathfrak{D}\bar{\chi} = 0$, the two equations imply

$$\mathfrak{D}\chi_R = \mathfrak{D}\chi_I = 0.$$

Lemma 5.11, applied using (83), gives

$$\mathfrak{L}\chi_R = \mathfrak{L}\chi_I = 0.$$

Thus $Z_R := \bar{\partial}^\# \chi_R$ and $Z_I := \bar{\partial}^\# \chi_I$ are holomorphic.

For a real function f , write $\bar{\partial}^\# f = X - iJX$. Comparing real parts in

$$g(X - iJX, \cdot) = \bar{\partial}f = \frac{1}{2}(df + iJdf)$$

gives $g(X, \cdot) = \frac{1}{2}df$. Hence $-JX = \text{Im } \bar{\partial}^\# f$ is Hamiltonian. A Hamiltonian real holomorphic field on a Kähler manifold is Killing. Therefore

$$\mathfrak{w} := \text{Im } Z_R, \quad \mathfrak{k} := -\text{Im } Z_I$$

are Hamiltonian Killing fields. They commute with $\mathcal{K}$ because χ was chosen $\mathcal{K}$ -invariant. They also extend holomorphically across D . Indeed, (83) bounds their g -norms. In a compactification chart (q, z) the Poincaré asymptotics give

$$g_{q\bar{q}} \sim \frac{1}{|q|^2 |\log |q||^2}, \quad g_{z\bar{z}} \sim 1.$$

Thus the vertical coefficient of either holomorphic field is $O(|q| \log |q|)$ and its horizontal coefficient is bounded. Both coefficients have removable singularities, and the vertical one vanishes on D . Finally,

$$Y_{\text{real}} = \text{Re}(Z_R + iZ_I) = J \text{Im } Z_R - \text{Im } Z_I = \mathfrak{k} + J\mathfrak{w}.$$

□

It remains to show that $\overline{\mathfrak{D}}\chi = 0$. Set

$$F := \overline{\mathfrak{D}}\chi.$$

Because $\mathfrak{D}\chi = 0$, Lemma 5.10 gives

$$F = (s_g)^j \chi_j - (s_g)_j \chi^j.$$

Using $\mathfrak{L}s_g = \mathfrak{L}\chi = 0$, differentiation of this identity yields

$$\bar{\partial}^\# F = [\bar{\partial}^\# s_g, \bar{\partial}^\# \chi].$$

The bracket vanishes by the choice of Y . Hence $\bar{\partial}F = 0$, so F is holomorphic.

The derivative estimate (83) and the local formula for $\overline{\mathfrak{D}}$ show that F is bounded. It therefore extends holomorphically across D in every compactification chart. Since $\overline{M}$ is compact, $F \equiv c$ for some constant c . To determine c , use φ_ε above and the formal adjoint identity for the conjugate operator:

$$\int_M \varphi_\varepsilon F d\text{vol}_g = \int_M \langle \overline{\mathfrak{L}}\varphi_\varepsilon, \overline{\mathfrak{L}}\chi \rangle d\text{vol}_g.$$

The right-hand side is supported in the transition shell and is $O(\varepsilon)$ by (84) and (83). The left-hand side tends to $c \text{Vol}_g(M)$. Thus $c = 0$.

We have proved $\overline{\mathfrak{D}}\chi = 0$. Proposition 5.12 now proves Theorem 5.1.

Theorem 5.13. *Every Hermitian non-Kähler gravitational instanton asymptotic to ALF-A, cyclic quotient of ALF-A, or rational AF, whose extremal field has closed orbits, is holomorphically and isometrically toric.*

It remains to compare our decay convention with the one used by Biquard–Gauduchon [9]. Their definition is stated with rate one, whereas Section 3 gives every positive rate strictly smaller than one.

Lemma 5.14. *Let (M, h, J) be a toric Hermitian non-Kähler gravitational instanton with one of the ALF/AF expansions obtained in Theorem 3.23. Then the classification arguments of [9, Sections 4–8] apply for every decay exponent $\delta_0 > 0$.*

Proof. The torus found above preserves the complex structure and the asymptotic data, and its Lie algebra contains the collapsing field. Thus the compatibility condition in [9, Definition 1.2] is satisfied. Choose the first toric Hamiltonian coordinate to be $x_1 = s_g$, corresponding to the canonical field $\mathcal{K}$; the angular basis need not be integral. Theorem 3.25 gives $x_1 = \rho^{-1} + O'(\rho^{-1-\delta_0})$. Since $g = x_1^2 h$, substitution in the toric action–angle expressions gives the analogues of formulas (62)–(64) in [9, Section 4]: the Poincaré leading terms are the same, while their $O(x_1)$ remainders become $O'(x_1^{\delta_0})$ in the same weighted frame.

This replacement does not affect their global uniformization. The analogue of their formula (64) still has a multiplicative $1 + o(1)$ error, so the Tod coordinate $z + i\rho$ has the same linear growth at infinity. The proof of [9, Proposition 4.1] uses precisely this growth, together with Schwarz reflection, to show that the coordinate is affine. Moreover, the O' -control preserves the strict derivative signs of the model at infinity. The maximum-principle argument in their convexity subsection therefore still gives positivity and convexity of the boundary function.

The later sections of [9] use this uniformization, the convex piecewise-linear boundary data, and the exact Delzant smoothness conditions, but no further rate-one estimate. Their reconstruction and classification therefore apply. The polynomial hypothesis $\delta_0 > 0$ is used precisely in making all the weighted errors tend to zero with their derivatives. This is the only modification of their rate-one proof. □

In Appendix A we rule out the noncyclic quotient of ALF-A situation. Together with Theorem 5.13 and Lemma 5.14, we have obtained the classification of all Hermitian non-Kähler ALF/AF gravitational instantons. The resulting list is as follows.

Theorem 5.15. *Any Hermitian non-Kähler ALF/AF gravitational instanton must be one of the following:*

- the Kerr metrics;
- the Chen-Teo metrics;
- the reversed Taub-NUT metric, i.e. the Taub-NUT metric with orientation opposite to the hyperkähler orientation;
- the Taub-bolt metric with both orientations.

Combining with the classification of ends Theorem 3.23, we conclude the following classification of Hermitian non-Kähler gravitational instantons.

Theorem 5.16. *A Hermitian non-Kähler gravitational instanton belongs to one of the following classes:*

- ALE with $\Upsilon \subset U(2)$, which corresponds to a special Bach-flat Kähler orbifold in the terminology of [50].
- ALF-A, in which case it must be the reversed Taub-NUT metric or the Taub-bolt metric with both orientations.
- AF, in which case it must be the Kerr metrics or the Chen-Teo metrics.

APPENDIX A. THE THREE-POINT SPHERICAL CASE

This appendix classifies the compactified complex surfaces that can occur in the noncyclic quotient of ALF-A case. There are three orbifold points in the added divisor, and it is a spherical orbifold. We first obtain a finite list of possible compactified pairs. The final subsection then shows that none of the pairs in the list supports the conformal Kähler metric of a gravitational instanton.

Suppose that a Hermitian non-Kähler gravitational instanton (M, h) is asymptotic to a noncyclic quotient of an ALF-A model. Its natural compactification $(\overline{M}, D)$ has three quotient points on $D \simeq \mathbb{P}^1$, and their orders form a spherical triple.

Throughout the appendix, set

$$A := -(K_{\overline{M}} + D).$$

Here $\overline{M} \setminus D$ is smooth, A is ample, and the $\mathbb{C}^*$ -action generated by the extremal field extends to $\overline{M}$ and fixes D pointwise. Let

$$l : \widetilde{M} \longrightarrow \overline{M}$$

be the minimal resolution, write $\widetilde{D}$ for the proper transform of D , and write E_i for the exceptional chain over the i -th orbifold point.

All three quotient points are cyclic. To see this locally, choose a coordinate chart at a point of D in which the lift of D is a complex line. Each orbifold group is a finite subgroup of $U(2)$. It preserves the tangent line of D and therefore also its orthogonal complement, so it lies in $U(1) \times U(1)$. This forces the group to be cyclic. Choosing its generator so that the tangent character along D is ζ , the germ is

$$\frac{1}{m}(1, a) := \mathbb{C}^2/\mu_m, \quad \zeta \cdot (z, w) = (\zeta z, \zeta^a w), \quad (a, m) = 1,$$

with the lifted divisor $\{w = 0\}$. Here m is the order and a is the residue.

Write (m_1, m_2, m_3) for the orders of these three local orbifold cyclic groups, up to permutation. Positivity of the orbifold Euler characteristic

$$\delta := \chi_{\text{orb}}(D) = -1 + \frac{1}{m_1} + \frac{1}{m_2} + \frac{1}{m_3} > 0$$

gives exactly the following spherical triples:

spherical type	(m_1, m_2, m_3)	local surface germs
D_n (dihedral)	$(2, 2, n), n \geq 2$	$\frac{1}{2}(1, 1), \frac{1}{2}(1, 1), \frac{1}{n}(1, a_3)$
T (tetrahedral)	$(2, 3, 3)$	$\frac{1}{2}(1, 1), \frac{1}{3}(1, a_2), \frac{1}{3}(1, a_3)$
O (octahedral)	$(2, 3, 4)$	$\frac{1}{2}(1, 1), \frac{1}{3}(1, a_2), \frac{1}{4}(1, a_3)$
I (icosahedral)	$(2, 3, 5)$	$\frac{1}{2}(1, 1), \frac{1}{3}(1, a_2), \frac{1}{5}(1, a_3)$

Each residue is coprime to its order. Thus a notation such as $(2, 3, 3)$ below records the three orders. The local surface groups are the three cyclic groups displayed in the last column.

The *log-anticanonical degree of the pair* is $A^2 > 0$. Put $L := l^*A$. For an irreducible curve $C \subset \widetilde{M}$, we also call $\deg_A(C) := L \cdot C$ its *log-anticanonical degree*. It is zero on every component of the exceptional chains E_i and is strictly positive on every other curve, because A is ample.

A.1. The Hitchin-Thorpe type identity. We first isolate the metric identity used below. It is stronger than the usual Hitchin–Thorpe inequality because conformal Kähler geometry determines the self-dual Weyl energy.

Let (M, h, J) be a Hermitian non-Kähler Ricci-flat four-manifold with one noncyclic quotient of ALF- A end. Its complex structure admits a natural orbifold compactification $(\overline{M}, D)$ for which A is ample and the singular Hermitian metric of Propositions 4.4–4.5 has continuous plurisubharmonic local potentials. Give M the complex orientation, and let

$$\tau(M) := \text{sign Im}(H_c^2(M; \mathbb{R}) \longrightarrow H^2(M; \mathbb{R})).$$

We use $\chi(M)$ for the Euler characteristic of a sufficiently large compact core. For a large compact truncation M_R , orient $Y_R = \partial M_R$ by the outward-normal convention. One has the following signature formula (see [30])

$$\tau(M_R) = \frac{1}{12\pi^2} \int_{M_R} (|W_h^+|^2 - |W_h^-|^2) d\text{vol}_h - \frac{1}{2} \eta(Y_R) + \mathcal{T}_R,$$

where η is the eta invariant and $\mathcal{T}_R$ is a boundary term. The boundary term tends to zero below, and we define

$$\Theta_M := \frac{1}{2} \lim_{R \rightarrow \infty} \eta(Y_R).$$

With the outward-boundary orientation fixed in this definition, the limiting signature formula is

$$(85) \quad \tau(M) + \Theta_M = \frac{1}{12\pi^2} \int_M (|W_h^+|^2 - |W_h^-|^2) d\text{vol}_h.$$

Proposition A.1. *We have*

$$(86) \quad A^2 = 2\chi(M) + 3(\tau(M) + \Theta_M).$$

Proof. Put $g = s^2 h$, where $s = s_g > 0$ is the scalar curvature of the Kähler metric g . Let Ω be the curvature of the Hermitian metric $e^{-2\log s} g_{-K}$ on $A|_M$, normalized so that $[\Omega/(2\pi)] = c_1(A)$. LeBrun’s curvature computation [45], already used in (76), gives

$$\Omega(\cdot, J\cdot) = \frac{s}{6} g + s^{-2} (|ds|^2 g - ds \otimes ds - Jds \otimes Jds).$$

Its two Hermitian eigenvalues are

$$\frac{s}{6}, \quad \frac{s}{6} + s^{-2} |ds|^2,$$

and hence

$$(87) \quad \Omega \wedge \Omega = 2 \left(\frac{s^2}{36} + \frac{|ds|^2}{6s} \right) d\text{vol}_g.$$

The scalar-flatness of $h = s^{-2}g$ is equivalent to

$$6s\Delta_g s - 12|ds|^2 + s^3 = 0.$$

Substitution in (87) gives

$$(88) \quad \Omega \wedge \Omega = \left(\frac{s^2}{12} + \frac{\Delta_g s}{6} \right) d\text{vol}_g.$$

Since the local potential of the Hermitian metric is continuous, it follows that

$$\int_M \Omega \wedge \Omega = 4\pi^2 A^2.$$

The ALF expansion gives

$$\text{Area}_g(\{\rho = R\}) = O(R^{-1}), \quad |\partial_n s| = O(R^{-1}),$$

so the boundary flux of ds is $O(R^{-2})$. Integrating (88) therefore yields

$$(89) \quad \int_M s^2 d\text{vol}_g = 48\pi^2 A^2.$$

On a Kähler surface, $|W_g^+|^2 = s^2/24$. The L^2 -norm of the Weyl tensor is conformally invariant in dimension four, and thus

$$(90) \quad \int_M |W_h^+|^2 d\text{vol}_h = 2\pi^2 A^2.$$

For our Ricci-flat ALF metric, since we have the metric expansion, the resulting Gauss–Bonnet and signature formulas are

$$(91) \quad \chi(M) = \frac{1}{8\pi^2} \int_M (|W_h^+|^2 + |W_h^-|^2) d\text{vol}_h,$$

$$(92) \quad \tau(M) + \Theta_M = \frac{1}{12\pi^2} \int_M (|W_h^+|^2 - |W_h^-|^2) d\text{vol}_h.$$

See for example [30].

Finally, substitute (90) in (91)–(92) and eliminate the anti-self-dual Weyl energy. The result is (86). $\square$

A.2. Equivariant resolution graphs. Recall that m and a denote, respectively, the order and residue of the orbifold group above. When the singular point lies on D , we take $D = \{w = 0\}$ and order the exceptional curves in the minimal resolution from $\tilde{D}$ outward. The integer a below is chosen relative to this ordering. The local representation lies in $\text{SU}(2)$ exactly when $a = m - 1$; the type $\frac{1}{m}(1, 1)$ is genuinely $\text{U}(2)$ unless $m = 2$.

We use the negative continued-fraction convention

$$\frac{m}{a} = [b_1, \dots, b_r] := b_1 - \frac{1}{b_2 - \frac{1}{\ddots - 1/b_r}}.$$

Thus the cyclic singularity $\frac{1}{m}(1, a)$ has exceptional chain $[-b_1, \dots, -b_r]$. In particular,

$$\frac{1}{m}(1, 1) \longleftrightarrow [-m], \quad \frac{1}{m}(1, m-1) \longleftrightarrow \underbrace{[-2, \dots, -2]}_{m-1 \text{ curves}}.$$

For a point $p_i \in D$, the exceptional set $E_i = l^{-1}(p_i)$ is a connected chain of negative curves. We call this ordered chain the *arm* at p_i . The arm contains only components contracted by l , and its *order* and *residue* are those of the cyclic point obtained by contracting it; equivalently they are the

integers (m, a) in $m/a = [b_1, \dots, b_r]$. We call an arm *rooted* if all its components are created by blow-ups over its point on the fixed curve, and *inherited* if it contains the proper transform of a curve already on the starting surface. A *one-sided* sequence first blows up the intersection of the fixed curve with a transverse orbit closure and then always blows up the outer endpoint, between the current connector and that orbit closure.

The last exceptional curve in such a construction, lying between the arm and the proper transform of the orbit closure, is the (-1) -*connector*; it is not a component of the arm. A (-1) -*separator* is instead a (-1) -curve inserted at an inner node, separating two nonempty pieces that one would need to retain in the same arm. A *nonboundary curve* means a curve not contained in $\tilde{D} \cup \text{Exc}(l)$.

We next explain the equivariant minimal model reduction. By the same proof of Proposition 4.15, $\tilde{M}$ is rational, so its smooth minimal model is $\mathbb{P}^2$ or a Hirzebruch surface H_n with $n = 0$ or $n \geq 2$. Every (-1) -curve is $\mathbb{C}^*$ -invariant and can be contracted equivariantly.

Choose an equivariant birational morphism

$$\sigma : \tilde{M} \longrightarrow S$$

to a smooth minimal rational surface and factor its inverse into blow-ups at fixed points. Such a morphism gives one of the following two kinds of presentation.

If σ does not contract $\tilde{D}$, its image is a pointwise fixed curve and the pair $(S, \sigma(\tilde{D}))$ is one of

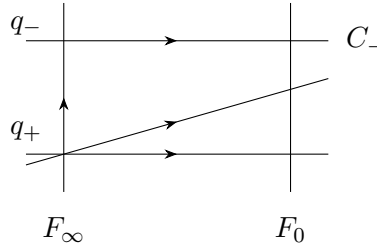
$$(93) \quad (\mathbb{P}^2, \text{ a fixed line}), \quad (H_n, \text{ a fixed section}), \quad (H_n, \text{ a fixed ruling fibre}),$$

with $n \geq 0, n \neq 1$. We call this a *D-preserving presentation*.

If σ contracts $\tilde{D}$, then its image is a fixed point of the action on S ; we call this a *fixed-point presentation*.

These alternatives exhaust minimal model presentations, but need not be disjoint at the level of pairs: the same pair can admit one presentation of each kind.

In the fixed ruling fibre model with $n > 0$, write $\pi : H_n \rightarrow \mathbb{P}^1$, $F_0 = \pi^{-1}(0)$, and $F_\infty = \pi^{-1}(\infty)$; let F denote the ruling class, and write $C_- := C_\infty$ for the negative section. We use standard coordinates in which the action is $t \cdot (z, w) = (tz, w)$ near F_0 . The fibre F_∞ is invariant as a set, rather than fixed pointwise, has square zero, and contains exactly two fixed points: the negative section C_- joins F_0 to $q_- \in F_\infty$, while every closure of a nonspecial one-dimensional orbit is a section in the class $C_- + nF$, hence has square n , and ends at the other point $q_+ \in F_\infty$. We call these the *positive orbit sections*.



The following elementary observation is the main restriction on arms created after a pointwise fixed curve is already present. It does not apply to the two chains that may be created before $\tilde{D}$ appears in the fixed-point case.

Lemma A.2. *An exceptional arm created by blow-ups over a point of an already existing fixed curve is an A_{m-1} -chain of (-2) -curves. In the D -preserving case, at most one arm can contain a curve inherited from the surface in (93); if it does, then the minimal model can be chosen to be H_n with fixed fibre, and that arm is the single negative section $[-n]$ of H_n .*

Proof. First note that a rational curve C disjoint from $\tilde{D} \cup \text{Exc}(l)$ cannot have $C^2 \leq -2$. Indeed, l is an isomorphism near C , and adjunction gives

$$l^*A \cdot C = -K_{\tilde{M}} \cdot C = 2 - 2p_a(C) + C^2 \leq 2 + C^2 \leq 0,$$

contrary to the ampleness of A .

Let S be the orbit closure through a point p of the fixed curve. After the first blow-up over p , there is a (-1) -curve between the proper transforms of the fixed curve and S . Suppose inductively that the local configuration is

$$\tilde{D} - E_1(-2) - \cdots - E_j(-2) - F(-1) - S.$$

Blowing up the outer fixed point $F \cap S$ changes F into the next (-2) -component and creates a new (-1) -connector. In contrast, a blow-up at an inner node inserts a (-1) -separator between two nonempty parts of the intended exceptional set. If this separator belongs to the exceptional chain of l , then that chain is not minimal, contradicting that $\tilde{M}$ is the minimal resolution of $\overline{M}$. If it does not, connectedness of the exceptional chain prevents the part beyond the separator from belonging to it; the first curve in that part has square at most -2 and is disjoint from $\tilde{D} \cup \text{Exc}(l)$, contradicting the preceding adjunction estimate. Further inner blow-ups leave a new (-1) -separator in the same interval. Hence every center is successively the outer one, and the resulting arm is

$$[-2, \dots, -2].$$

If it has j components, its continued fraction is $(j+1)/j$; hence its order is $m = j+1$ and it is the A_{m-1} -arm asserted in the lemma.

It remains to consider an arm containing the proper transform of an invariant curve C already present on the starting surface. On $\mathbb{P}^2$ such an orbit closure has $C^2 = 1$; for a fixed section of H_n it is a ruling fibre with $C^2 = 0$; and, when the fixed curve is a ruling fibre, every inherited orbit section other than the negative section has $C^2 = n \geq 0$. To make any such C a component of a minimal resolution arm one must lower C^2 to at most -2 . Along the path from the fixed curve to C , or at the opposite endpoint of C , the requisite repeated blow-ups necessarily leave a (-1) -separator. If it is retained in $\text{Exc}(l)$, the resolution is not minimal; if it is omitted, connectedness excludes the portion beyond it, whose first curve is then disjoint from $\tilde{D} \cup \text{Exc}(l)$ and has square at most -2 . Both alternatives are impossible. Thus none of these nonnegative inherited curves occurs in an arm.

In the fixed-fibre model the only negative invariant curve meeting F_0 is the negative section $C_-^2 = -n$. Blowing up $F_0 \cap C_-$ inserts a (-1) -separator. Blowing up $q_- = C_- \cap F_\infty$ and then contracting the proper transform F_∞ is the elementary transformation $H_n \rightarrow H_{n+1}$. By such elementary transformations, there is a minimal model H_n with fixed fibre where there is no further blow-up at q_- . After choosing this normalized model, no blow-up at q_- is present, so its negative section itself is the single arm $[-n]$. Since it is the only negative invariant curve meeting F_0 , it is the unique inherited arm. $\square$

A.2.1. D -preserving models. For later convenience, we include the case of $(H_1, \text{ a fixed ruling fibre})$ in the following proposition.

Proposition A.3. *Assume that D contains three cyclic quotient points of orders at least two and that their orbifold Euler characteristic is positive. Then the following are the only possibilities before imposing the Hitchin-Thorpe identity.*

- (1) *If the starting pair is $(\mathbb{P}^2, \text{ a fixed line})$, all three arms have order two.*
- (2) *A fixed-section model on H_n is impossible.*

- (3) In the fixed-fibre model on H_n one has $n \geq 2$. The negative section is the inherited arm $[-n]$ and the other two arms are A_{r-1} and A_{s-1} , where

$$2 \leq r, s \leq n + 1.$$

There is at most one further blow-up at the remaining isolated fixed point q_+ . In the cases that survive below, this blow-up is allowed.

Proof. In the $\mathbb{P}^2$ case, all three arms are rooted by Lemma A.2. An order- m rooted arm requires m successive one-sided blow-ups. The surviving orbit line then has square $1 - m$ and log-anticanonical degree $3 - m$. It lies outside the boundary, so ampleness forces $m = 2$. This gives the triple $(2, 2, 2)$. The three surviving orbit lines meet at the isolated fixed point outside the fixed line and each has log-anticanonical degree one. Blowing up that point would make all three degrees zero, so no additional blow-up is possible.

If the fixed curve is a section of H_n , a ruling fibre starts with square zero. All of the arms are rooted by Lemma A.2. Even a rooted A_1 -arm requires two blow-ups and leaves a nonboundary fibre of square -2 and log-anticanonical degree zero. This excludes the fixed-section case.

Now take the fixed curve to be a ruling fibre F_0 of H_n . Since

$$-(K_{H_n} + F_0) \cdot C_- = 1 - n,$$

the negative section must be part of the exceptional set when $n \geq 2$. By Lemma A.2, it is the unique inherited arm. The other two arms are rooted at distinct points of F_0 . Their first blow-ups lower the square of F_0 twice. There can be no blow-up over F_0 besides these two points: its exceptional curve E would be a nonboundary (-1) -curve meeting $\tilde{D}$, and

$$l^*A \cdot E = -K_{\tilde{M}} \cdot E - \tilde{D} \cdot E - \Delta \cdot E \leq 1 - 1 = 0,$$

contrary to the ampleness of A . Hence the only blow-ups centered on F_0 are the first blow-ups creating the two rooted arms. Since $F_0^2 = 0$, they give $\tilde{D}^2 = -2$.

An order- r rooted arm is produced by r blow-ups along its positive orbit section. That section starts with square n and ends with square $n - r$; equivalently, its log-anticanonical degree is $n + 2 - r$. Hence $r \leq n + 1$, and similarly $s \leq n + 1$. The cases $n = 0$ and $n = 1$ give no new cases. Indeed, for $n = 0$ even a rooted A_1 -arm leaves its orbit section with square -2 and degree zero. For $n = 1$, the negative section is a (-1) -curve, contracting it gives the $\mathbb{P}^2$ fixed-line presentation above.

All positive orbit sections meet at $q_+ \in F_\infty$. We now compute the log-anticanonical degree of F_∞ . Since C_- is the contracted $[-n]$ -arm, the log pullback differs from $-(K_{H_n} + F_0)$ by $-(1 - 1/n)C_-$. Thus, up to the rooted-arm corrections over F_0 , which are disjoint from F_∞ , l^*A is

$$\left(1 + \frac{1}{n}\right)C_- + (n + 1)F.$$

Indeed, $-(K_{H_n} + F_0) = 2C_- + (n + 1)F$, and requiring zero intersection with C_- determines the correction coefficient to be $1 - 1/n$. Since $C_- \cdot F_\infty = 1$ and $F \cdot F_\infty = 0$, the required degree is $1 + 1/n$. Let G be the exceptional curve of a blow-up at q_+ and let L' be the new log-anticanonical class. After this blow-up we have

$$L' \cdot F'_\infty = \frac{1}{n}, \quad L' \cdot G = 1.$$

Since $n \geq 2$, G is not pointwise fixed and has exactly two fixed points. Any further blow-up at these two points is impossible: the proper transform of G then has degree zero, and at the F'_∞ -endpoint that fibre would also have degree $1/n - 1 < 0$. Thus at most one blow-up at q_+ is possible. After the first blow-up at q_+ , the two positive-section degrees become $n + 1 - r$ and $n + 1 - s$. Hence this optional blow-up also requires $r, s \leq n$. $\square$

A.2.2. *Fixed-point models.* On the minimal model, the $\mathbb{C}^*$ -action is contained in a standard two-dimensional torus. Under this toric action, the minimal model is represented by a fan. Every isolated fixed point corresponds to a cone between two adjacent rays in the standard fan, and its two invariant local axes extend to the adjacent rays of that fan.

An isolated fixed point is a *source* if nearby points converge to it as $t \rightarrow 0$, a *sink* if they converge to it as $t \rightarrow \infty$, and a *saddle* otherwise. In equivariant local coordinates the action is

$$(94) \quad t \cdot (x, y) = (t^p x, t^q y), \quad (p, q) = 1.$$

We denote the two invariant axes in the minimal model as $U = \{x = 0\}$ and $V = \{y = 0\}$, which are two $\mathbb{C}^*$ -orbit closures. At a source both weights are positive, at a sink both are negative, and at a saddle they have opposite signs. A pointwise fixed exceptional curve can be extracted only from a source or sink, as we will explain below. Replacing t by t^{-1} , we henceforth take $p, q > 0$.

We denote the two rays in the fan corresponding to U and V as $u = (1, 0)$ and $v = (0, 1)$. There is a minimal sequence of blow-ups at the source to make the fixed curve appear. Denote the primitive ray

$$(95) \quad \lambda = pu + qv.$$

A single blow-up at the source changes the weights in its two charts to

$$(96) \quad (p, q - p), \quad (p - q, q).$$

To obtain the fixed curve, we need to reduce one of the two weights to zero. Thus the required blow-ups are precisely the subtractive Euclidean algorithm. Equivalently, the minimal sequence of blow-ups corresponds to the unique minimal regular subdivision of the cone $\langle u, v \rangle$ containing λ . Write its rays in the two ordered strings

$$u = u_0, u_1, \dots, u_r = \lambda, \quad \lambda = v_0, v_1, \dots, v_s = v,$$

with determinant one for every adjacent pair. The divisors associated with $u_1, \dots, u_{r-1}$ and $v_1, \dots, v_{s-1}$ form two possibly empty chains. For each intermediate ray r_i , its two neighbors satisfy $r_{i-1} + r_{i+1} = b_i r_i$ with $b_i \geq 2$, so the corresponding divisor has square $-b_i$. These are exactly the two Hirzebruch–Jung strings for the cones on the two sides of λ . The ray λ is inserted last, as the sum of its adjacent rays, and its divisor is therefore a pointwise fixed (-1) -curve. It is the first appearance of the curve that becomes $\tilde{D}$.

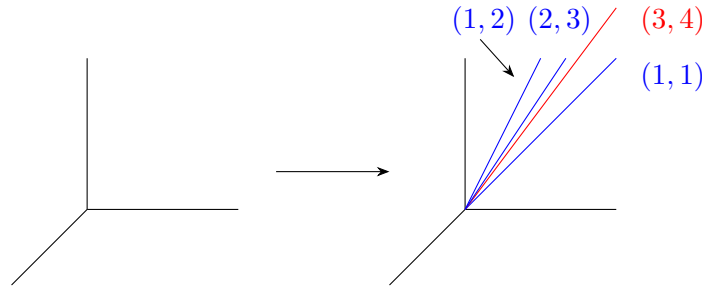


FIGURE 5. An example of source extraction with weights $(3, 4)$ on $\mathbb{P}^2$. Here $\lambda = (3, 4)$.

Choose a blow-up factorization from the minimal model to $\tilde{M}$. Blow-ups with centers disjoint from the above process commute with it, so we arrange that the preceding minimal sequence is performed first, stopping when the fixed divisor appears. We call this process the *source extraction*. All exceptional curves that appear in this process, except the fixed curve, have self-intersection at most -2 . Until it is complete, the entire local configuration is toric. All later blow-ups are treated only after $\tilde{D}$ has appeared.

Put

$$B = \tilde{D} \cup E_1 \cup E_2 \cup E_3, \quad L = l^*A.$$

The two chains between λ and the first rays outside B will be called the *pre-D chains*; after adjoining U or V when that axis is contracted by l , they are the two *pre-D arms*. A pre-D chain may be empty. We say that an original axis is *absorbed* when its proper transform is the outermost component of the corresponding pre-D arm; otherwise that axis is a nonboundary curve outside B .

We first make the following normalization. Suppose that the model is H_N with a $\mathbb{C}^*$ -action. Here we allow $N = 1$, so that the discussion includes the case of $\mathbb{P}^2$ with fixed point, where after one blow-up at the fixed point it becomes H_1 . One can blow up a fixed point of a ruling fibre and contract the proper transform of that fibre. When $N > 0$, if the point lies on C_- this changes H_N to H_{N+1} ; if it does not, it changes H_N to H_{N-1} . This operation changes H_0 to H_1 . On tangent weights this is one of the subtractions in (96). Now suppose the source in H_N lies on a positive-section or is in H_0 . There has to be a blow-up at the source, since such blow-up is contained in the source extraction process. One can then perform the above elementary transformation: after the blow-up at the source, we contract the proper transform of the ruling fibre. We then iterate these elementary transformations when the source is on positive-sections, and eventually there are the following two possibilities:

- Either we end with a model H_N , where the source becomes a fixed curve. This is the situation of Proposition A.3.
- Or we end with a model H_N with $N \neq 0$, where the source lies on the negative-section.

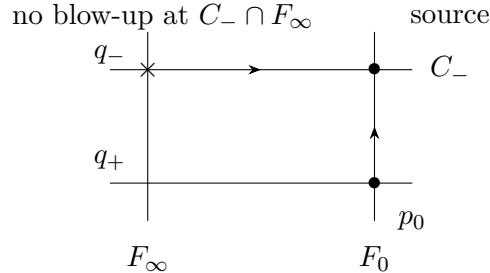
Hence in the following we may suppose the model is an H_N with the source contained in the negative-section. Here we allow the possibility $N = 1$.

We may further normalize to remove any blow-up at q_- . Recall

$$q_- = C_- \cap F_\infty$$

is the fixed point of C_- opposite the source. If q_- is blown up, then the proper transform F'_∞ is a (-1) -curve, and contracting it gives the elementary transformation $H_N \rightarrow H_{N+1}$. If E denotes the exceptional curve of the blow-up, then its image is a ruling fibre of H_{N+1} , and the same intermediate surface is obtained from H_{N+1} by blowing up the fixed point where this new ruling meets the positive-section. Thus this elementary transformation moves the blow-up from $C_- \cap F_\infty$ to the positive-section, disjoint from the two source axes F_0 and C_- . We may iterate this operation to remove any blow-up at q_- .

Hence in the following, we work under this normalization. That is, the model is chosen to be H_N with $N > 0$, the source is at $C_- \cap F_0$, and there is no blow-up at $C_- \cap F_\infty$. See the following configuration.



Lemma A.4. *There is a model for which the source extraction is the minimal regular subdivision above and each original axis is either outside B or absorbed as the outermost component of its pre-D arm. Every third arm, and every arm component created after $\tilde{D}$ appears, is a one-sided A-chain.*

Proof. Write the logarithmic pullback as

$$l^*(K_{\overline{M}} + D) = K_{\tilde{M}} + \tilde{D} + \Delta, \quad \Delta = \sum c_{ij} E_{ij}, \quad 0 \leq c_{ij} < 1.$$

If $C \not\subset B$ is an invariant irreducible curve, adjunction gives

$$(97) \quad L \cdot C \leq 2 + C^2 - \tilde{D} \cdot C - \Delta \cdot C.$$

Thus an invariant curve outside B with square at most -2 contradicts ampleness; a (-1) -curve outside B that meets $\tilde{D}$ also has nonpositive degree, which cannot happen.

The minimal regular subdivision containing the primitive ray λ is unique. An extra blow-up at an inner edge of either pre- D chain leaves a (-1) -separator: including it makes the cyclic resolution nonminimal, whereas omitting it separates a curve of square at most -2 from B and contradicts (97). Once $\tilde{D}$ exists, Lemma A.2 applies and gives precisely the asserted rooted A -chains.

It remains to consider the two original axes. By our normalization, the model is H_N with the source contained in the negative-section. The two axes are $U = F_0$ and $V = C_-$. Suppose that the final arm extends beyond V at its other fixed point. Since there is no blow-up at q_- , at least two more blow-ups are needed at q_+ to introduce a curve beyond V with square at most -2 . But this leaves a nonboundary curve of square at most -2 inside, which is impossible.

Suppose instead that the final arm extends beyond U . To ensure this, there is either a blow-up at q_+ or a blow-up at the fixed point on F_0 opposite to the source, p_0 . If there is no blow-up at p_0 , then at least two more blow-ups are necessary at q_+ . For the same reason above this cannot happen, so there must be blow-ups at p_0 . After a blow-up at p_0 we may contract the proper transform of F_0 , which gives a new model H_{N-1} . This elementary transformation can be repeated until there is no blow-up at p_0 , but possibly we might end with H_0 , violating the normalization we made above. There is necessarily a blow-up at the source. If eventually the minimal model is H_0 , then in the above configuration, C_- and F_0 have square zero. After the source blow-up, we may contract C_- , which gives a model H_1 . In this model H_1 , both the normalization and the statements of the lemma hold. $\square$

As in our above normalization, we allow H_1 as a model in the following. Write F_0 for a ruling fibre and C_- for the negative section, and place the source at $F_0 \cap C_-$. So the two axes are F_0 and C_- . Let u and v be their respective fan rays. The adjacent rays on their other sides are $-v$ and $-u + Nv$. For an entry of order $m > 1$ and an invertible class $c \pmod{m}$, let $\langle c \rangle_m \in \{1, \dots, m-1\}$ denote its standard representative. Then, for $\lambda = pu + qv$, the order and residue of each pre- D arm are read from the determinant and continued fraction as follows:

$$(98) \quad \begin{array}{c|cc} & \text{axis outside the arm} & \text{axis absorbed into the arm} \\ \hline F_0\text{-side} & (q, \langle -p^{-1} \rangle_q) & (p, \langle q^{-1} \rangle_p) \\ C_-\text{-side} & (p, \langle -q^{-1} \rangle_p) & (Np + q, \langle p^{-1} \rangle_{Np+q}). \end{array}$$

Each entry is (m, a) for the oriented cyclic type $\frac{1}{m}(1, a)$, equivalently for the continued fraction m/a read from $\tilde{D}$ outward. An order-one entry means that the corresponding pre- D chain is empty.

A.3. The eta invariant. For coprime integers a, m with $m > 0$, let

$$s(a, m) = \sum_{j=1}^{m-1} \left(\left(\frac{j}{m} \right) \right) \left(\left(\frac{aj}{m} \right) \right)$$

be the Dedekind sum³, where $((x)) = x - [x] - \frac{1}{2}$ for $x \notin \mathbb{Z}$ and $((x)) = 0$ for $x \in \mathbb{Z}$. We use

$$(99) \quad s(1, m) = \frac{(m-1)(m-2)}{12m}, \quad s(m-1, m) = -s(1, m).$$

³The use of Dedekind sums is suggested by ChatGPT; this formulation of eta invariants also appears in earlier literature.

Put $B = \tilde{D} \cup E_1 \cup E_2 \cup E_3$ and orient $Y = \partial N(B)$ as the boundary of a small neighborhood of B . This is a smooth Seifert three-manifold, although its base is an orbifold. If $b = \tilde{D}^2$ and the i -th arm is

$$[-b_{i1}, \dots, -b_{i\ell_i}], \quad \frac{m_i}{a_i} = [b_{i1}, \dots, b_{i\ell_i}],$$

then we write the *Seifert data* as

$$Y = Y(b; (m_1, a_1), (m_2, a_2), (m_3, a_3)).$$

Its rational degree is

$$(100) \quad d = b + \sum_{i=1}^3 \frac{a_i}{m_i}.$$

Equivalently, d is the self-intersection D^2 after the arms are contracted. These Seifert data are read from the resolution graph.

For every graph considered below, $b \leq -2$. Moreover, for each of the four spherical triples the maximum possible value of $\sum_i a_i/m_i$ is strictly less than two. Thus $d < 0$.

Choose a Seifert connection θ with $\theta(Y_S) = 1$ for the primitive fibre generator Y_S of period 2π . With our orientation,

$$\int_Y \theta \wedge d\theta = -4\pi^2 d,$$

and let γ be the round orbifold metric on the base. Since the eta invariant is unchanged by an overall rescaling, the metrics

$$g_{Y,\epsilon} = \epsilon^{-2} \pi^* \gamma + \theta^2$$

give the same adiabatic limit as $\pi^* \gamma + \epsilon^2 \theta^2$. For $d < 0$, define the adiabatic limit by

$$\eta_{\text{ad}}(Y) := \frac{1}{2} \lim_{\epsilon \rightarrow 0} \eta(Y, g_{Y,\epsilon}).$$

The eta invariant is computed by Ouyang's formula, as restated in [11, equation (18), Theorem 5.2, and equation (27)], which gives

$$(101) \quad \eta_{\text{ad}}(Y) = 1 + \frac{d}{3} + 4 \sum_{i=1}^3 s(a_i, m_i).$$

In the notation of that reference, we specialize

$$(\alpha_i, \rho_i, \beta_i) = (m_i, 1, a_i).$$

Its Rademacher–Dedekind sum is therefore our $s(a_i, m_i)$; see also Ouyang's original computation [55].

Finally, a compact core is $\tilde{M} \setminus N(B)$, whose outward boundary is $-Y$. The eta invariant changes sign when the orientation is reversed. Comparing with the definition of Θ_M in the preceding subsection therefore gives

$$(102) \quad \Theta_M = \eta_{\text{ad}}(-Y) = -\eta_{\text{ad}}(Y).$$

We now express the Hitchin–Thorpe identity using only the three local cyclic types.

Lemma A.5. *Suppose the three arms have types $\frac{1}{m_i}(1, a_i)$. Then Proposition A.1 is equivalent to*

$$(103) \quad \sum_{i=1}^3 s(a_i, m_i) = 0.$$

Proof. Write $X = \widetilde{M}$, $D_0 = \widetilde{D}$, $b = D_0^2$, and

$$\frac{m_i}{a_i} = [b_{i1}, \dots, b_{i\ell_i}], \quad \ell = \sum_i \ell_i, \quad S = \sum_i s(a_i, m_i).$$

Let Q_i be the negative intersection matrix of the i -th arm and let v_i have entries

$$(v_i)_j = (K_X + D_0) \cdot E_{ij} = b_{ij} - 2 + \delta_{j1}.$$

If $z_i = -Q_i^{-1}v_i$, then

$$l^*(K_{\overline{M}} + D) = K_X + D_0 + \sum_{i,j} (z_i)_j E_{ij}.$$

It follows from adjunction on D_0 that

$$(104) \quad A^2 = K_X^2 - 4 - b - \sum_i v_i^t Q_i^{-1} v_i.$$

Let a_i^* be the inverse of a_i modulo m_i . Direct inversion of the Hirzebruch–Jung intersection matrix gives

$$\frac{a_i}{m_i} - v_i^t Q_i^{-1} v_i = \sum_j (b_{ij} - 2) + \frac{a_i + a_i^*}{m_i}.$$

The classical continued-fraction form of Dedekind reciprocity, obtained by iterating the reciprocity, gives

$$12s(a_i, m_i) = \sum_j (b_{ij} - 3) + \frac{a_i + a_i^*}{m_i};$$

it also follows immediately by induction on the continued fraction. Using

$$d = b + \sum_i \frac{a_i}{m_i}$$

in (104), we obtain

$$(105) \quad A^2 = K_X^2 + \ell - 4 - d + 12S.$$

In both minimal model cases one has $b \leq -2$: in the fixed-point case at least one arm is created after D_0 appears, while the D -preserving possibilities were computed above. For each spherical triple, $\sum_i a_i/m_i < 2$. Hence $d < 0$, and $B = D_0 \cup E_1 \cup E_2 \cup E_3$ is a negative-definite tree. Since X is rational,

$$e(X) = 12 - K_X^2, \quad \tau(X) = K_X^2 - 8.$$

The tree B has $1 + \ell$ components, Euler characteristic $\ell + 2$, and signature $-(1 + \ell)$. Additivity therefore gives

$$(106) \quad \chi(M) = 10 - K_X^2 - \ell, \quad \tau(M) = K_X^2 + \ell - 7.$$

Equations (101) and (102) give

$$\Theta_M = -1 - \frac{d}{3} - 4S.$$

Consequently

$$(107) \quad 2\chi(M) + 3(\tau(M) + \Theta_M) = K_X^2 + \ell - 4 - d - 12S.$$

Comparing (105) and (107) proves the assertion. $\square$

Lemma A.6. *In the fixed-fibre case of Proposition A.3, the Hitchin–Thorpe identity is equivalent to*

$$(108) \quad s(1, n) = s(1, r) + s(1, s).$$

The same condition holds whether or not the optional blow-up at q_+ is performed.

Proof. First consider the case where there is no blow-up at q_+ . The central weight is $\tilde{D}^2 = -2$. The inherited arm $[-n]$ has continued fraction $n/1$, while the rooted arms satisfy

$$\underbrace{[2, \dots, 2]}_{r-1} = \frac{r}{r-1}, \quad \underbrace{[2, \dots, 2]}_{s-1} = \frac{s}{s-1}.$$

Thus the Seifert data are

$$(109) \quad (-2; (n, 1), (r, r-1), (s, s-1)).$$

Thus

$$d = \frac{1}{n} - \frac{1}{r} - \frac{1}{s} < 0.$$

Set

$$T = s(1, n) - s(1, r) - s(1, s).$$

The pullback of A to the iterated blow-up of H_n is

$$(110) \quad \left(1 + \frac{1}{n}\right) C_- + (n+1)F - \frac{r-1}{r} \sum_{j=1}^r e_j - \frac{s-1}{s} \sum_{j=1}^s f_j,$$

where e_j, f_j are total-transform exceptional classes. Squaring gives

$$(111) \quad \begin{aligned} A^2 &= n + 6 + \frac{1}{n} - r - s - \frac{1}{r} - \frac{1}{s} \\ &= 3 - d + 12T. \end{aligned}$$

The boundary divisor is a negative-definite tree, and direct additivity of Euler characteristic and signature gives

$$\chi(M) = 3, \quad \tau(M) = 0.$$

Equations (101)–(102) give

$$\Theta_M = -1 - \frac{d}{3} - 4T.$$

Consequently the right-hand side of (86) is

$$2\chi(M) + 3(\tau(M) + \Theta_M) = 3 - d - 12T.$$

Comparison with (111) gives $T = 0$.

A blow-up at q_+ changes (A^2, χ, τ) by $(-1, +1, -1)$ and does not change the eta invariant. It therefore subtracts one from both sides of (86), and the same condition remains. $\square$

By our normalization, the model is taken to be H_N , with $N > 0$, and the source is placed at $F_0 \cap C_-$. There is no blow-up at $C_- \cap F_\infty$. We arrange the blow-up factorization so that the source extraction and the construction of the three arms are performed first. We call a configuration obtained by performing admissible blow-ups supported over the two axes $F_0 \cup C_-$ a *base configuration*. Here supported over means that each center lies in the successive total transform of $F_0 \cup C_-$; this includes the source extraction and the construction of the three arms. The possible subsequent blow-ups will be classified in Lemma A.8.

Proposition A.7. *Every base configuration is one of the following:*

S	source weights	arms	resulting pair
H_1 or H_0	$(1, 2)$ or $(3, 1)$	$[-2], [-3], [-2, -2]$	the D -preserving $(2, 3, 3)$ pair
H_0	$(2, 1)$	$[-2], [-2], [-2]$	the D -preserving $(2, 2, 2)$ pair
H_0	$(3, 2)$	$[-3], [-2, -2], [-2]$	a second, non- D -preserving $(2, 3, 3)$ pair
H_2	$(2, 1)$	$[-2], [-2, -3], [-2]$	a non- D -preserving $(2, 2, 5)$ pair.

Proof. Write

$$u = F_0, \quad v = C_-, \quad \lambda = pu + qv, \quad K = Np + q,$$

with the fan conventions preceding (98). The first blow-up of the source extraction is centered at $F_0 \cap C_-$. Hence the proper transform of the negative section satisfies

$$(C'_-)^2 \leq -N - 1 \leq -2.$$

If C_- were not absorbed, Lemma A.4 would put C'_- outside B , contradicting (97). Thus C_- is necessarily absorbed. Consequently there are only two cases, according as F_0 is outside B or absorbed. Set

$$R(x, y) := \frac{x}{y} + \frac{y}{x} + \frac{1}{xy}.$$

Only C_- is absorbed. The neighbors of the F_0 -ray u after the source extraction are $-v$ and $\lceil p/q \rceil u + v$. Therefore

$$(F'_0)^2 = -\left\lceil \frac{p}{q} \right\rceil.$$

Nor can F'_0 be blown up at its opposite endpoint: since $(F'_0)^2 \leq -1$, that blow-up would leave outside B a curve of square at most -2 . Since F'_0 is outside B , (97) excludes $p > q$. If $p = q$, primitivity gives $p = q = 1$, and F'_0 is then a (-1) -curve outside B meeting $\tilde{D}$, which is excluded by the same inequality. Consequently

$$(113) \quad 0 < p < q.$$

In particular $q \geq 2$. By (98), the two pre- D arm orders are q and K .

Let t be the order of the rooted third arm. The inversion invariance and oddness of Dedekind sums reduce (103) to

$$-s(p, q) + s(p, K) - s(1, t) = 0.$$

Since $K \equiv q \pmod{p}$, subtraction of the Dedekind reciprocity formulas for (p, K) and (p, q) gives

$$(114) \quad R(p, K) - R(p, q) = t + 2/t - 3.$$

We solve this equation over the spherical triples (q, K, t) , where $K = Np + q > q$. In the dihedral case $(q, K, t) = (2, m, 2)$, it becomes

$$(m - 2)(p^2 - (2m - 1)) = 0.$$

Here $m > 2$, so $p^2 = 2Np + 3$, or $p(p - 2N) = 3$. Its only positive solution is

$$(p, q, N, K, t) = (3, 2, 1, 5, 2),$$

which violates (113). In the tetrahedral case, $(q, K, t) = (2, 3, 3)$ and $Np = 1$, giving the solution

$$(p, q, N, K, t) = (1, 2, 1, 3, 3).$$

The octahedral and icosahedral placements are exhausted by the following substitutions; the last two columns are the two sides of (114):

$(q, K; t)$	(p, N)	$R(p, K) - R(p, q)$	$t + 2/t - 3$
$(2, 3; 4)$	$(1, 1)$	$2/3$	$3/2$
$(2, 4; 3)$	$(1, 2)$	$3/2$	$2/3$
$(3, 4; 2)$	$(1, 1)$	$5/6$	0
$(2, 3; 5)$	$(1, 1)$	$2/3$	$12/5$
$(2, 5; 3)$	$(1, 3), (3, 1)$	$12/5, 0$	$2/3$
$(3, 5; 2)$	$(1, 2), (2, 1)$	$26/15, 2/3$	0

Thus the tetrahedral tuple is the unique possibility in this case.

For this tuple the starting surface is H_1 with source weights $(1, 2)$. After contracting the (-1) -curve we obtain the minimal model $\mathbb{P}^2$ with source weights $(3, 2)$. For the source weights $(3, 2)$, the

source extraction gives the two pre- D arms $[-2]$ and $[-3]$. The remaining order is $t = 3$; after $\tilde{D}$ appears, the one-sided construction at a third point of $\tilde{D}$ gives the rooted A_2 -arm $[-2, -2]$. Hence the configuration has arms

$$[-2], \quad [-3], \quad [-2, -2].$$

This graph is isomorphic to the D -preserving $(2, 3, 3)$ pair of Proposition A.3. It is the first row of (112).

Both F_0 and C_- are absorbed. First consider a blow-up at the fixed point of F_0 opposite the source; it commutes with the source extraction. Since F_0 is the outermost component of its arm, the exceptional curve of such a blow-up lies outside B . A second blow-up over that endpoint would leave an outside curve of square at most -2 , contrary to (97). Thus at most one such modification occurs. If it occurs, it inserts the ray

$$h = u - v$$

between $-v$ and u . Contracting u then gives the elementary transformation $H_N \rightarrow H_{N-1}$. In the new model v is the absorbed negative section, h is the outside ruling which is not absorbed, and

$$\lambda = ph + (p + q)v, \quad Np + q = (N - 1)p + (p + q).$$

If $N \geq 2$, this is precisely the preceding one-absorbed-axis case. The only new case is therefore the case of $H_1 \rightarrow H_0$.

Put $a = v$ and $b = h = u - v$ on H_1 . After u is contracted, the four rays $\{\pm a, \pm b\}$ form the fan of H_0 ; the a -axis is absorbed and the b -axis is outside B . Since $u = a + b$,

$$\lambda = (p + q)a + pb.$$

Write the resulting H_0 source weights as

$$(P, Q) = (p + q, p).$$

The two pre- D arms have order P , with residues Q^{-1} and $-Q^{-1}$ modulo P . Their Dedekind sums cancel, so (103) reduces to $s(t - 1, t) = 0$. Since $t > 1$, (99) gives $t = 2$. Sphericity of $(P, P, 2)$ then gives $P = 2$ or $P = 3$.

The square of the outside b -axis is $-[Q/P]$. By (97) it must be -1 ; since $P > 1$ and $(P, Q) = 1$, this is equivalent to $0 < Q < P$. Primitivity now leaves exactly

$$(P, Q) = (2, 1), \quad (3, 1), \quad (3, 2).$$

Their pre- D arms are, respectively,

$$([-2], [-2]), \quad ([-2, -2], [-3]), \quad ([-3], [-2, -2]),$$

and the rooted third arm is $[-2]$. For $(P, Q) = (2, 1)$ all three arms agree, and the pair is the D -preserving $(2, 2, 2)$ pair with model H_0 . For $(P, Q) = (3, 1)$, this is the D -preserving $(2, 3, 3)$ pair with model H_0 , which is isomorphic to the first row, where the model is H_1 with source weights $(1, 2)$. So after identifying the $(P, Q) = (3, 1)$ case with the first row, we have exactly the two H_0 rows of (112).

In the remaining case no modification can be performed at the opposite fixed points of C_- and F_0 . After the source extraction, the proper transform of the absorbed F_0 -axis has square $-[p/q]$. As a component of a minimal Hirzebruch–Jung arm it has square at most -2 ; hence

$$0 < q < p.$$

The two pre- D arm orders are p and $K = Np + q$. Using (98), inversion invariance, and $K \equiv q \pmod{p}$, the Dedekind condition is

$$s(K, p) + s(p, K) - s(1, t) = 0.$$

Reciprocity gives

$$(115) \quad \frac{p}{K} + \frac{K}{p} + \frac{1}{pK} = t + \frac{2}{t}.$$

Here $(p, K) = 1$ and $K > p$. The spherical possibilities are

triple	$(p, K; t)$	left side	right side
$(2, 2, n)$	$(2, n; 2)$	$\frac{n^2+5}{2n}$	3
$(2, 3, 3)$	$(2, 3; 3)$	$\frac{7}{3}$	$\frac{11}{3}$
$(2, 3, 4)$	$(2, 3; 4), (3, 4; 2)$	$\frac{7}{3}, \frac{13}{6}$	$\frac{9}{2}, 3$
$(2, 3, 5)$	$(2, 3; 5), (2, 5; 3), (3, 5; 2)$	$\frac{7}{3}, 3, \frac{7}{3}$	$\frac{27}{5}, \frac{11}{3}, 3$.

The first row gives $n = 1$ or 5 , and $K > p$ leaves $n = 5$; every other row fails. Therefore

$$(p, q, N, K, t) = (2, 1, 2, 5, 2).$$

The source extraction on H_2 has rays

$$u, \quad \lambda = 2u + v, \quad u + v, \quad v,$$

with relevant self-intersections

$$u^2 = -2, \quad \lambda^2 = -1, \quad (u + v)^2 = -2, \quad v^2 = -3.$$

The two pre- D arms are consequently $[-2]$ and $[-2, -3]$. Two one-sided blow-ups at a third point of $\tilde{D}$ give the rooted $[-2]$ -arm and change $\tilde{D}^2$ from -1 to -2 . This is the H_2 row of (112).

We have therefore exhausted all possible base configurations. $\square$

For each ruled model in Proposition A.7, let C_+ be the invariant section disjoint from C_- and put

$$q_+ = C_+ \cap F_\infty.$$

This is the unique fixed point outside the successive total transform of the two axes. Proposition A.7 exhausts all admissible blow-ups supported over those axes. Conversely, blowing up/down over q_+ leaves the boundary orbifold data, and hence the spherical and Dedekind conditions, unchanged. Thus every remaining possibility is obtained from a base configuration by blow-ups over q_+ . We call this the *remote case*.

Lemma A.8. *For each row of (112), the blow-up of q_+ is admissible, but no second blow-up over q_+ is compatible with ampleness. Up to isomorphisms, the associated base and remote pairs are as follows:*

$$(116)$$

S	source weights	base configuration	remote pair
H_0	$(2, 1)$	D -preserving $(2, 2, 2)$	D -preserving remote $(2, 2, 2)$
H_1	$(1, 2)$	D -preserving $(2, 3, 3)$	$common\ D$ -preserving remote $(2, 3, 3)$
H_0	$(3, 2)$	$second, non$ - D -preserving $(2, 3, 3)$	
H_2	$(2, 1)$	non - D -preserving $(2, 2, 5)$	non - D -preserving remote $(2, 2, 5)$.

Proof. First notice that there is at most one blow-up at q_+ . Two or more blow-ups over q_+ produce nonboundary curves with square at most -2 , which is impossible.

It remains to identify the remote pair obtained from the $H_0, (3, 2)$ row and $H_1, (1, 2)$ row. This can be seen explicitly by tracing the blow-ups. $\square$

Theorem A.9. *Under the hypotheses of this appendix, a three-point spherical compactification of a Hermitian non-Kähler gravitational instanton is one of seven pairs. They consist of two $(2, 2, 2)$ pairs, three $(2, 3, 3)$ pairs, and two $(2, 2, 5)$ pairs described below.*

Proof. There are already the seven pairs from Proposition A.7 and Lemma A.8. It remains to check that the D -preserving cases do not contribute new candidates.

Among D -preserving presentations, Proposition A.3 and Lemma A.6 leave only

$$(2, 2, 2), \quad (2, 3, 3).$$

Indeed, with $s_m = s(1, m)$,

$$s_2 = 0, \quad s_3 = \frac{1}{18}, \quad s_4 = \frac{1}{8}, \quad s_5 = \frac{1}{5}.$$

Thus $(2, 2, k)$ forces $k = 2$; the tetrahedral case requires the inherited order-three arm; and the octahedral and icosahedral triples fail.

For $(2, 2, 2)$, start with H_2 and a fixed ruling fibre, retain the negative section as one arm, and create two rooted A_1 -arms. This gives $\text{Bl}_4 H_2$. The unique remote blow-up at q_+ gives $\text{Bl}_5 H_2$, also presented as the triple-root $\text{Bl}_6 \mathbb{P}^2$ model. These are isomorphic to the $H_0, (2, 1)$ cases in Proposition A.7 and Lemma A.8, respectively.

For the D -preserving $(2, 3, 3)$ pair, start with H_3 , retain the negative section $[-3]$, and create rooted A_1 - and A_2 -arms. This gives $\text{Bl}_5 H_3$. Its remote q_+ -blow-up gives $\text{Bl}_6 H_3$. These are isomorphic to the $H_1, (1, 2)$ cases in Proposition A.7 and Lemma A.8, respectively. $\square$

For reference, their numerical data are as follows. A superscript “nr” denotes the non- D -preserving $(2, 3, 3)$ presentation. Here $d = D^2$ is the orbifold normal degree, and Y is the oriented boundary of a neighborhood of the resolved boundary divisor.

marked resolution	Seifert data	d	A^2	(χ, τ)	Θ_M
$\text{Bl}_4 H_2$ (222)	$(-2; (2, 1), (2, 1), (2, 1))$	$-\frac{1}{2}$	$\frac{7}{2}$	$(3, 0)$	$-\frac{5}{6}$
$\text{Bl}_5 H_2$ (222)	$(-2; (2, 1), (2, 1), (2, 1))$	$-\frac{1}{2}$	$\frac{7}{2}$	$(4, -1)$	$-\frac{13}{6}$
$\text{Bl}_5 H_3$ (233)	$(-2; (3, 1), (2, 1), (3, 2))$	$-\frac{1}{2}$	$\frac{5}{2}$	$(3, 0)$	$-\frac{5}{6}$
$\text{Bl}_6 H_3$ (233)	$(-2; (3, 1), (2, 1), (3, 2))$	$-\frac{1}{2}$	$\frac{5}{2}$	$(4, -1)$	$-\frac{13}{6}$
$\text{Bl}_5 H_0$ (233) ^{nr}	$(-2; (3, 1), (2, 1), (3, 2))$	$-\frac{1}{2}$	$\frac{7}{2}$	$(3, 0)$	$-\frac{5}{6}$
$\text{Bl}_4 H_2$ (225)	$(-2; (2, 1), (5, 3), (2, 1))$	$-\frac{2}{3}$	$\frac{22}{3}$	$(2, 1)$	$-\frac{13}{15}$
$\text{Bl}_5 H_2$ (225)	$(-2; (2, 1), (5, 3), (2, 1))$	$-\frac{2}{3}$	$\frac{22}{3}$	$(3, 0)$	$-\frac{13}{15}$

We denote these seven rows, in their displayed order, by

$$222_b, \quad 222_r, \quad 233_b, \quad 233_r, \quad 233_b^{\text{nr}}, \quad 225_b, \quad 225_r.$$

Here b denotes that it is a base configuration and r its remote blow-up.

A.4. The extremal scalar-curvature obstruction. We now test the seven pairs by the scalar curvature of the finite-volume extremal Kähler metric $g = s_g^2 h$. The compact calculation motivating this test is due to LeBrun–Simanca [48, Section 3.3]; its ALE version is used in [50, Section 7].

Let X be the primitive generator of the effective circle action, normalized to have period 2π , and choose its moment map x by

$$(117) \quad \iota_X \omega_g = -dx, \quad x = 0 \text{ at } D, \quad x = \ell \text{ at the opposite maximum.}$$

For an actual instanton the extremal field $\mathcal{K} = J\nabla s_g$ is a positive multiple of X . Since the convention above gives $X = J\nabla x$ and $s_g \rightarrow 0$ at D , this means

$$(118) \quad s_g = cx, \quad c = \frac{2\pi}{T_{\mathcal{K}}} > 0.$$

For each of the seven actions, the opposite maximum is an isolated fixed point. At an interior saddle p_j , order the local coordinates so that

$$(119) \quad e^{i\theta} \cdot (z_1, z_2) = (e^{ir_j\theta} z_1, e^{-iq_j\theta} z_2), \quad r_j, q_j > 0, \quad (r_j, q_j) = 1,$$

and write $x_j = x(p_j)$. Equivariant Darboux coordinates may be chosen so that

$$x - x_j = \frac{1}{2}(r_j|z_1|^2 - q_j|z_2|^2).$$

Proposition A.10. *Put*

$$\varepsilon_j := \frac{1}{r_j} - \frac{1}{q_j}.$$

For the finite-volume Kähler metric associated with a hypothetical instanton,

$$(120) \quad \int_M s_g d\text{vol}_g = 8\pi^2 \left[\delta\ell + \sum_j \varepsilon_j (\ell - x_j) \right],$$

$$(121) \quad \int_M x s_g d\text{vol}_g = 4\pi^2 \left[\delta\ell^2 + \sum_j \varepsilon_j (\ell^2 - x_j^2) \right].$$

Equivalently,

$$(122) \quad \int_M \left(x - \frac{\ell}{2} \right) s_g d\text{vol}_g = 4\pi^2 \sum_j \varepsilon_j x_j (\ell - x_j).$$

Here $\delta = \chi_{\text{orb}}(D)$.

Proof. On a chamber of regular values, use the local ansatz

$$g = W dx^2 + W^{-1} \eta^2 + W e^\varphi (du^2 + dv^2), \quad \eta(X) = 1.$$

We use the following two area forms in this calculation:

$$\omega_x = W e^\varphi du \wedge dv, \quad \widehat{\omega}_x = e^\varphi du \wedge dv.$$

If ρ_x is the Ricci form of the orbifold metric $e^\varphi(du^2 + dv^2)$, the local computation leading to [48, equation (3.16)] gives

$$(123) \quad s_g d\text{vol}_g = \left(2\rho_x - \frac{\partial^2 \widehat{\omega}_x}{\partial x^2} \right) \wedge dx \wedge \eta.$$

This identity is local.

For every regular value, let

$$\Sigma_x := x^{-1}(x)/S^1$$

be the orbifold Kähler reduction, and set

$$\mathcal{U}(x) := \int_{\Sigma_x} \widehat{\omega}_x.$$

Orbifold Gauss–Bonnet and integration along the circle give, for every affine function f ,

$$(124) \quad \int_M f(x) s_g d\text{vol}_g = 2\pi \int_0^\ell f(x) (4\pi \chi_{\text{orb}}(\Sigma_x) - \mathcal{U}''(x)) dx.$$

There is no atomic contribution from $\mathcal{U}''$ at a saddle by a local computation.

At the cusp, x is a positive multiple of s_g and (63)–(64) give

$$W^{-1} = O(x^2), \quad e^\varphi = O(x^2), \quad \partial_x e^\varphi = O(x).$$

After integration in the finite orbifold charts, these estimates give

$$\mathcal{U}(x) = O(x^2), \quad \mathcal{U}'(x) = O(x).$$

At the isolated maximum, the standard equivariant normal form gives

$$W^{-1} = O(\ell - x), \quad \int_{\Sigma_x} \omega_x = O(\ell - x).$$

Since $\widehat{\omega}_x = W^{-1}\omega_x$, it follows that

$$\mathcal{U}(x) = O((\ell - x)^2), \quad \mathcal{U}'(x) = O(\ell - x).$$

Thus the $\mathcal{U}''$ term in (124) integrates to zero for $f = 1$ and $f = x$.

Below a saddle of weights $(r_j, -q_j)$, its local quotient has cone order q_j ; above it, the cone order is r_j . Consequently

$$(125) \quad \chi_{\text{orb}}(\Sigma_x) = \delta + \sum_j \varepsilon_j \mathbf{1}_{\{x > x_j\}}.$$

Substituting this step into (124) proves (120) and (121). Their centered combination is (122). $\square$

We next compute the volume moments. Define the reduced symplectic area

$$\mathcal{V}(x) := \int_{\Sigma_x} \omega_x.$$

If the absolute values of the two weights at the isolated maximum are P, Q , the Duistermaat–Heckman formula implies that across a critical value x_j we have

$$\mathcal{V}'(x_j+) - \mathcal{V}'(x_j-) = -\frac{2\pi}{r_j q_j}.$$

Near a maximum of weights $(-P, -Q)$, the reduced area is $2\pi(\ell - x)/(PQ)$. Summing the slope jumps while integrating backward from the maximum gives

$$(126) \quad \mathcal{V}(x) = 2\pi \left[\frac{\ell - x}{PQ} - \sum_j \frac{(x_j - x)_+}{r_j q_j} \right].$$

Here $(u)_+ := \max\{u, 0\}$. Equivalently,

$$\mathcal{V}'' = -2\pi \sum_j \frac{1}{r_j q_j} \delta_{x_j}, \quad \mathcal{V}(x) = \frac{2\pi}{PQ}(\ell - x) \quad \text{near } \ell.$$

Put

$$(127) \quad y_j := \frac{x_j}{\ell}, \quad \gamma_j := \frac{1}{r_j q_j}, \quad \mathbf{B}_k := \frac{1}{PQ} - \sum_j \gamma_j y_j^k \quad (k \geq 1).$$

The reduced area at the cusp is $\mathcal{V}(0) = 2\pi\ell\mathbf{B}_1$. Since (126) is concave, positivity of the reduced areas is equivalent to

$$(128) \quad 0 < y_j < 1, \quad \mathbf{B}_1 > 0.$$

Fiber integration of $d \text{vol}_g = \omega_g^2/2$ now gives

$$(129) \quad T_k := \int_M x^k d \text{vol}_g = \frac{4\pi^2 \ell^{k+2}}{(k+1)(k+2)} \mathbf{B}_{k+2} \quad (k = 0, 1, 2).$$

Indeed, both integrals in (126) use

$$\int_0^a x^k (a - x) dx = \frac{a^{k+2}}{(k+1)(k+2)}.$$

For $k = 1, 2$, set

$$(130) \quad C_k := \delta + \sum_j \varepsilon_j (1 - y_j^k).$$

Proposition A.10 becomes

$$(131) \quad S_0 := \int_M s_g d\text{vol}_g = 8\pi^2 \ell C_1, \quad S_1 := \int_M x s_g d\text{vol}_g = 4\pi^2 \ell^2 C_2.$$

For an extremal metric whose extremal vector field is proportional to X , scalar curvature is affine in the moment map:

$$s_g = \alpha x + \beta.$$

Writing $\Delta = T_0 T_2 - T_1^2 > 0$, the equations $S_0 = \alpha T_1 + \beta T_0$ and $S_1 = \alpha T_2 + \beta T_1$ give

$$(132) \quad \alpha = \frac{S_1 T_0 - S_0 T_1}{\Delta}, \quad \beta = \frac{S_0 T_2 - S_1 T_1}{\Delta}.$$

For the conformal Kähler metric of a gravitational instanton, $s_g > 0$ on M and $s_g \rightarrow 0$ precisely at the added divisor. Thus $\beta = 0$ and $\alpha > 0$. Equations (129) and (131) show that

$$(133) \quad \beta = 0 \iff C_1 B_4 = C_2 B_3.$$

Under this condition,

$$(134) \quad \alpha = \frac{12C_1}{\ell^2 B_3}, \quad \int_M s_g^2 d\text{vol}_g = 48\pi^2 \frac{C_1^2 B_4}{B_3^2}.$$

Since (89) requires the last integral to equal $48\pi^2 A^2$, every candidate must satisfy

$$(135) \quad C_1 B_4 = C_2 B_3, \quad C_1^2 B_4 = A^2 B_3^2.$$

The discrete data entering these equations are read directly from the orbit graphs already used in the classification. We write a saddle of weights $(r, -q)$ as (r, q) in the following table; subscripts name its normalized critical value.

	case	δ	A^2	(P, Q)	(r_j, q_j)	$B_1 > 0$
(136)	222 _b	$\frac{1}{2}$	$\frac{7}{2}$	(1, 2)	$(1, 2)_{y_1}, (1, 2)_{y_2}$	$y_1 + y_2 < 1$
	222 _r	$\frac{1}{2}$	$\frac{5}{2}$	(1, 1)	$(1, 2)_{y_1}, (1, 2)_{y_2}, (1, 2)_{y_3}$	$y_1 + y_2 + y_3 < 2$
	233 _b	$\frac{1}{6}$	$\frac{7}{2}$	(1, 3)	$(1, 2)_{y_1}, (1, 3)_{y_2}$	$3y_1 + 2y_2 < 2$
	233 _r	$\frac{1}{6}$	$\frac{5}{2}$	(1, 2)	$(1, 2)_{y_1}, (1, 3)_{y_2}, (2, 3)_{y_3}$	$3y_1 + 2y_2 + y_3 < 3$
	233 _b ^{nr}	$\frac{1}{6}$	$\frac{7}{2}$	(2, 3)	$(1, 2)_{y_1}, (2, 3)_{y_2}$	$3y_1 + y_2 < 1$
	225 _b	$\frac{1}{5}$	$\frac{22}{5}$	(2, 5)	$(1, 2)_{y_1}$	$y_1 < \frac{1}{5}$
	225 _r	$\frac{1}{5}$	$\frac{17}{5}$	(2, 3)	$(1, 2)_{y_1}, (3, 5)_{y_2}$	$15y_1 + 2y_2 < 5$

The two necessary equations admit a uniform contradiction.

Proposition A.11. *None of the seven rows in (136) satisfies (135) under the condition (128).*

Proof. Put

$$k_j := q_j - r_j, \quad \sigma_+ := \frac{1}{P} + \frac{1}{Q}, \quad \Sigma_* := \sum_j \gamma_j k_j^2.$$

All the displayed saddles have $k_j > 0$. Define

$$\Lambda := PQ \sum_j \gamma_j \sqrt{\frac{k_j}{P+Q}}.$$

Direct substitution of the action data and the values of A^2 gives the following exact table.

case	σ_+	Σ_*	$A^2 + \Sigma_* = PQ\sigma_+^2$	Λ
222 _b	$\frac{3}{2}$	1	$\frac{9}{2}$	$\frac{2}{\sqrt{3}}$
222 _r	2	$\frac{3}{2}$	4	$\frac{3}{2\sqrt{2}}$
233 _b	$\frac{4}{3}$	$\frac{11}{6}$	$\frac{16}{3}$	$\frac{3}{4} + \frac{1}{\sqrt{2}}$
233 _r	$\frac{3}{2}$	2	$\frac{9}{2}$	$\frac{2(2+\sqrt{2})}{3\sqrt{3}}$
233 _{nr}	$\frac{5}{6}$	$\frac{2}{3}$	$\frac{25}{6}$	$\frac{4}{\sqrt{5}}$
225 _b	$\frac{7}{10}$	$\frac{1}{2}$	$\frac{49}{10}$	$\frac{5}{\sqrt{7}}$
225 _r	$\frac{5}{6}$	$\frac{23}{30}$	$\frac{25}{6}$	$\frac{3}{\sqrt{5}} + \frac{2}{5}\sqrt{\frac{2}{5}}$

In particular, every entry in the last column is strictly greater than one.

The quantities divided by below are positive. Indeed, $C_1 > 0$ because $\delta > 0$, $\varepsilon_j > 0$, and $0 < y_j < 1$; also $B_3 > B_1 > 0$. The two equations in (135) therefore first imply

$$(138) \quad C_1 C_2 = A^2 B_3, \quad C_2^2 = A^2 B_4.$$

For the first identity, multiply the zero-boundary equation by C_1 and compare it with the energy equation; division by B_3 gives $C_1 C_2 = A^2 B_3$. Multiply the zero-boundary equation by C_2 and use this first identity; division by B_3 gives $C_2^2 = A^2 B_4$.

Set $X_j = y_j^2$ and regard $\varepsilon = (\varepsilon_j)$ as a column vector. We have

$$(139) \quad \delta + \sum_j \varepsilon_j = \frac{1}{P} + \frac{1}{Q},$$

which is the orbifold Euler characteristic of a reduction just below the isolated maximum. From this,

$$C_2 = \sigma_+ - \sum_j \varepsilon_j X_j, \quad B_4 = \frac{1}{PQ} - \sum_j \gamma_j X_j^2.$$

Consequently the second equation in (138) is

$$(140) \quad \mathcal{J}(X) := \left(\sigma_+ - \sum_j \varepsilon_j X_j \right)^2 - A^2 \left(\frac{1}{PQ} - \sum_j \gamma_j X_j^2 \right) = 0.$$

Now $\varepsilon_j = \gamma_j k_j$. Using the middle identity in (137), direct completion of the square gives

$$(141) \quad \mathcal{J}(X) = (X - X^*)^T \left(A^2 \text{diag}(\gamma_j) + \varepsilon \varepsilon^T \right) (X - X^*), \quad X_j^* = \frac{k_j}{P + Q}.$$

The matrix is positive definite. Hence $\mathcal{J}$ has the unique zero $X = X^*$. Indeed, if $\Gamma = \text{diag}(\gamma_j)$ and $M_* = A^2 \Gamma + \varepsilon \varepsilon^T$, then $\Gamma X^* = \varepsilon / (P + Q)$ and $\varepsilon^T X^* = \Sigma_* / (P + Q)$. The identity $A^2 + \Sigma_* = PQ\sigma_+^2$ therefore gives $M_* X^* = \sigma_+ \varepsilon$, while its constant term gives (141).

At that zero, however, the reduced area inequality (128), after multiplication by PQ , would require

$$PQ \sum_j \gamma_j y_j < 1.$$

Its left side instead equals

$$PQ \sum_j \gamma_j \sqrt{X_j^*} = \Lambda > 1$$

by the last column of (137). Thus the unique zero of $\mathcal{J}$ lies outside (128). Under (128) one has $\mathcal{J} > 0$, contradicting (140). $\square$

Theorem A.12. *Under the hypotheses of this appendix, a Hermitian non-Kähler gravitational instanton cannot have a natural compactification with three quotient points on D whose orders form a spherical triple.*

Proof. Theorem A.9 reduces the case to the seven marked pairs in (136). A hypothetical instanton on any one of them must satisfy the zero-at-infinity and scalar energy equations (135). Proposition A.11 shows that no such Kähler class exists. $\square$

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